

Gasoline Profiles and Impact of Ethanol Blending in Asia

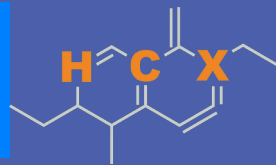
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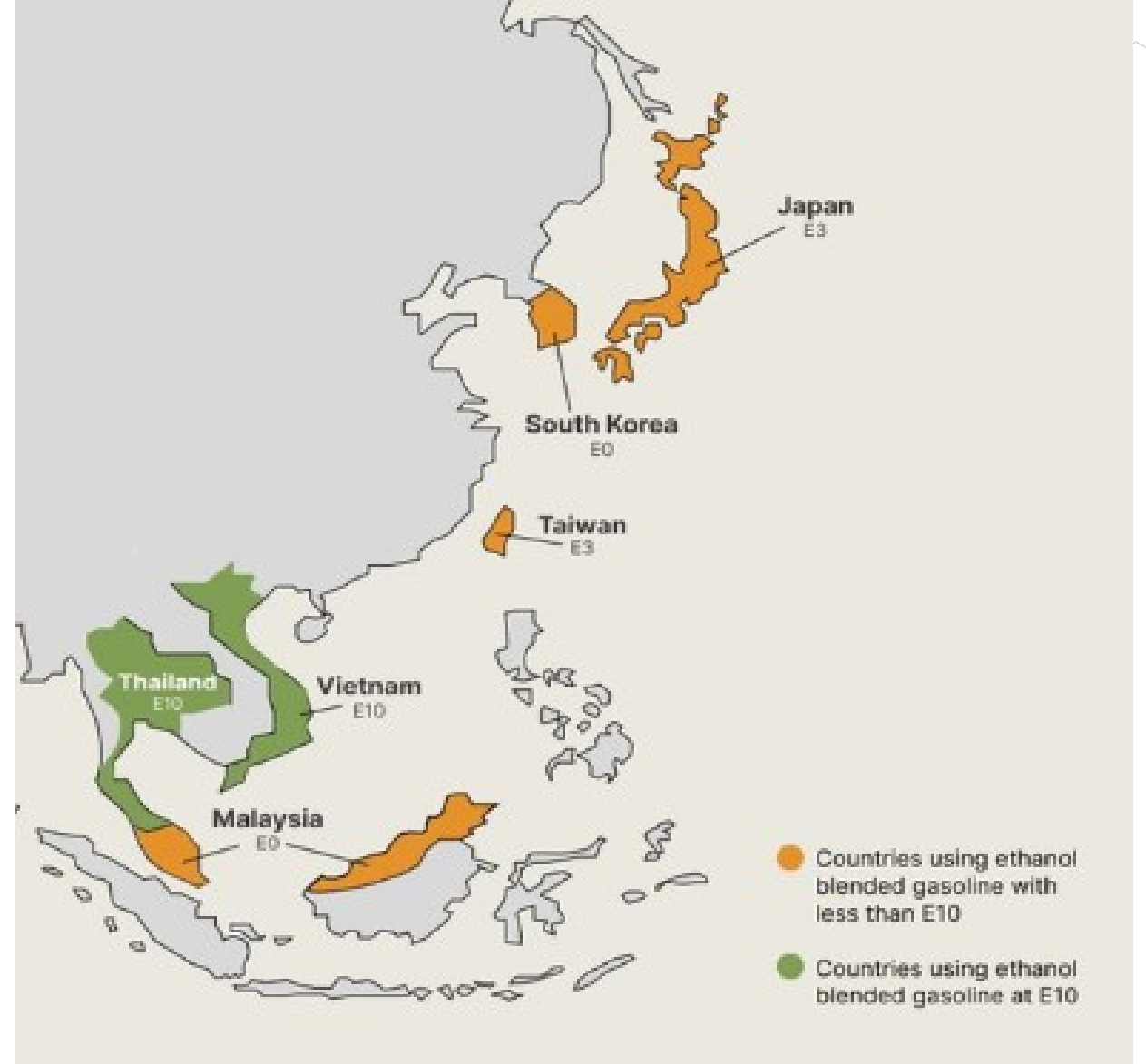


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Ethanol Blending in Asia

There are important fuel quality and environmental impact of vehicle emission challenges in these regions.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Asia to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.

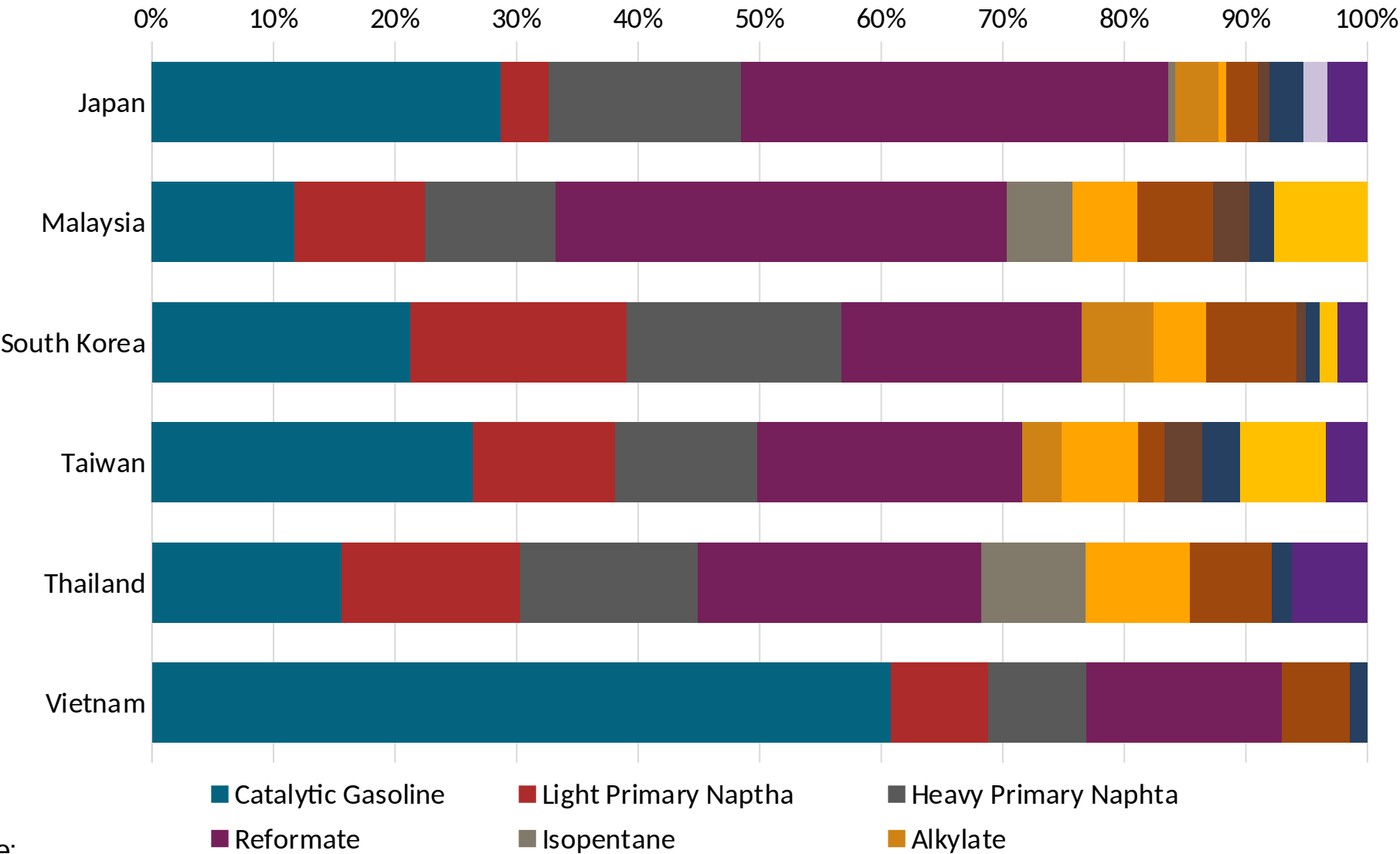


Seven countries with potential and additional use of ethanol were studied: 1) Gasoline market profiles; 2) Optimization of gasoline blends with ethanol and, 3) Environmental impact of gasolines blended with ethanol.

Gasoline Component Blending in Asia

Gasoline is a blend of a base gasoline and other components. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Asia are:

- Catalytic Gasoline
- Light Primary Naphta
- Heavy Primary Naphta
- Reformate
- Isopentane
- Alkylate
- Isomerate
- Hydrocracked Gasoline
- Coker Naphta
- n-Butane
- MTBE
- ETBE
- TAME
- Aromatics



Source:
Faro90

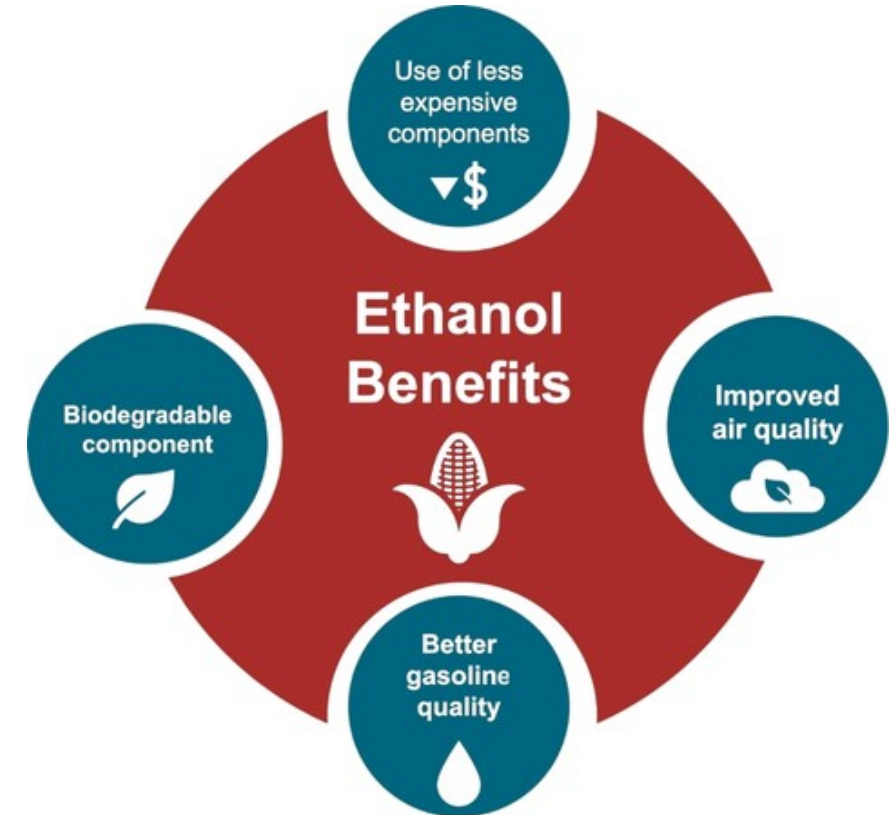
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component. The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives. To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

The blending model uses gasoline component spot average prices 2024 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.



Vehicle Emission Impact for Ethanol Gasoline Blending

The model used in this analysis takes as a reference the [International Vehicle Emissions Model \(IVE\)](#).

The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

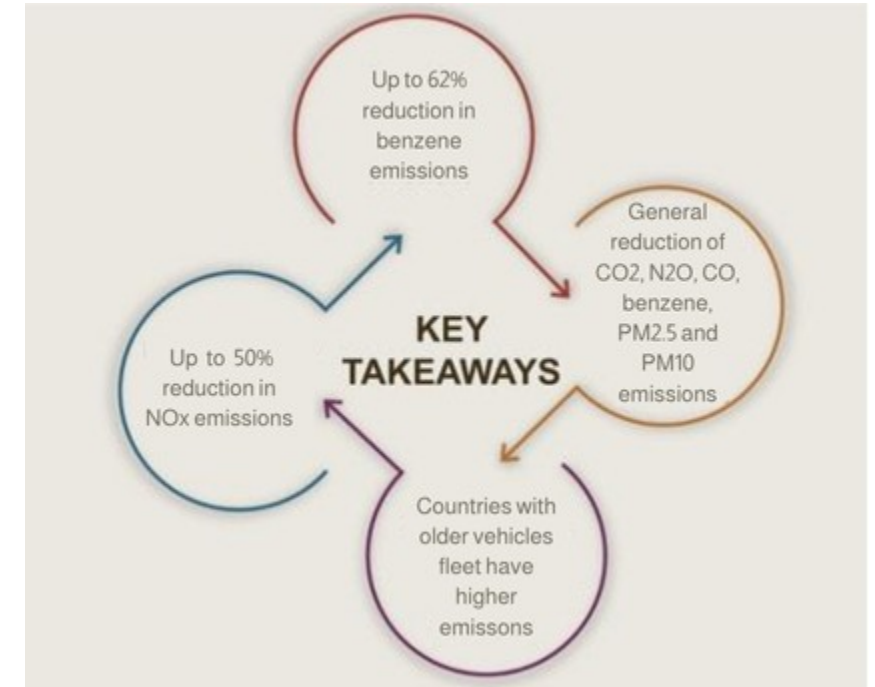
- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

**Source: Bureau of transportation statistics.*

Main Results



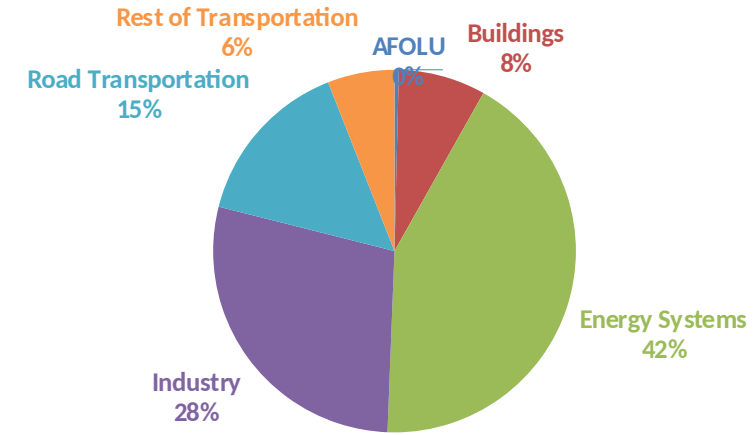
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

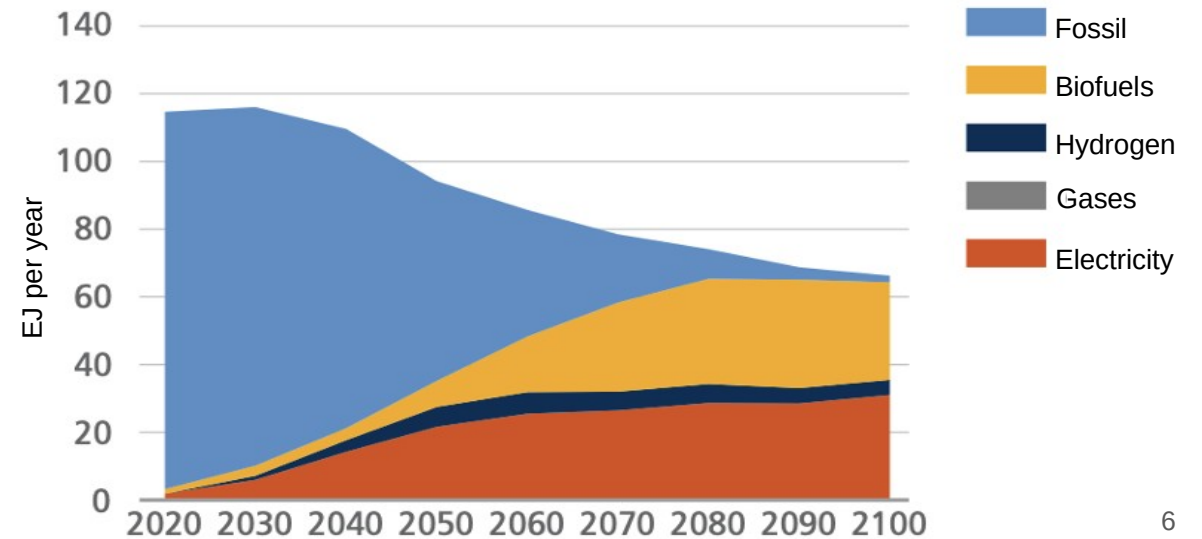
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE)**. Results are compared to the current country targets for 2035. Current 2024 country emission are those published by **IPCC**.

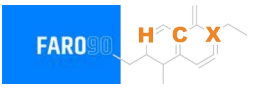
Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



Case Studies



1. [Japan](#)
2. [Malaysia](#)
3. [South Korea](#)
4. [Taiwan](#)
5. [Thailand](#)
6. [Vietnam](#)

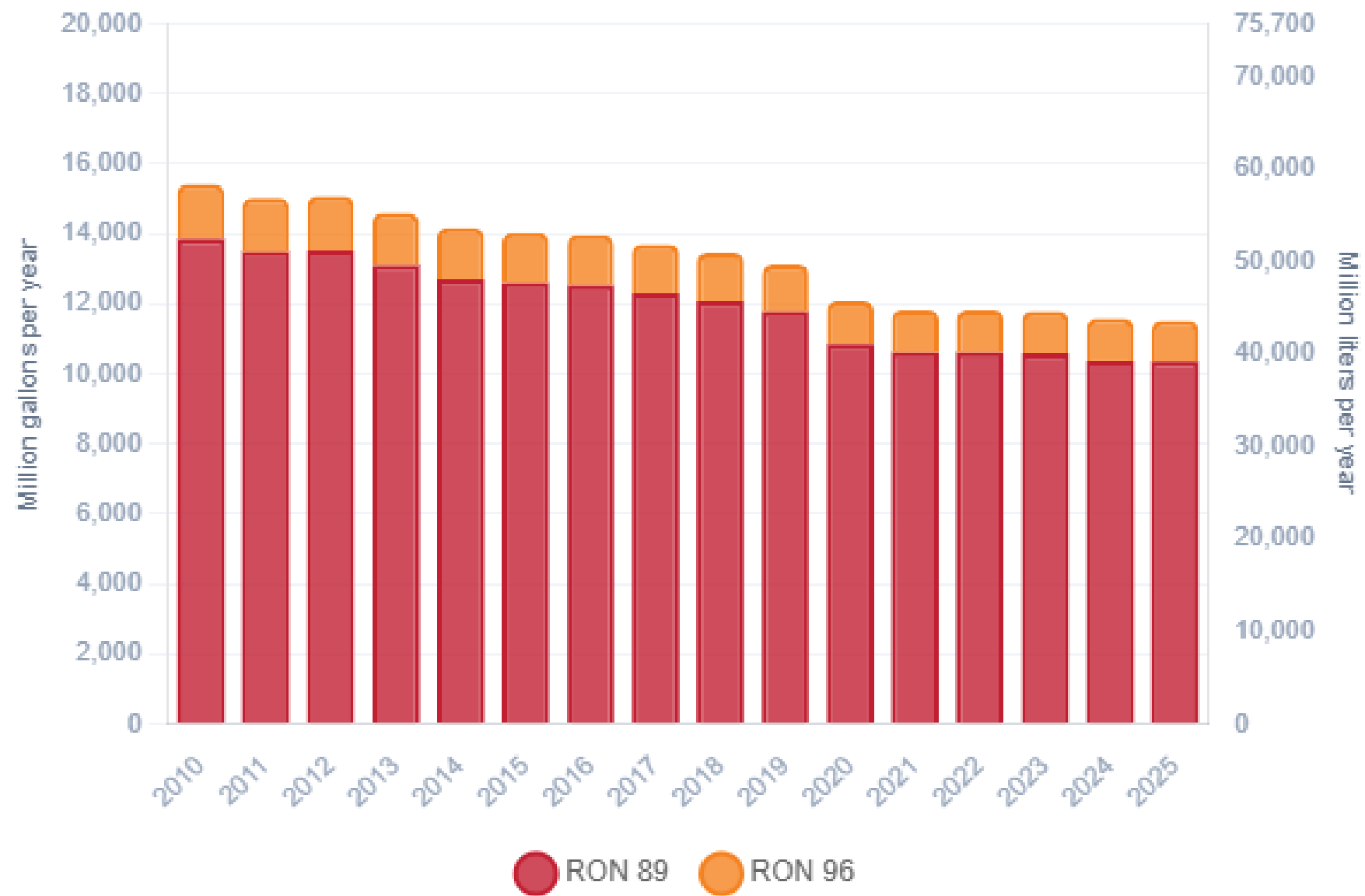
Ethanol Gasoline Blending in Japan



In 2025, gasoline consumption in Japan was 43,581 million liters (11,513 million gallons) and growth rate was -1.9%. Gasoline production and net imports were 42,543 and 877 million liters (11,107 and 231 million gallons) correspondingly. Japan complies with the JIS standards and offers two main grades: RON 89 and RON 96) with an average octane of 89.7 RON.

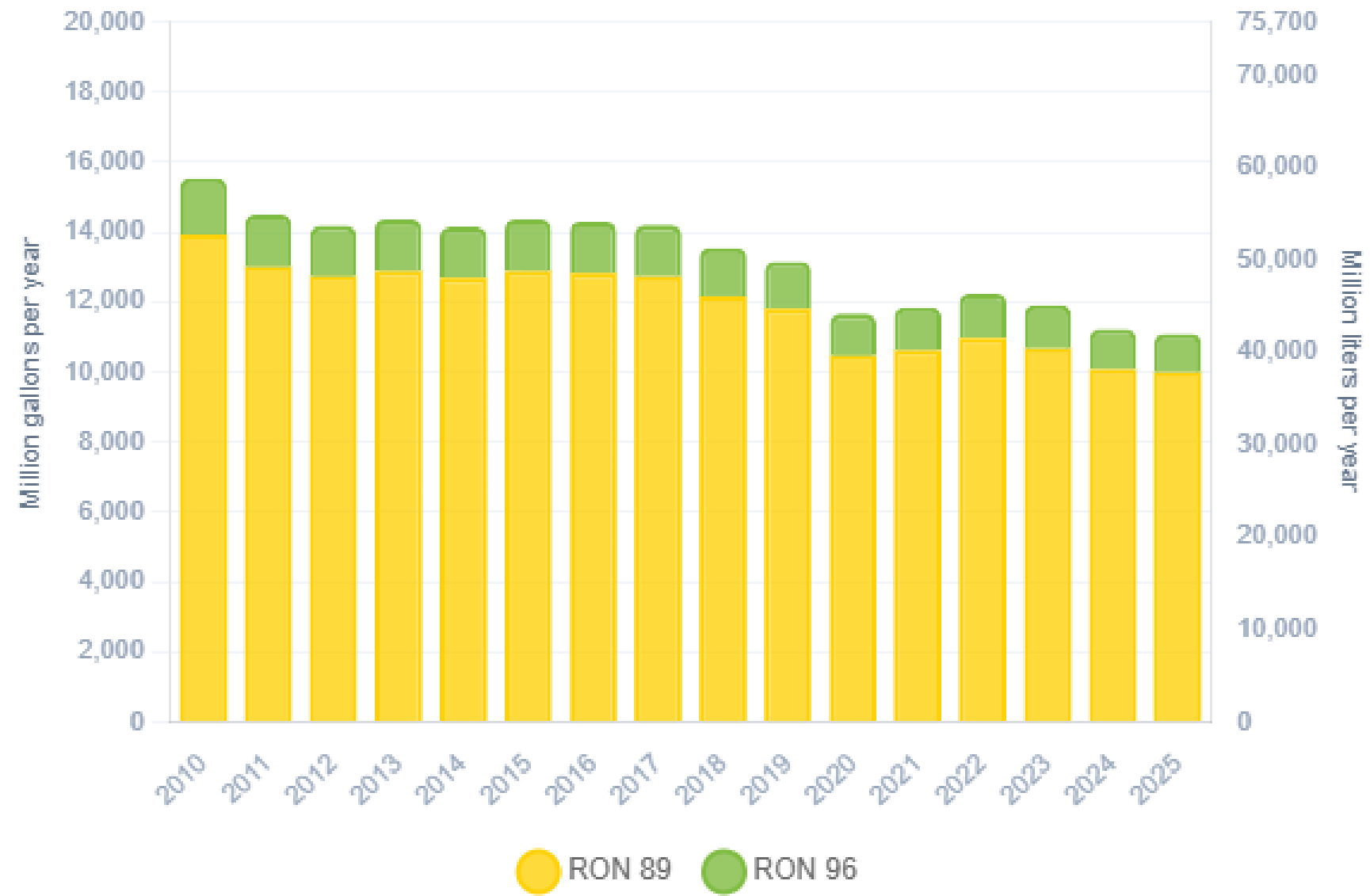
Japan currently allows E3. Last year's consumption was 745 million liters (197 million gallons) equivalent to 1.7%, primarily as the component ethyl tert-butyl ether (ETBE). There is no local production. Ethanol is imported as a blendstock for ETBE production and as ETBE. The Ministry of Economy, Trade and Industry (METI) aiming for E10 (10% ethanol) by 2030 and E20 (20% ethanol) by 2040. This initiative requires infrastructure improvements, as most existing gas stations are only equipped for the E3 blend.

Gasoline Demand in Japan



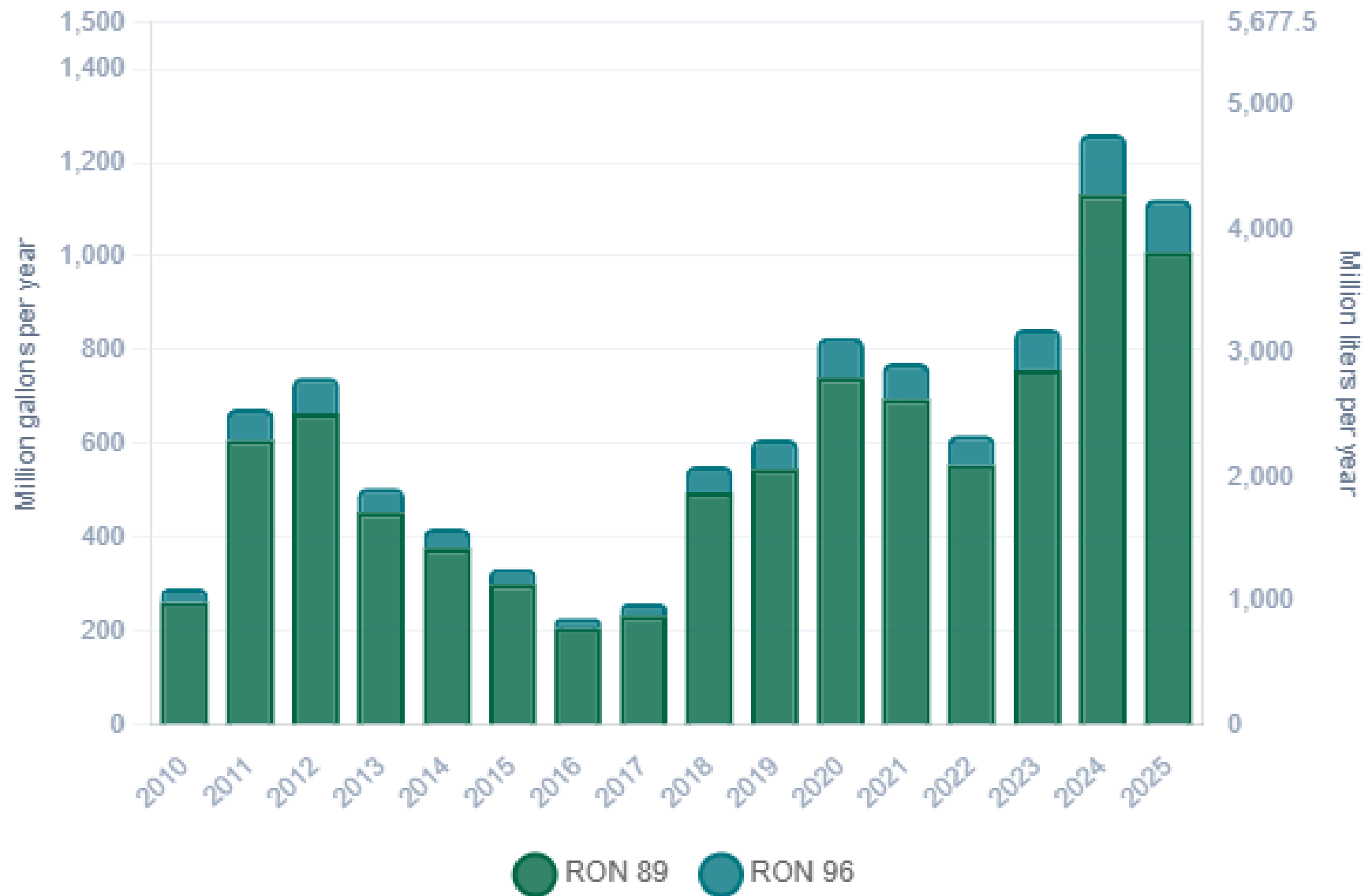
Source: METI

Gasoline Production in Japan



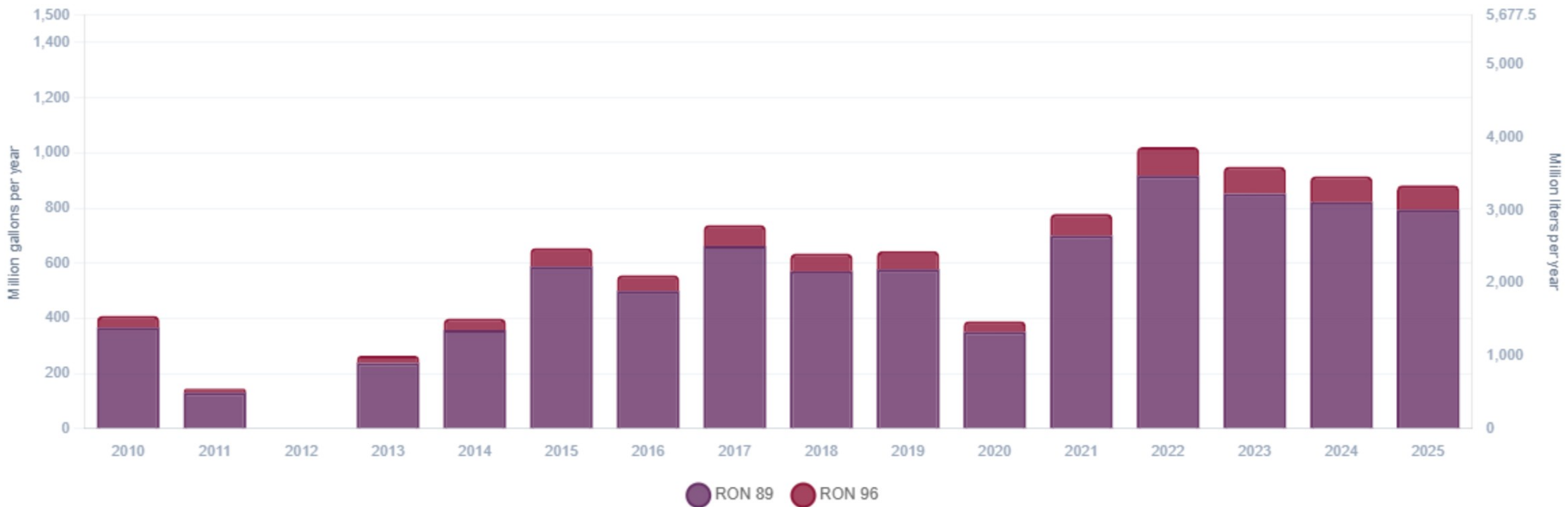
Source: METI

Gasoline Imports in Japan



Source: METI

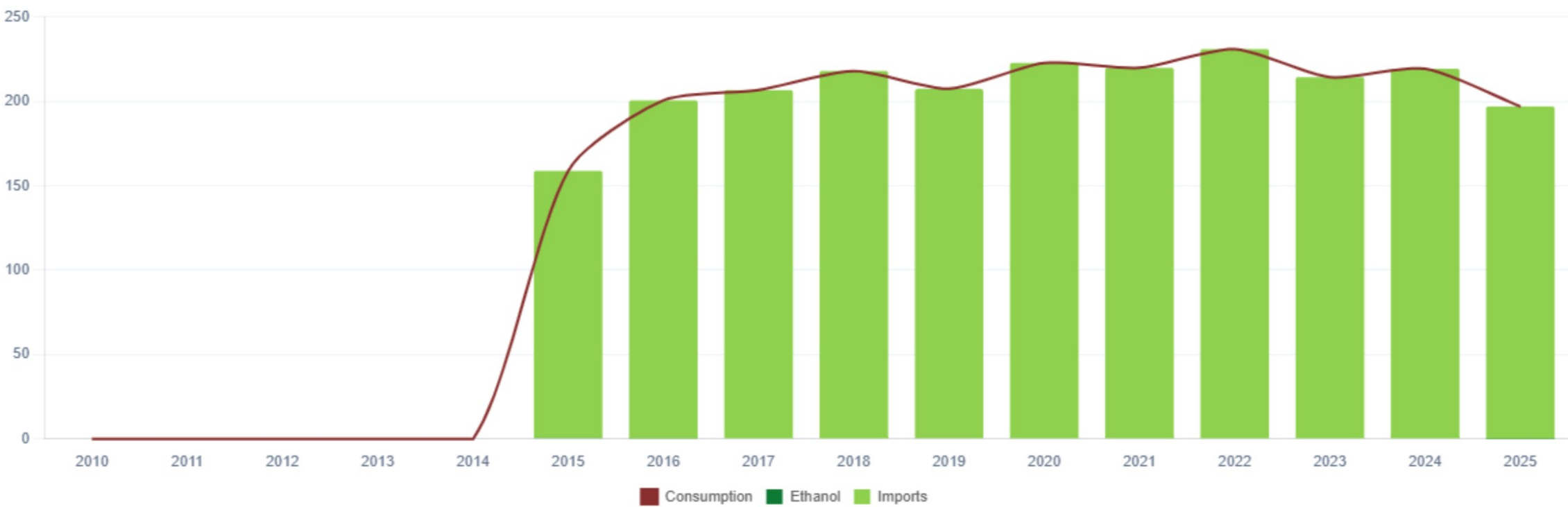
Gasoline Exports in Japan



Source: METI

Ethanol Balance in Japan

There is no recent local production of ethanol. Ethanol is supplied by imports of ethanol as a blendstock for ETBE production or as ETBE



Source: USDA, METI

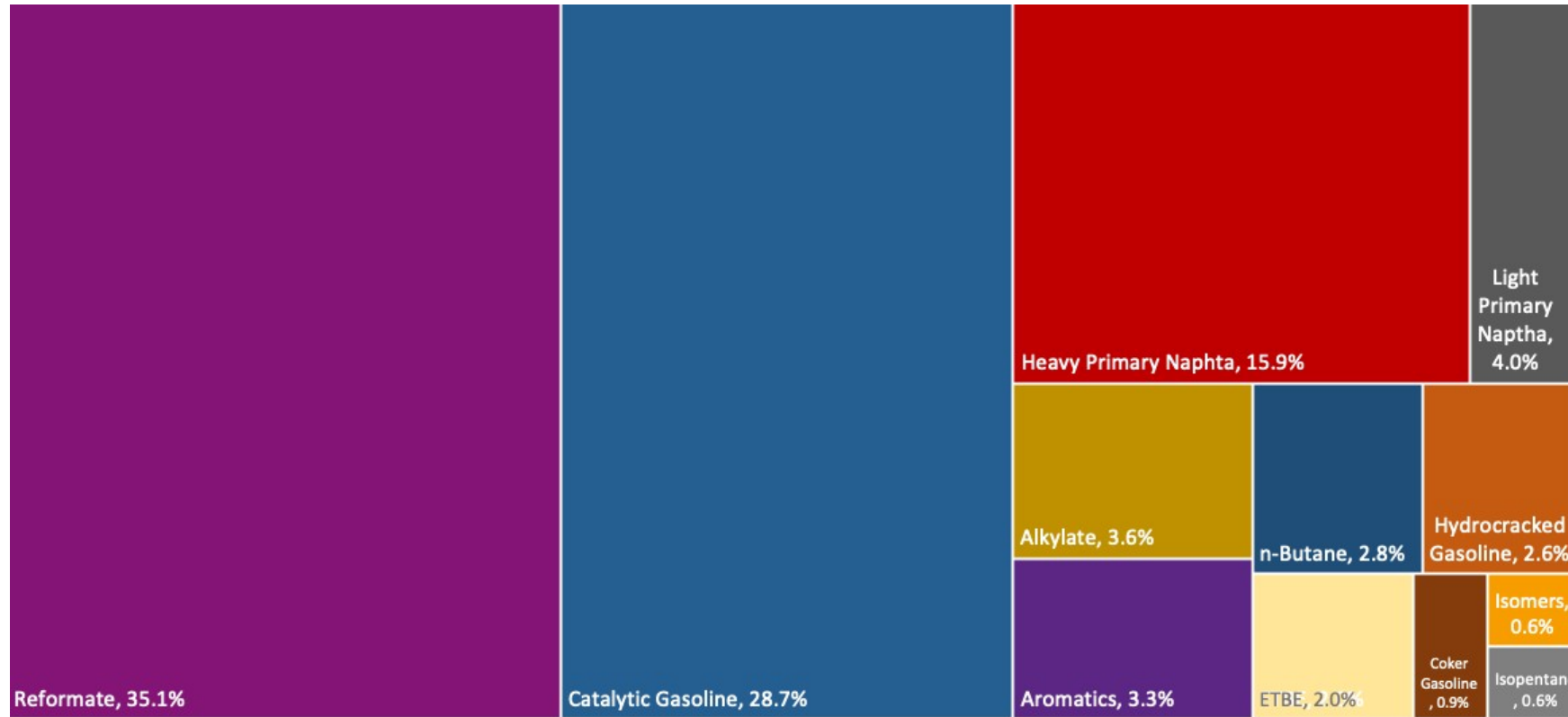
Gasoline Quality in Japan

Name	JIS K 2202: 2023 - Law on the Quality Control of Gasoline and Other Fuels							
Implementation Date	2023							
Applicability	Japan							
Grade	Regular		Premium		E10 Regular *		E10 Premium *	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	89.0		95.0		90.0		96.0	
MON								
AKI								
Benzene (%vol)		1.0		1.0		1.0		1.0
Aromatics (% vol)		-		-		-		-
Olefins (%vol)		-		-		-		-
Lead (g/L)		-		-		-		-
Manganese (mg/L)		-		-		-		-
Sulfur (ppm)		10		10		10		10
Oxygen (%w)		1.3		1.3		3.5		3.5
Ethanol (%vol)		3.0		3.0	3	10.0	3.0	10.0
MTBE (%vol)		7.0		7.0				
PVR 37.8°C (psi)	6.4	11.3	6.4	11.3	6.4	11.3	6.4	11.3
Carbon Intensity (gCOe/MJ)	0	88.74	-	88.74	0	88.74	-	88.74
Kerosene (%vol)	-	4.0	-	4.0	-	4.0	-	4.0

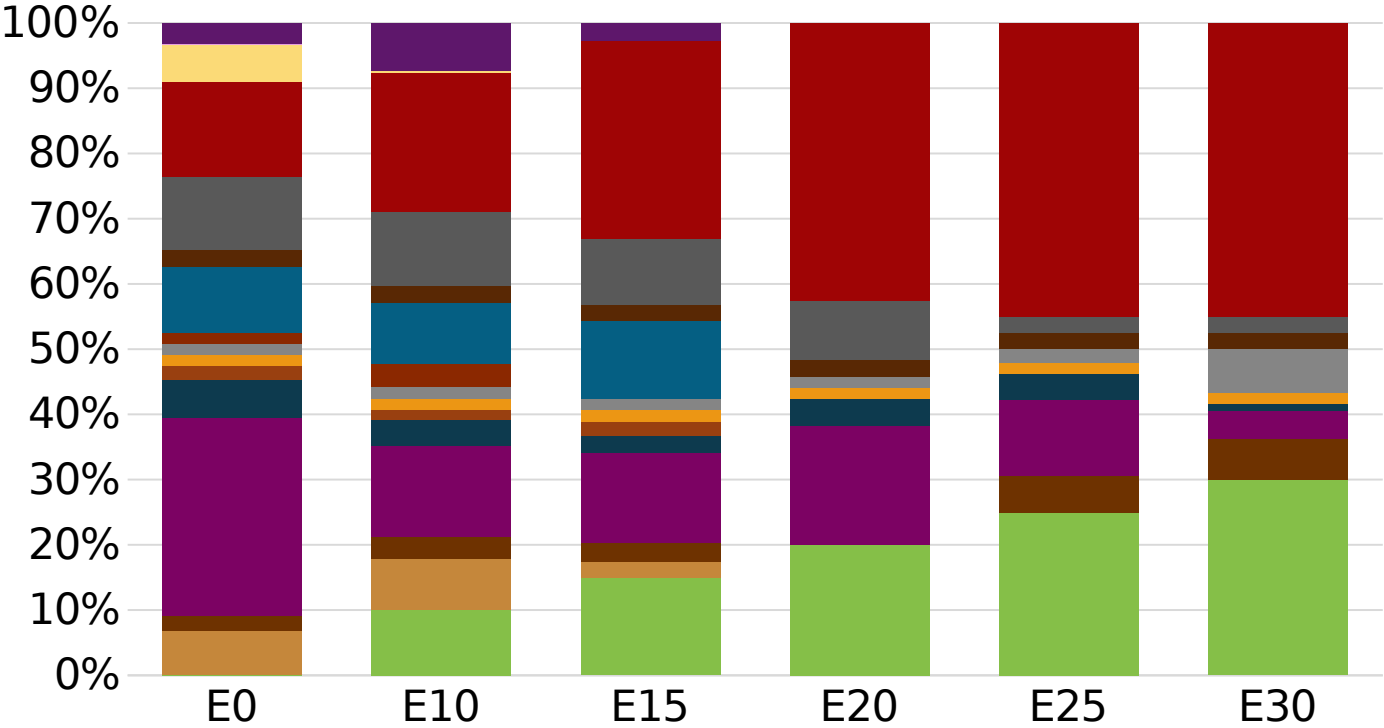
*Not Mandatory. A 55% Carbon Intensity Reduction target has been established and will be revised once LCA has been completed

Source: METI

Japan Available Blending Components

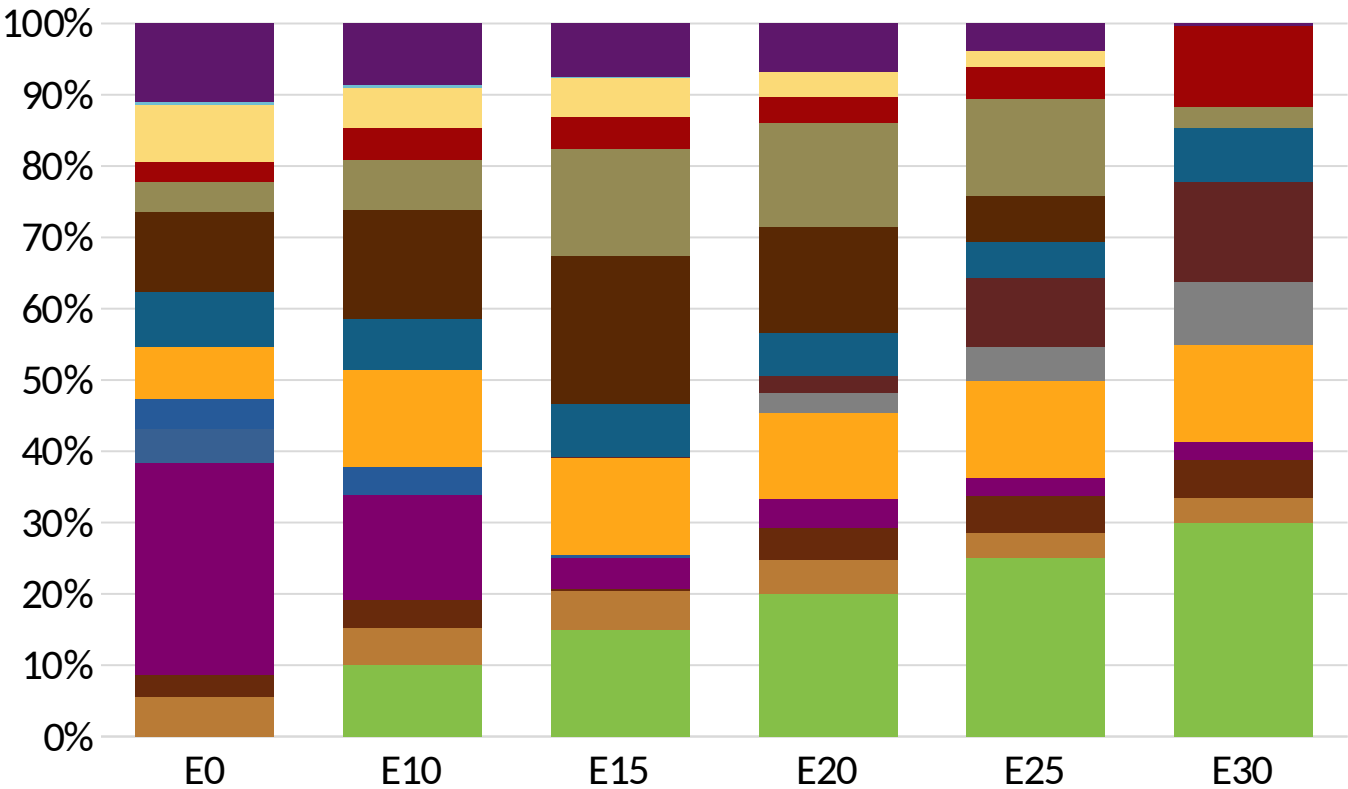


Japan – Gasoline 1 – Constant Octane



Average CIF 2025
Prices

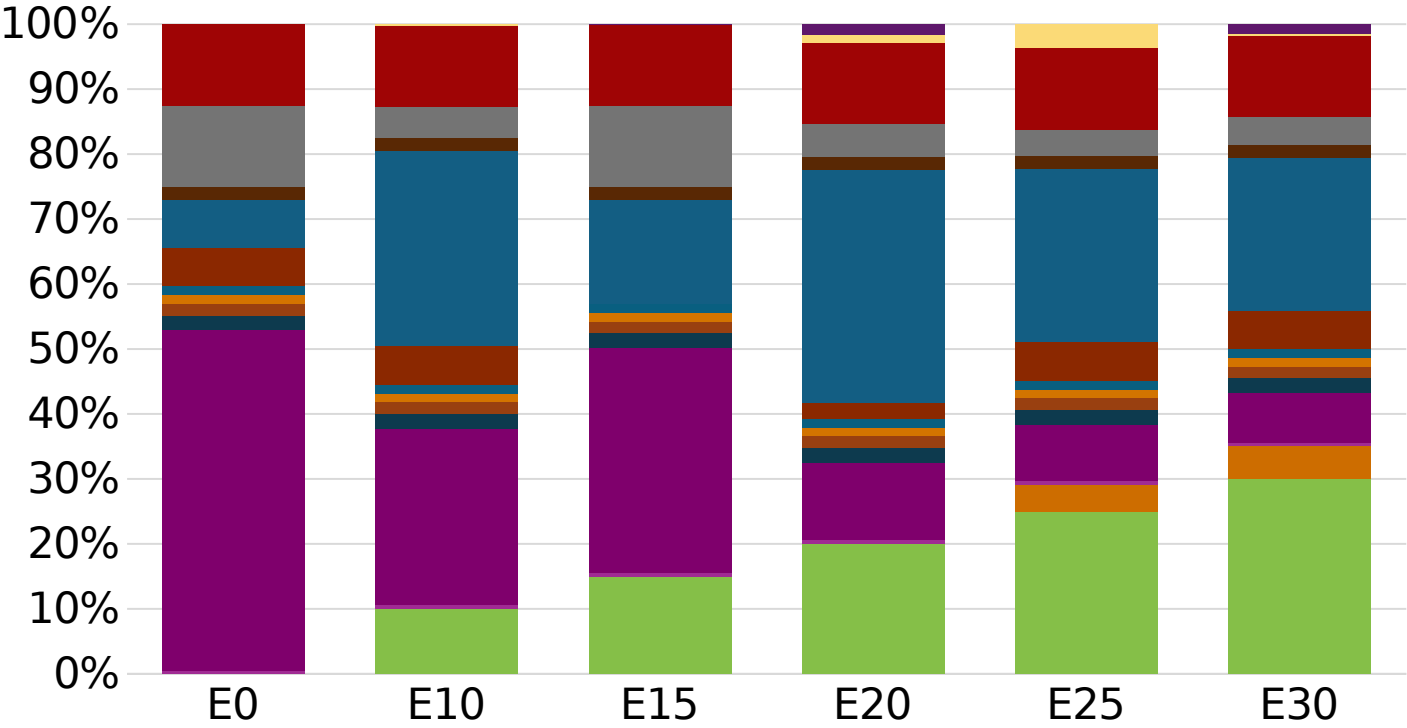
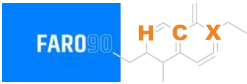
Japan - Gasoline 2 - Constant Octane



Octane (RON)	96.0	96.0	96.0	96.0	96.0	96.0
Price (US\$/gal)	\$2.20	\$2.13	\$2.05	\$1.97	\$1.88	\$1.77

Average CIF 2025 Prices

Japan – Gasoline 1 – Increased Octane

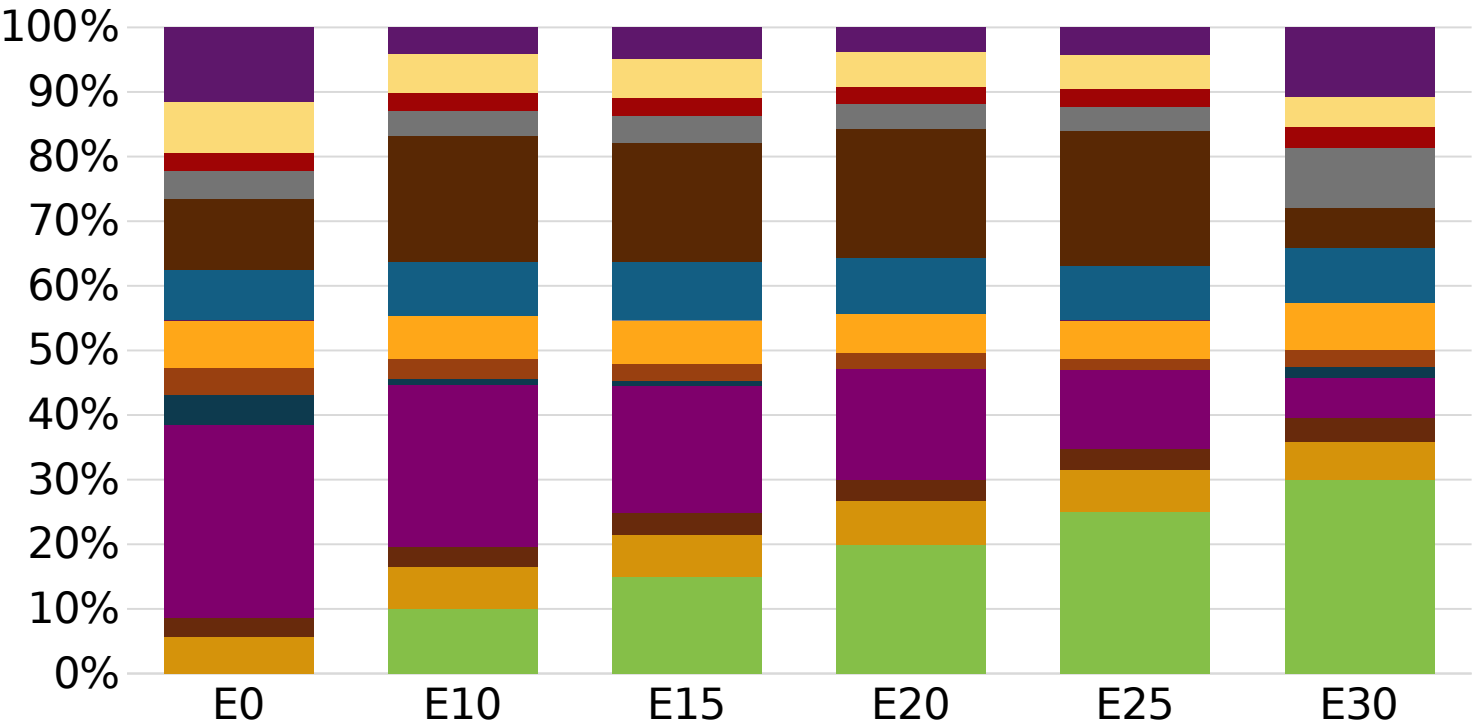


Average CIF 2025
Prices

Octane (RON)	89.2	93.3	95.2	97.1	98.8	100.0
Price (US\$/gal)	\$2.01	\$2.01	\$2.01	\$2.01	\$2.01	\$2.01

Source:
Faro90

Japan – Gasoline 2 – Increased Octane

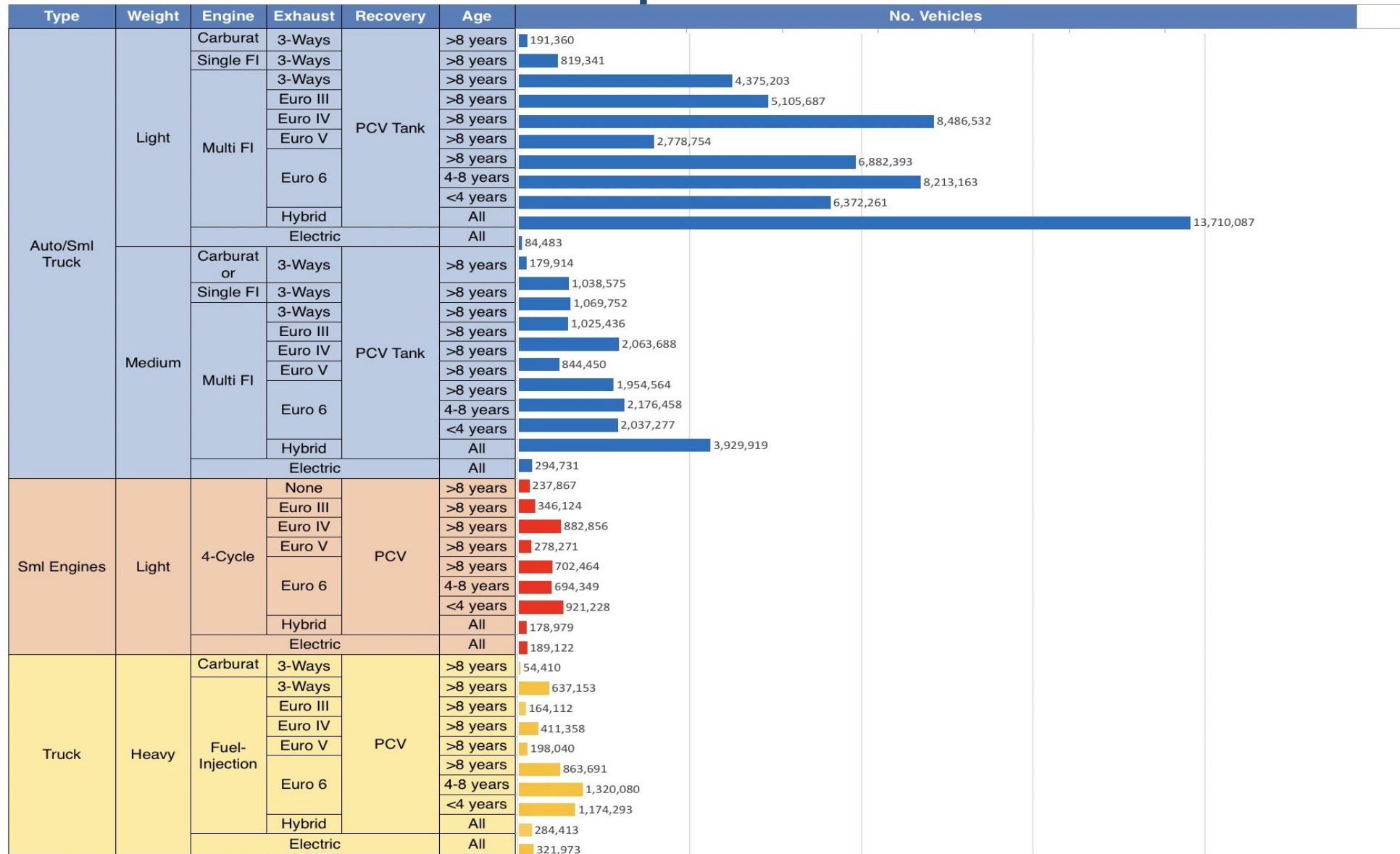


Average CIF 2025
Prices

Octane (RON)	96.0	97.6	99.1	100.5	102.0	103.2
Price (US\$/gal)	\$2.20	\$2.20	\$2.20	\$2.20	\$2.20	\$2.20

Source:
Faro90

Gasoline Vehicle Fleet - Japan

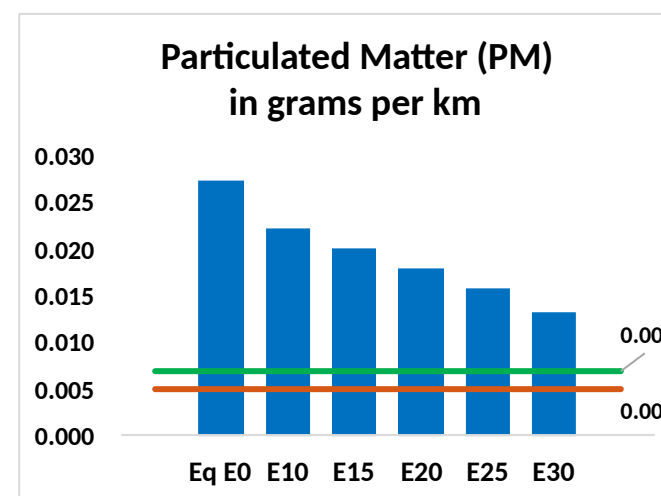
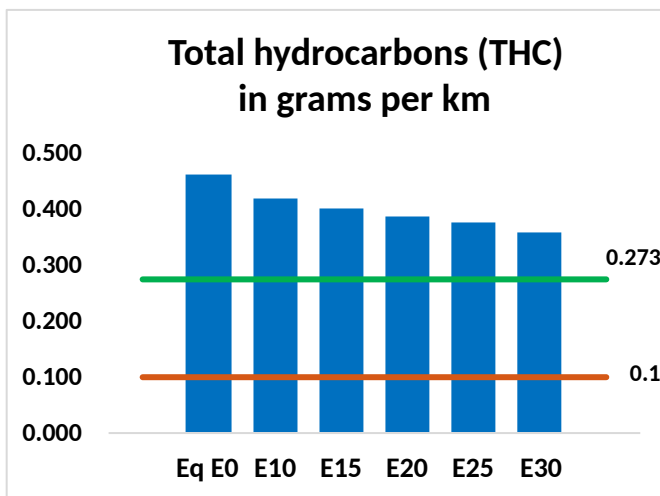
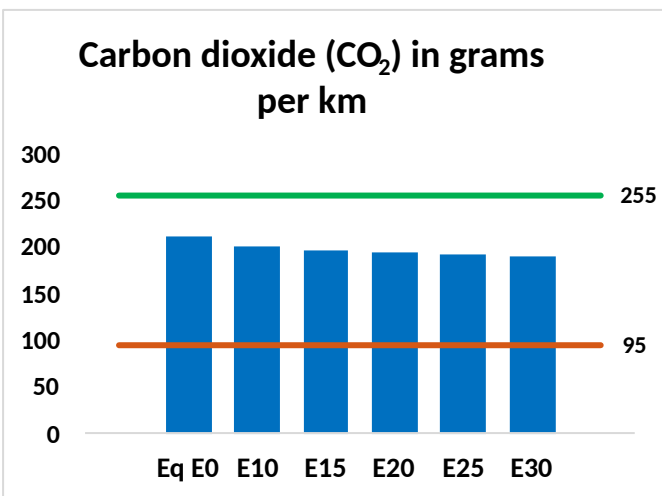
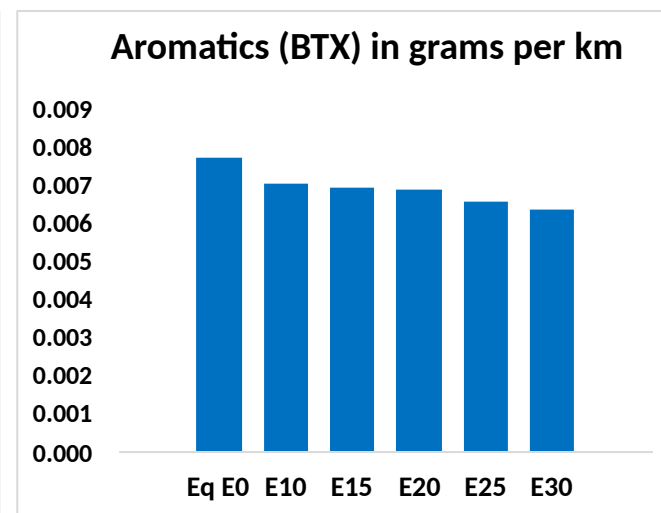
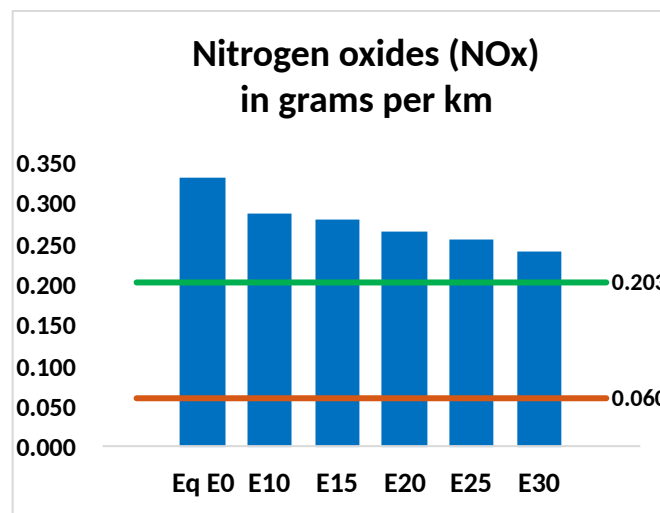
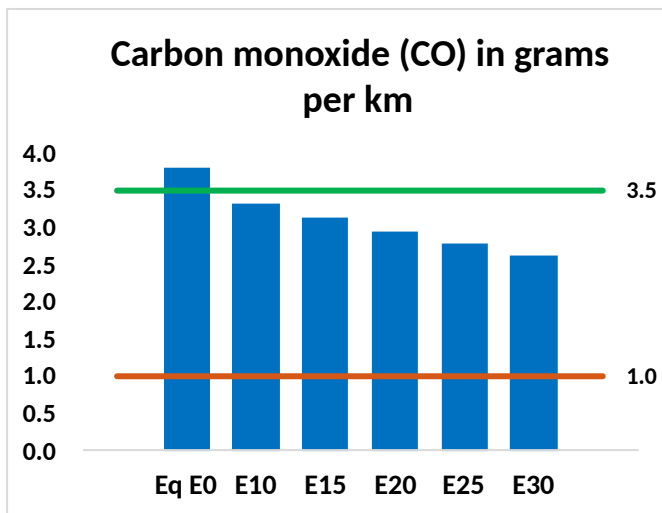
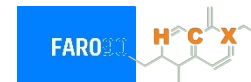


Vehicle Fleet: 89,079,267
Gasoline Vehicle Fleet: 81,722, 132

Source: AIRIA

Japan – Vehicular Emissions

— TIER III BIN 125 United States
— Euro 6 Standard

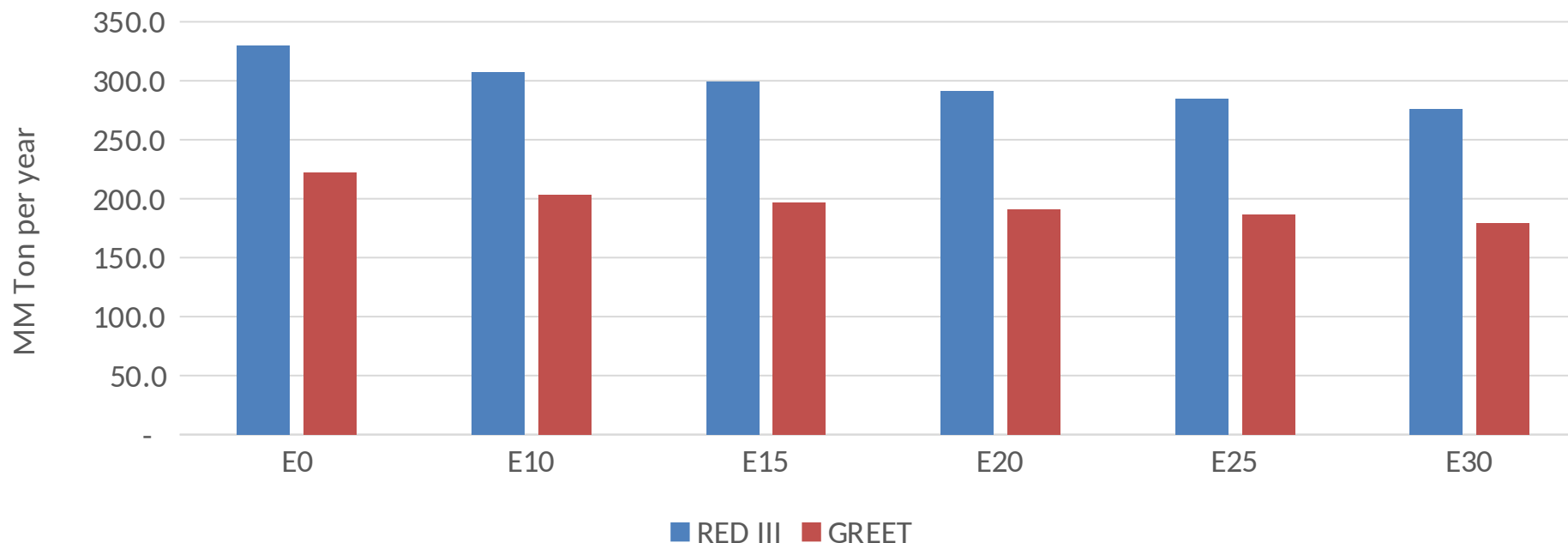


Japan – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	7,156,775	6,694,533	6,232,291	5,870,662	5,514,644	5,243,270	4,922,158	-13%	-23%
CO2	396,805,605	386,602,033	376,965,325	369,386,338	365,635,958	363,765,372	357,060,131	-5%	-8%
VOC	871,141	828,487	785,832	755,623	727,407	706,369	675,968	-10%	-16%
NOx	624,372	576,503	544,091	527,804	501,316	480,670	456,287	-13%	-20%
SOx	4,495	4,155	3,815	3,528	3,252	2,953	2,639	-15%	-28%
NH3	139,266	137,887	136,508	137,156	138,699	139,785	140,686	-2%	0%
N2O	8,599	8,552	8,518	8,559	8,736	8,833	8,933	-1%	2%
CH4	196,321	196,321	196,321	199,266	203,585	207,315	210,848	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	6,215	5,969	5,724	5,568	5,432	5,333	5,180	-8%	-13%
Acetaldehyde	14,149	16,239	23,827	36,286	49,436	56,490	66,749	68%	249%
Formaldehyde	48,367	47,024	54,816	64,366	67,433	74,599	81,209	13%	39%
Benzene	14,570	14,001	13,267	13,057	12,948	12,383	11,983	-9%	-11%
PM2.5	23,127	21,025	18,922	17,072	15,262	13,321	11,290	-18%	-34%
PM10	28,296	25,723	23,151	20,887	18,673	16,298	13,813	-18%	-34%

Japan – Gasoline Vehicle Fleet Life Cycle GHG Emissions

Life Cycle GHG Emission - Gasoline Vehicles



Country Emissions 2024 (MMT/year)	1,102.2
Target @ 2035 (MMT/year)	570.4
Delta	531.8

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	8.3%	11.2%	13.9%	16.1%	19.2%
RED II/III	0.0%	6.8%	9.3%	11.6%	13.6%	16.2%

% 2035 Target	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	3.5%	4.7%	5.8%	6.7%	8.0%
RED II/III	0.0%	4.2%	5.7%	7.2%	8.4%	10.1%

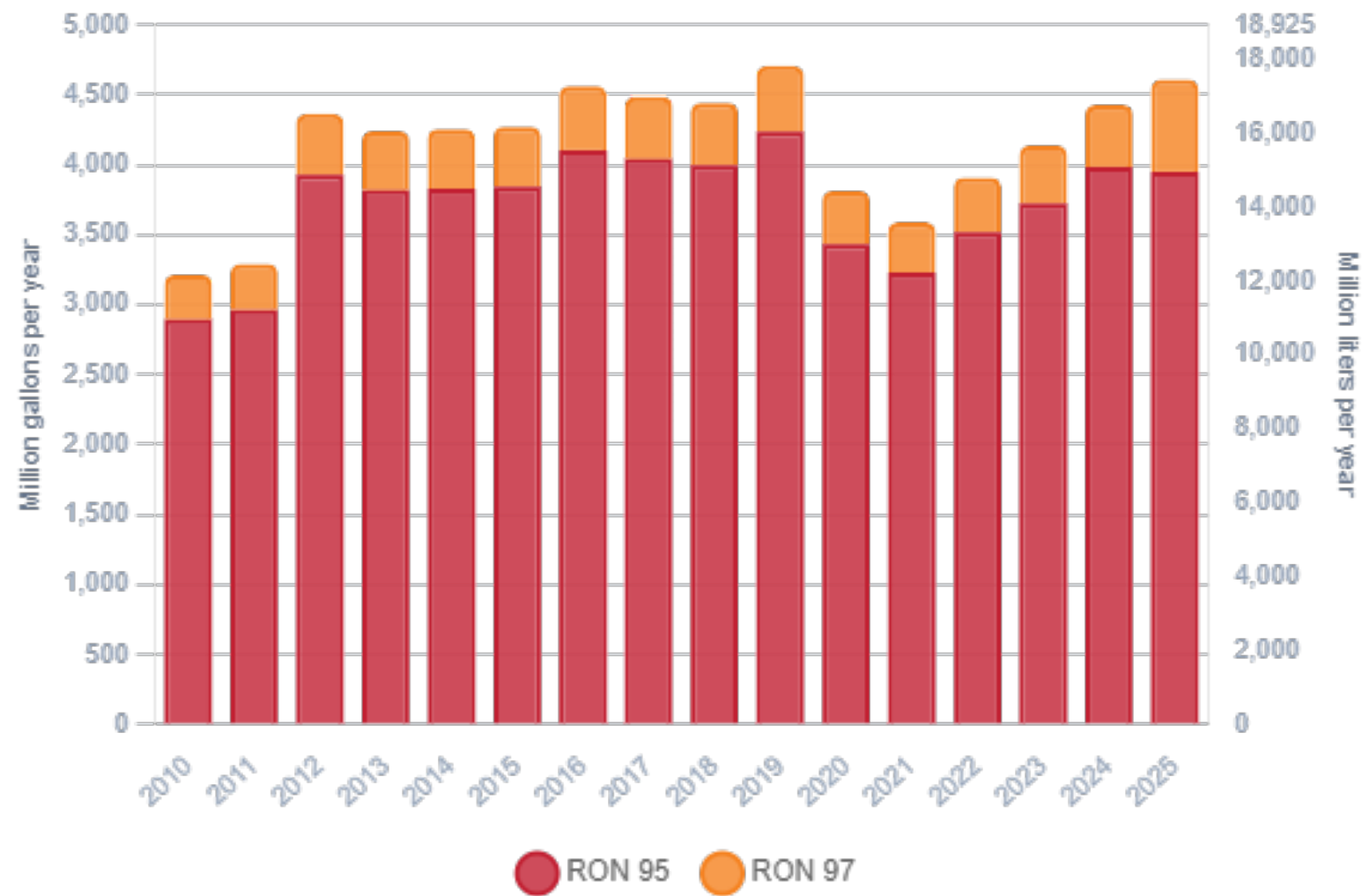
Ethanol Gasoline Blending in Malaysia



In 2025, gasoline consumption in Malaysia was 17,472 million liters (4,616 million gallons) and growth rate was 2.4%. Gasoline production and net imports were 6,864 and 10,608 million liters (1,814 and 2,803 million gallons) correspondingly. Malaysia offers two main grades: RON 95 and RON 97) with an average octane of 95.3 RON.

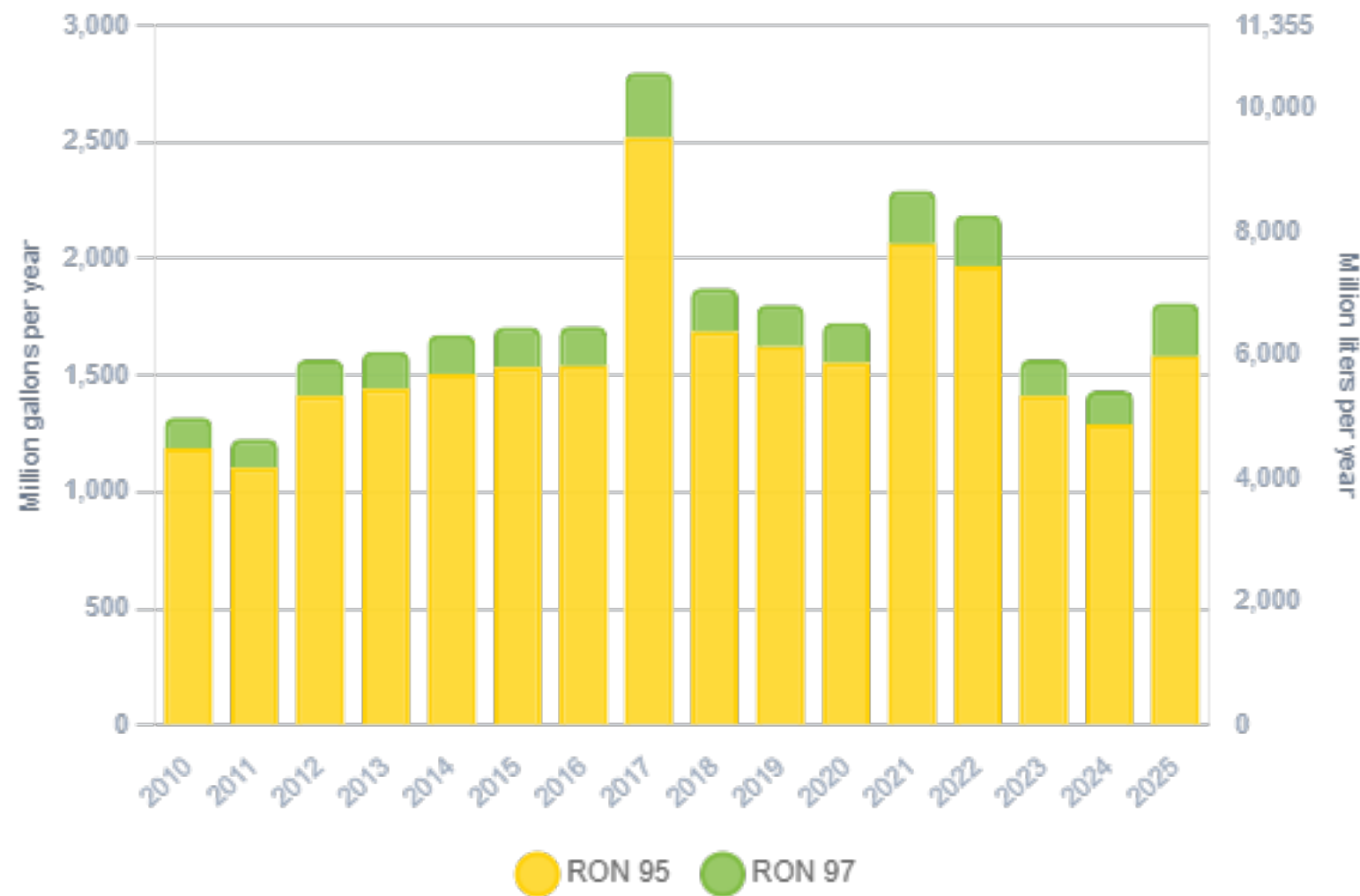
Government support for biofuels has so far targeted biodiesel. The Biofuel Industry Act of 2007 excluded ethanol as the source of alternative fuels under the National Biofuels Policy of 2006. As there is no policy support for ethanol and there is no production of fuel ethanol and minimal consumption. In 2025, ethanol consumption in Malaysia was 16 million liters (4 million gallons) representing 0.1% of gasoline demand.

Gasoline Demand in Malaysia



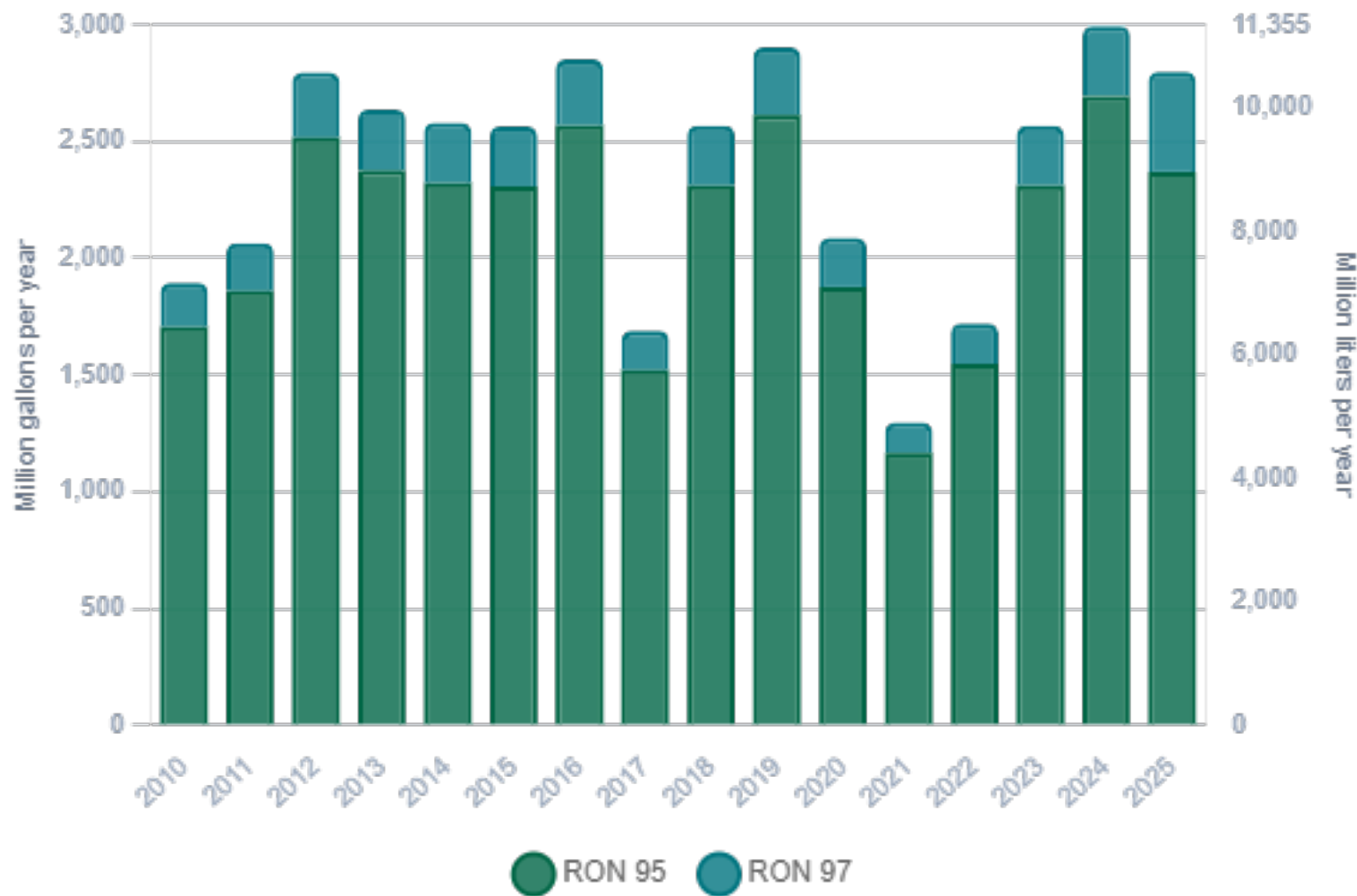
Source: EIA, Suruhanjaya Tenaga

Gasoline Production in Malaysia



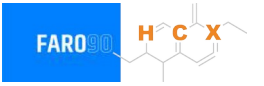
Source: EIA, Suruhanjaya Tenaga

Gasoline Net Imports in Malaysia



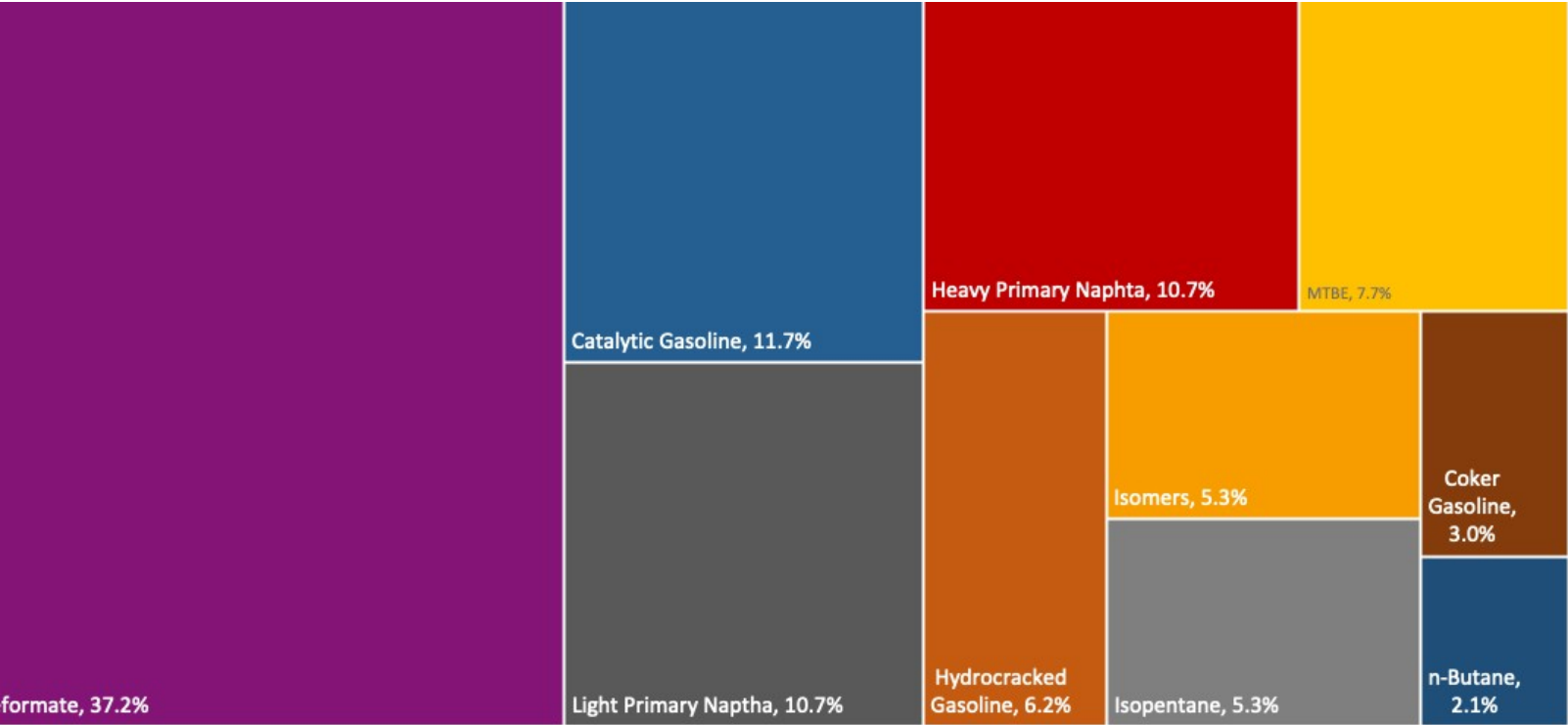
Source: EIA, Suruhanjaya Tenaga

Gasoline Quality in Malaysia



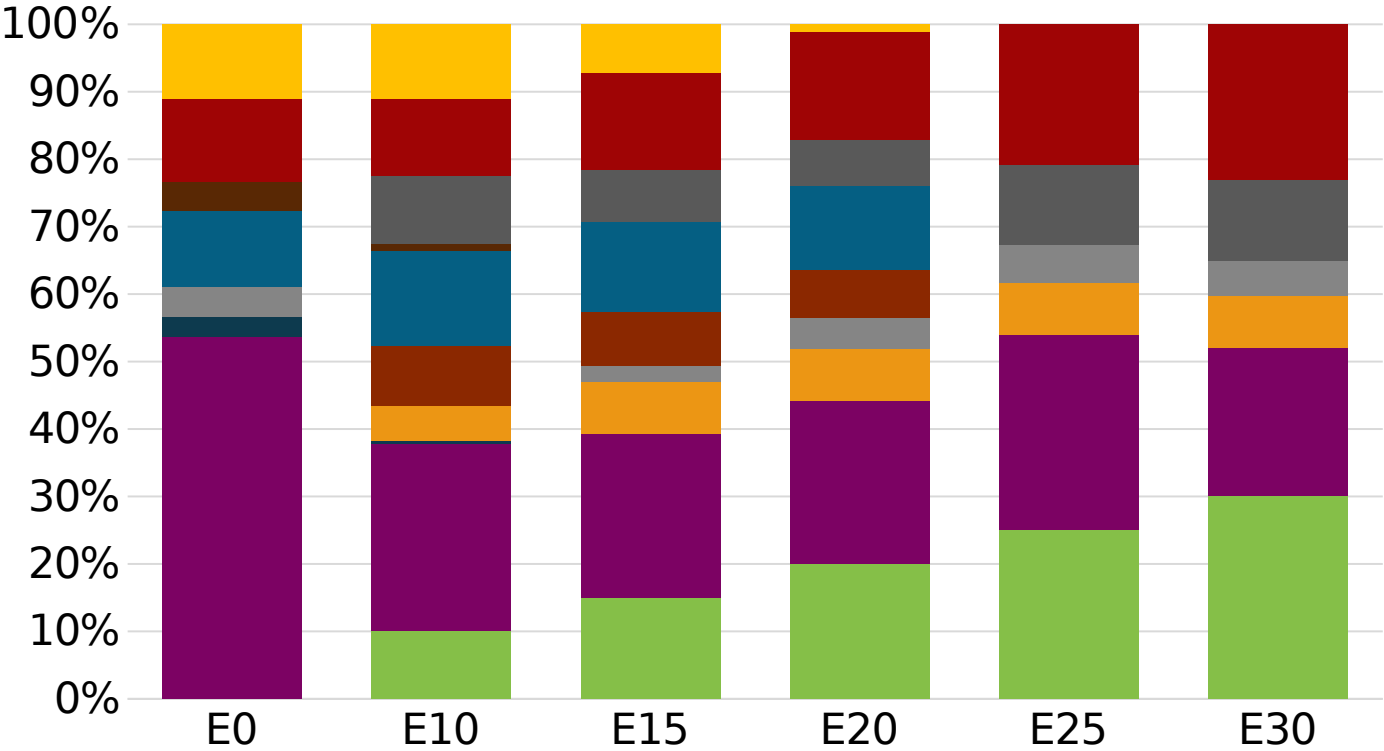
Source: Department of Standards Malaysia, Euro 4 Spec

Malaysia Available Blending Components



Source:
Faro90

Malaysia – Gasoline 95 – Constant Octane

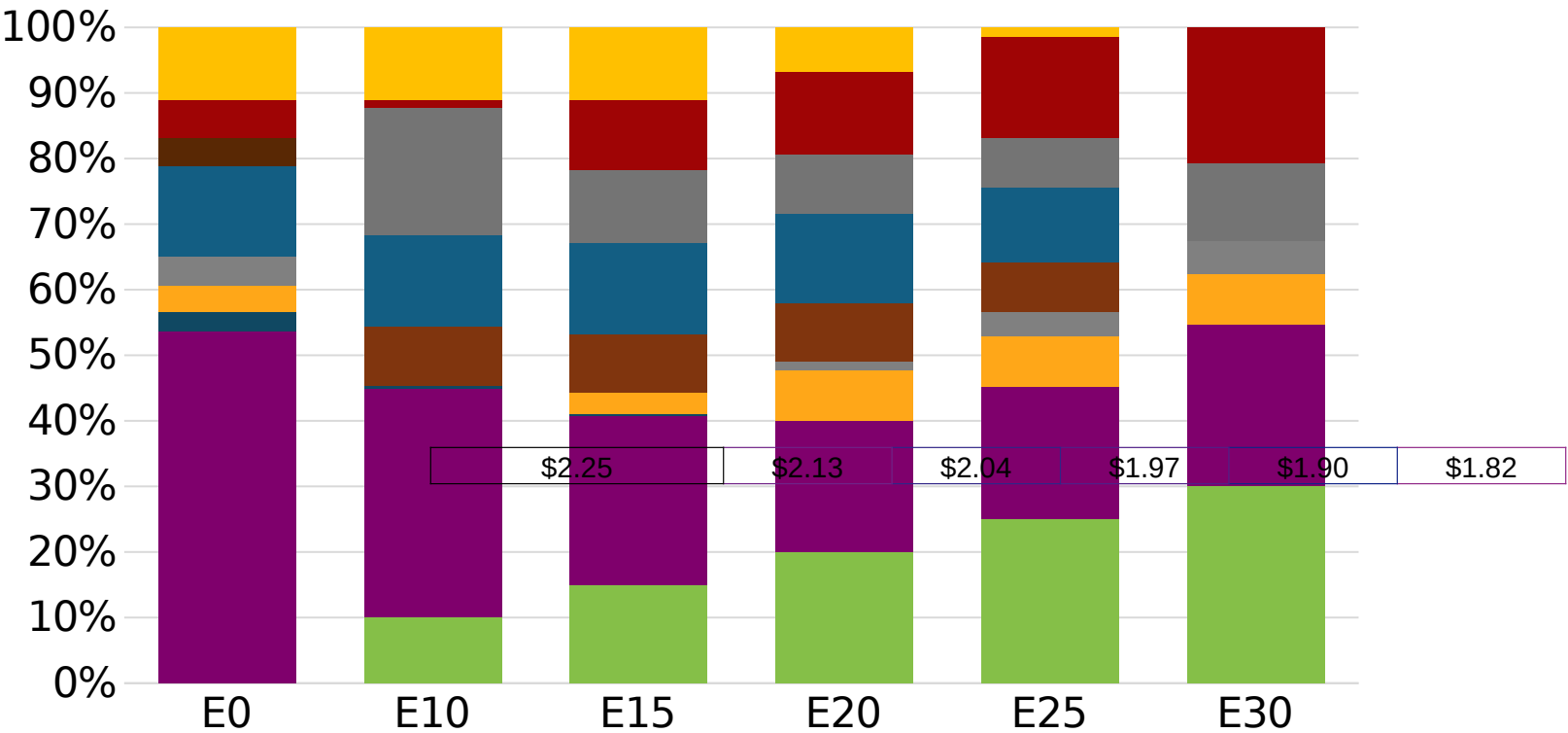


Average CIF 2025
Prices

Octane (RON)	95.0	95.0	95.0	95.0	95.6	96.6
Price (US\$/gal)	\$2.15	\$2.02	\$1.95	\$1.89	\$1.81	\$1.76

Source:
Faro90

Malaysia - Gasoline 97 - Constant Octane

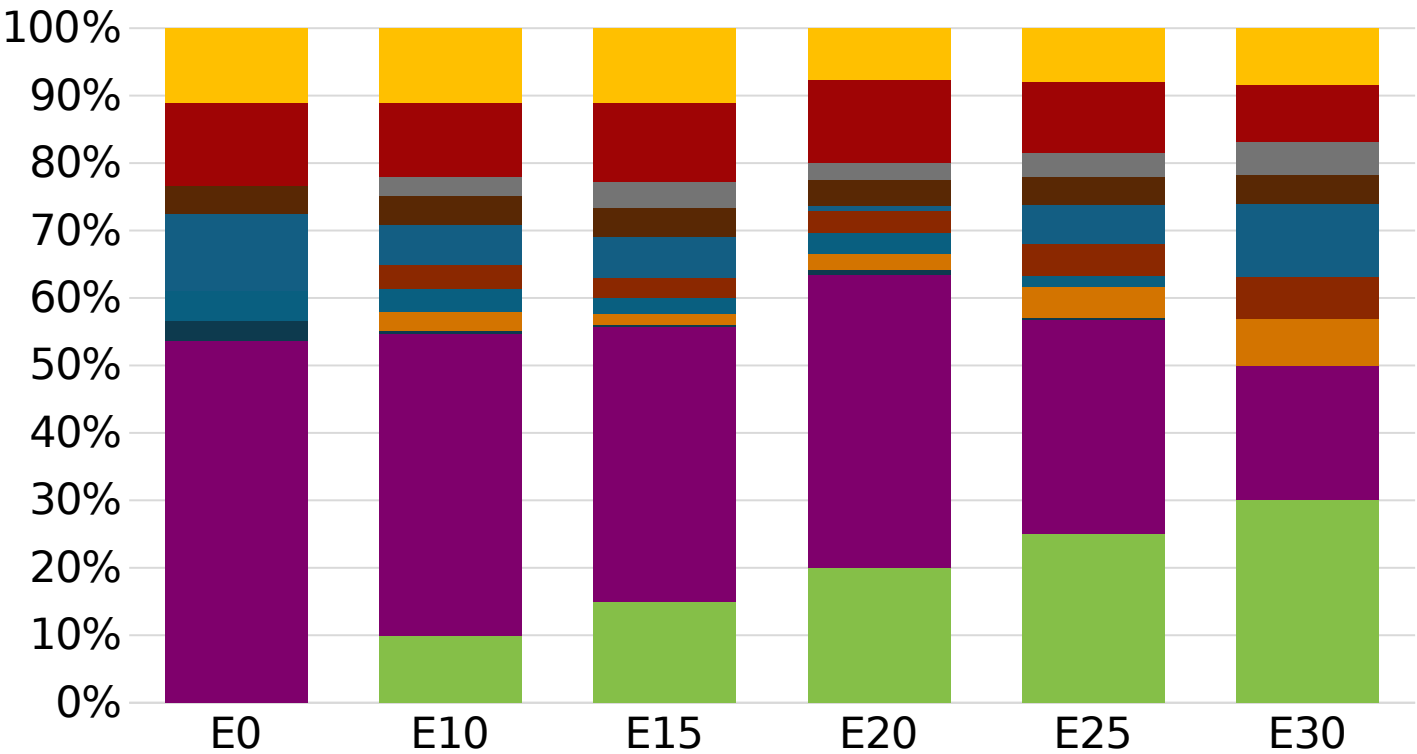


Average CIF 2025
Prices

Octane (RON)	97.0	97.0	97.0	97.0	97.0	97.5
Price (US\$/gal)	\$2.25	\$2.13	\$2.04	\$1.97	\$1.90	\$1.82

Source:
Faro90

Malaysia – Gasoline 95 – Increased Octane

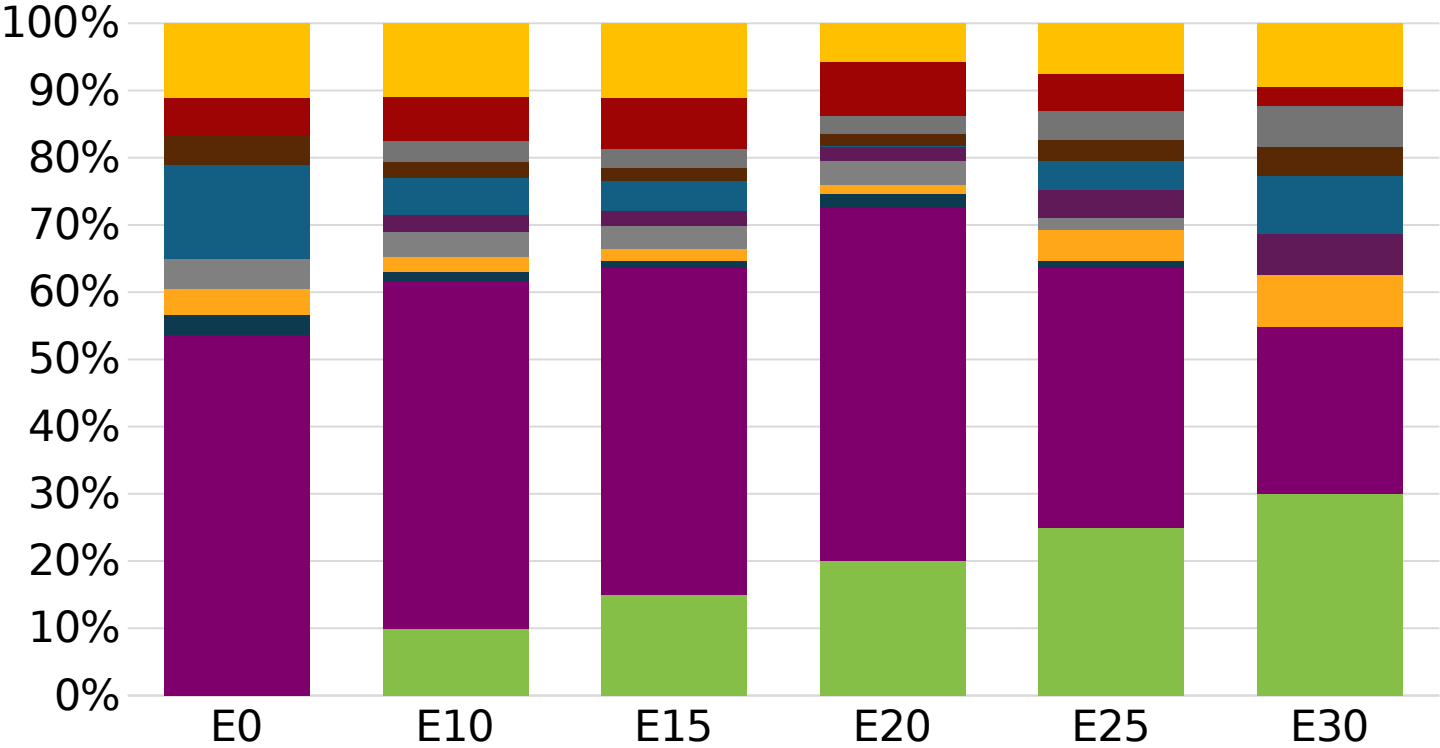


Average CIF 2025
Prices

Octane (RON)	95.0	97.8	99.4	100.5	101.2	103.2
Price (US\$/gal)	\$2.15	\$2.15	\$2.15	\$2.15	\$2.15	\$2.15

Source:
Faro90

Malaysia – Gasoline 97 – Increased Octane

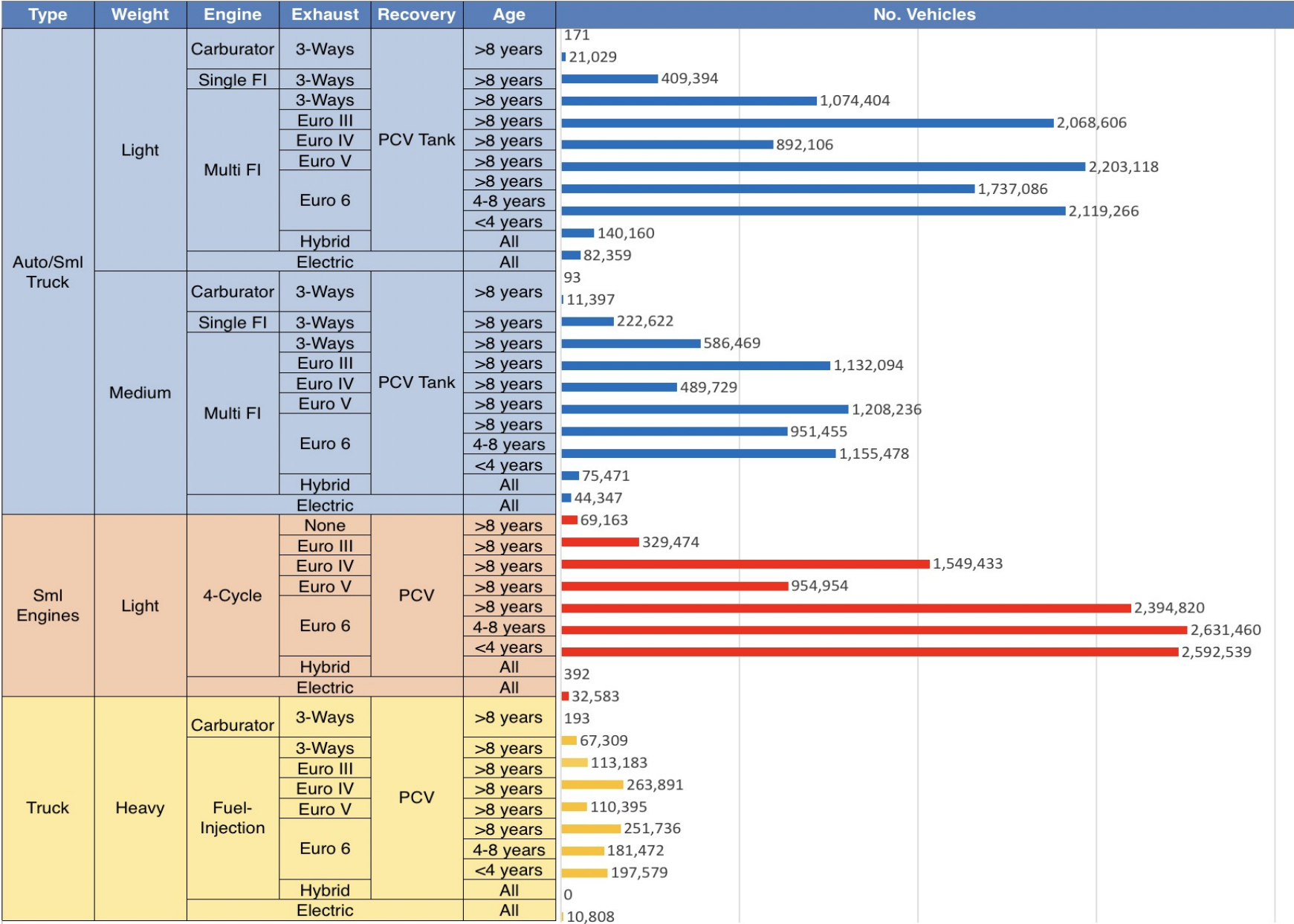


Average CIF 2025
Prices

Octane (RON)	97.0	99.8	101.3	102.1	102.9	104.9
Price (US\$/gal)	\$2.25	\$2.25	\$2.25	\$2.25	\$2.25	\$2.25

Source:
Faro90

Gasoline Vehicle Fleet - Malaysia

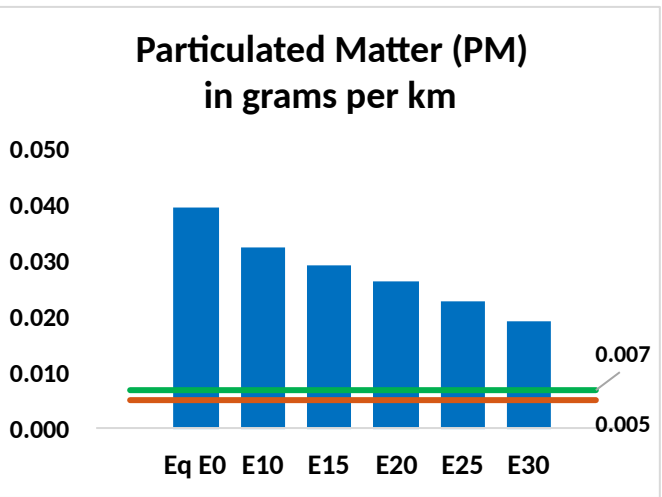
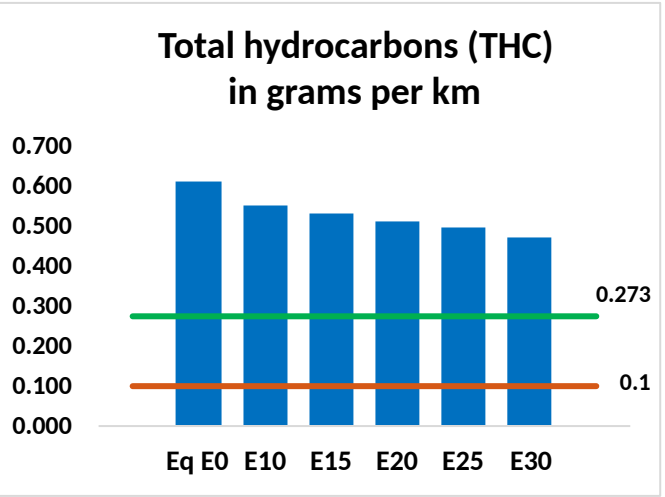
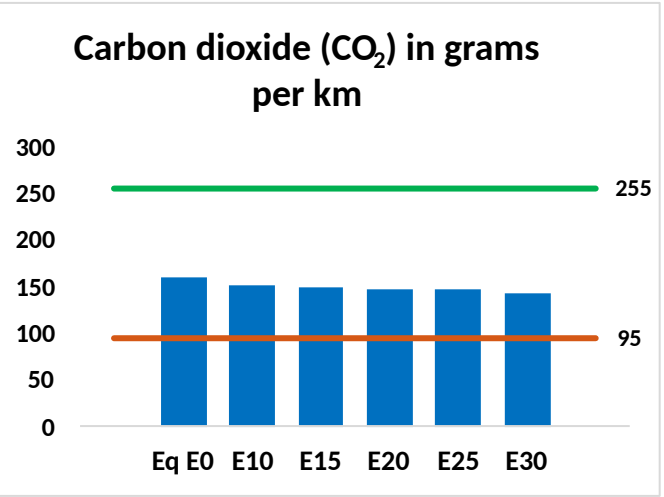
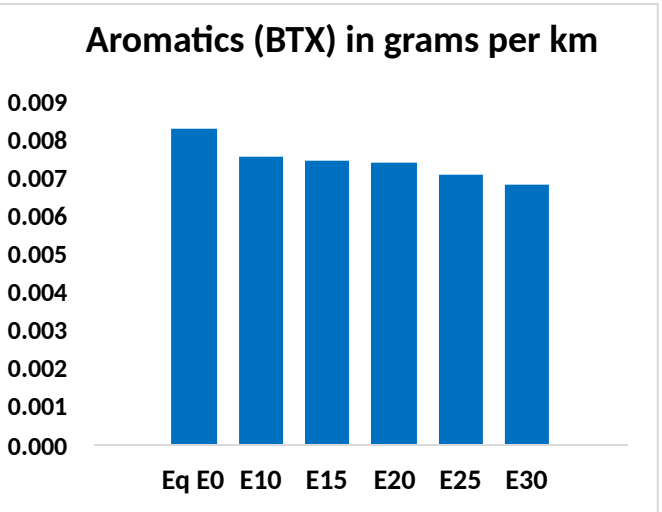
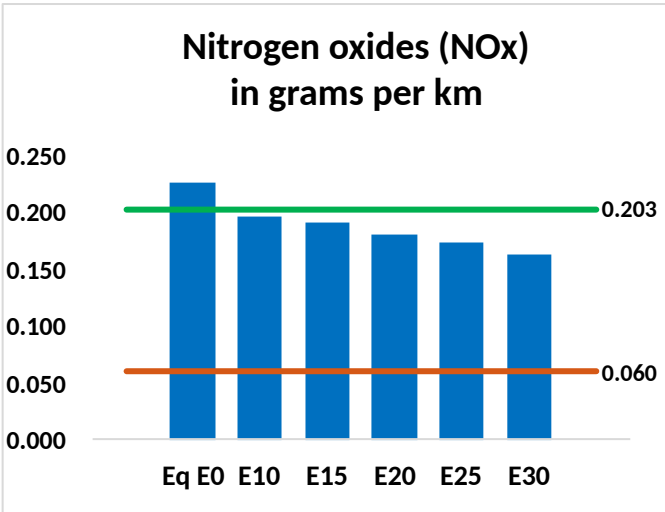
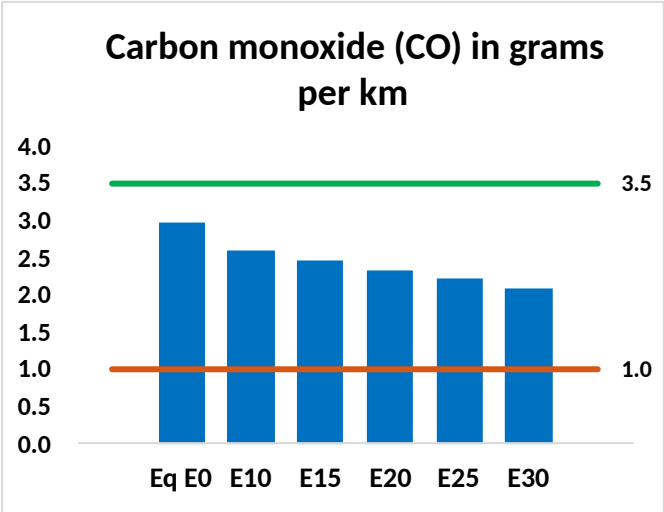
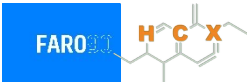


Vehicle Fleet: 28,376,474
Gasoline Vehicle Fleet: 25,554,393

Source: Data Gov Malaysia

Malaysia – Vehicular Emissions

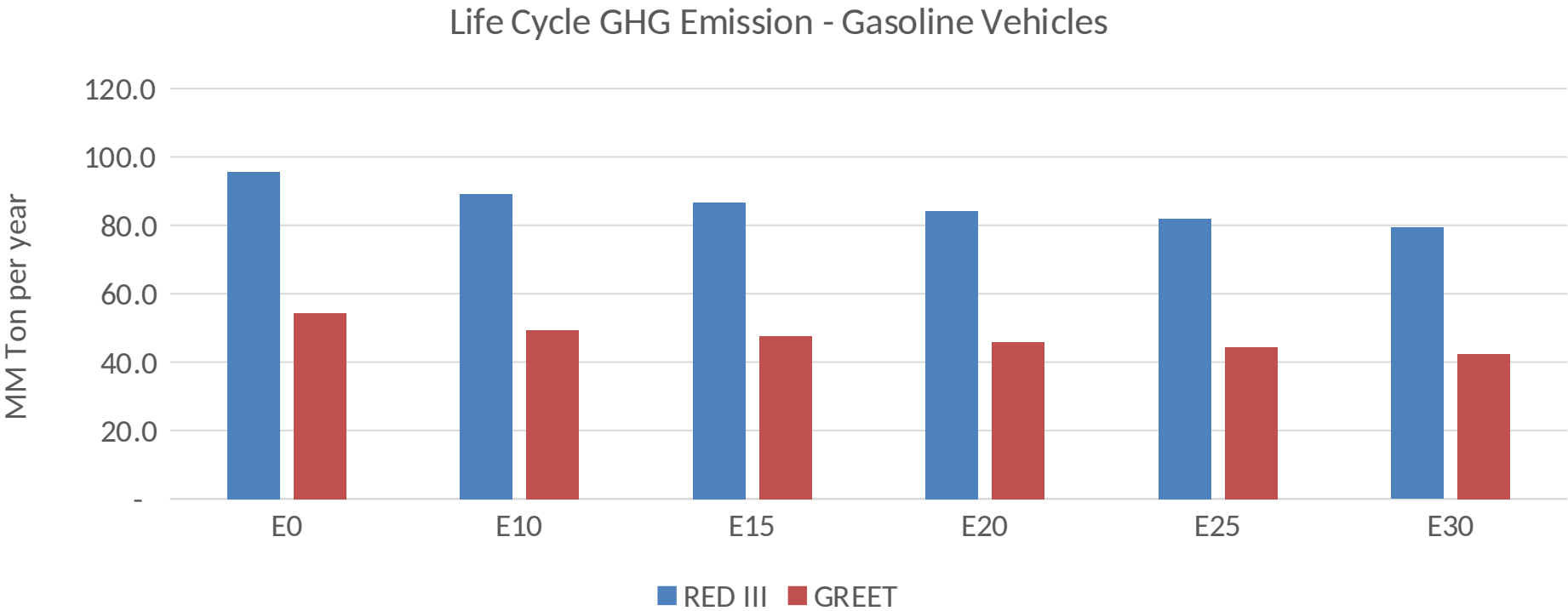
TIER III BIN 125 United States
Euro 6 Standard



Malaysia – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	1,787,897	1,676,608	1,565,319	1,479,227	1,394,939	1,330,835	1,253,790	-12%	-22%
CO2	96,171,362	93,698,384	91,362,794	89,525,921	88,616,964	88,187,848	86,562,292	-5%	-8%
VOC	367,966	349,915	331,865	319,071	307,115	298,200	285,323	-10%	-17%
NOx	136,847	126,355	119,107	115,389	109,428	104,743	99,234	-13%	-20%
SOx	1,106	1,022	939	868	800	727	649	-15%	-28%
NH3	36,424	36,063	35,703	35,872	36,276	36,560	36,795	-2%	0%
N2O	1,264	1,257	1,252	1,258	1,284	1,298	1,313	-1%	2%
CH4	88,011	88,011	88,011	89,331	91,267	92,940	94,524	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	2,124	2,035	1,946	1,888	1,836	1,798	1,741	-8%	-14%
Acetaldehyde	6,537	7,503	11,009	16,765	22,840	26,100	30,839	68%	249%
Formaldehyde	26,261	25,532	29,763	34,948	36,613	40,504	44,093	13%	39%
Benzene	5,017	4,822	4,569	4,496	4,459	4,264	4,127	-9%	-11%
PM2.5	6,229	5,663	5,096	4,598	4,110	3,588	3,041	-18%	-34%
PM10	17,715	16,105	14,494	13,077	11,690	10,204	8,648	-18%	-34%

Malaysia – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	407.7
Target @ 2035 (MMT/year)	285.4
Delta	122.3

Source: Greet, RED II/III, climateactiontracker.org, F90

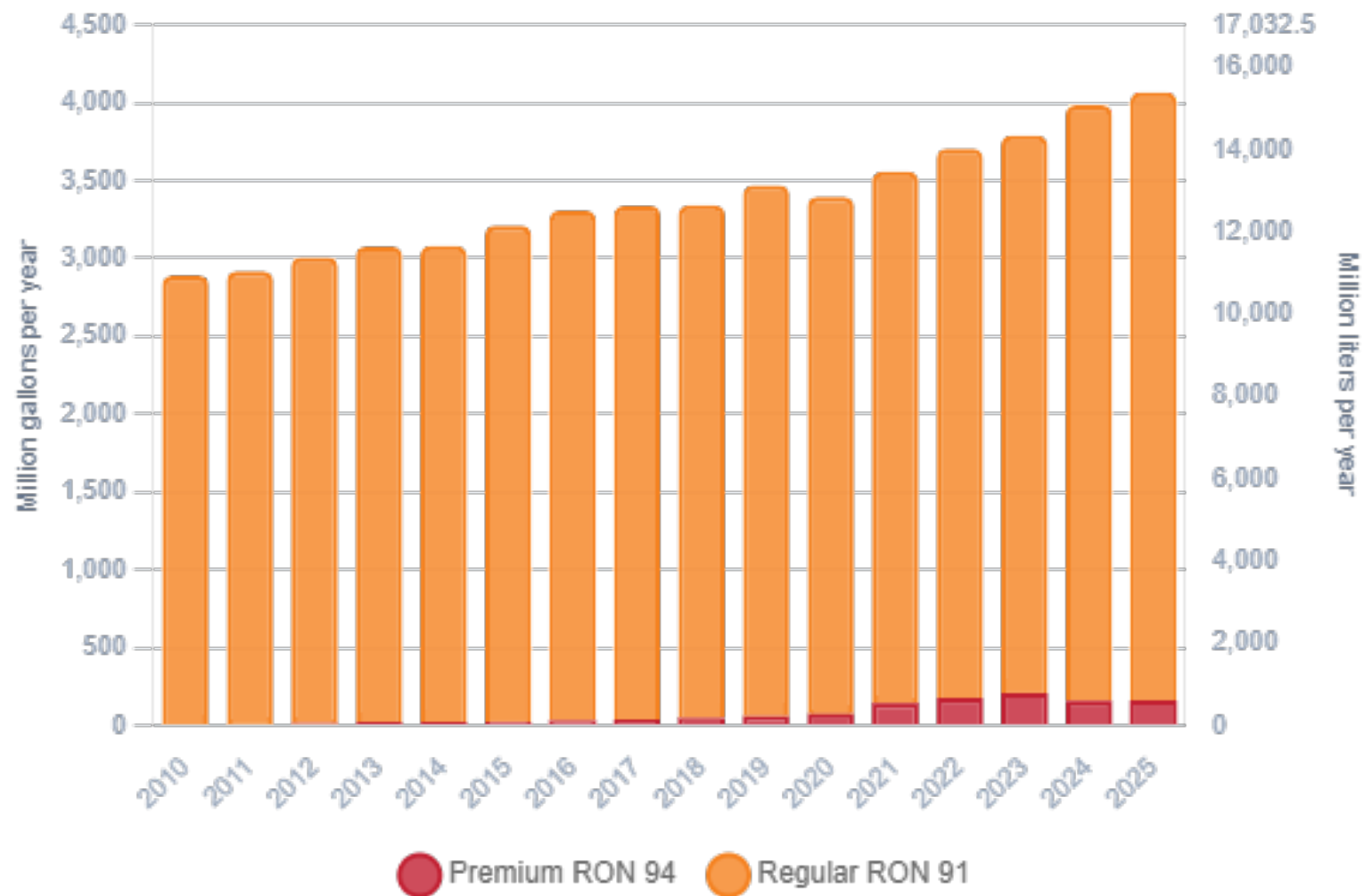
Ethanol Gasoline Blending in South Korea



In 2025, gasoline consumption in South Korea was 15,425 million liters (4,071 million gallons) and growth rate was 2.3%. Gasoline production and net imports were 32,383 and -17,715 million liters (8,555 and -4,680 million gallons) correspondingly. South Korea offers two main grades: RON 91 and RON 94) with an average octane of 91.1 RON.

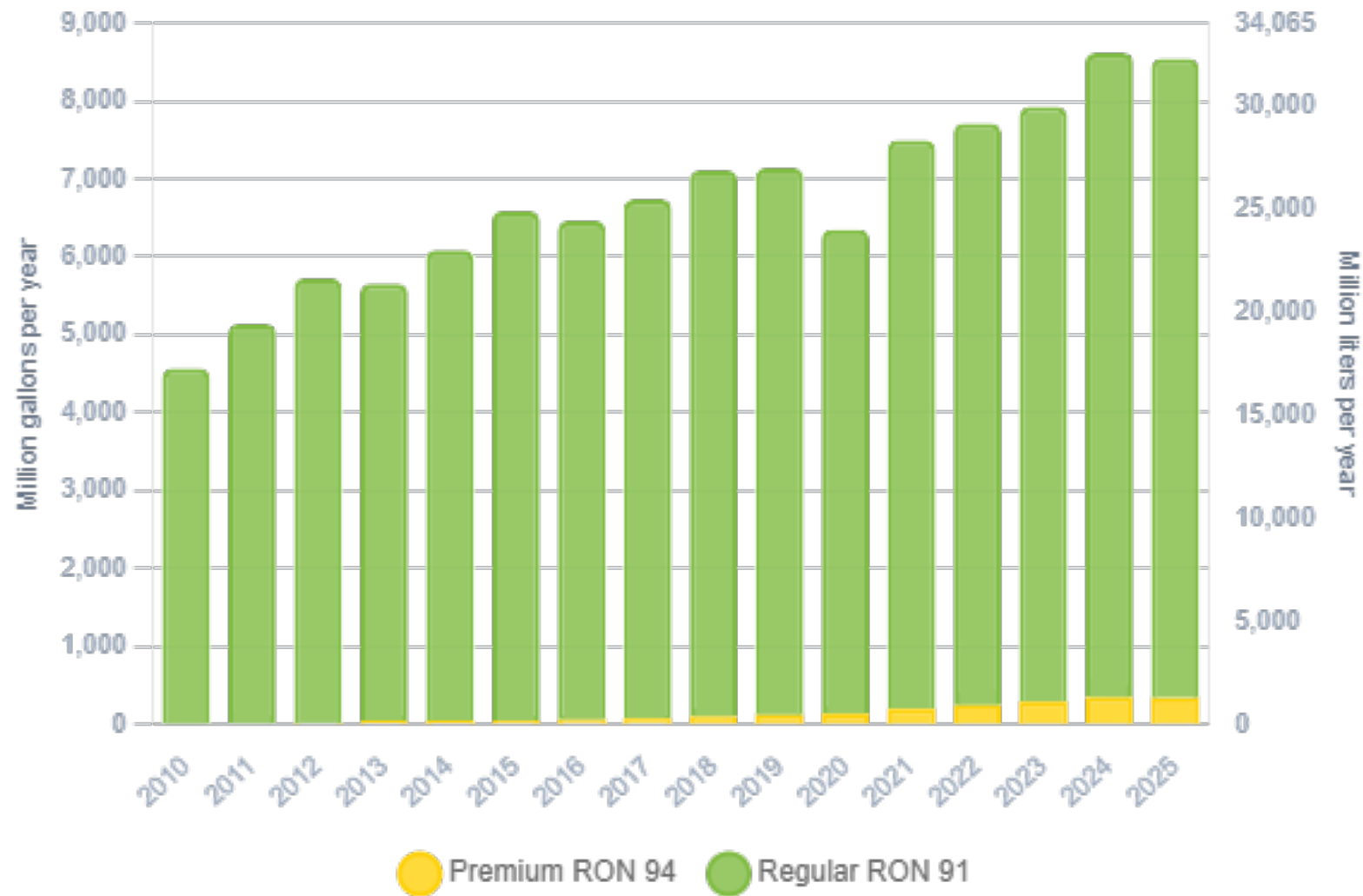
South Korea has implemented requirements for renewable fuels that are currently met with biodiesel blending. Although, there is an oxygen requirement, ethanol is not currently blended in gasoline. Efforts to implement pilot programs have been stalled because of a lack of consensus and reluctance of different actors in the petroleum industry.

Gasoline Demand in South Korea



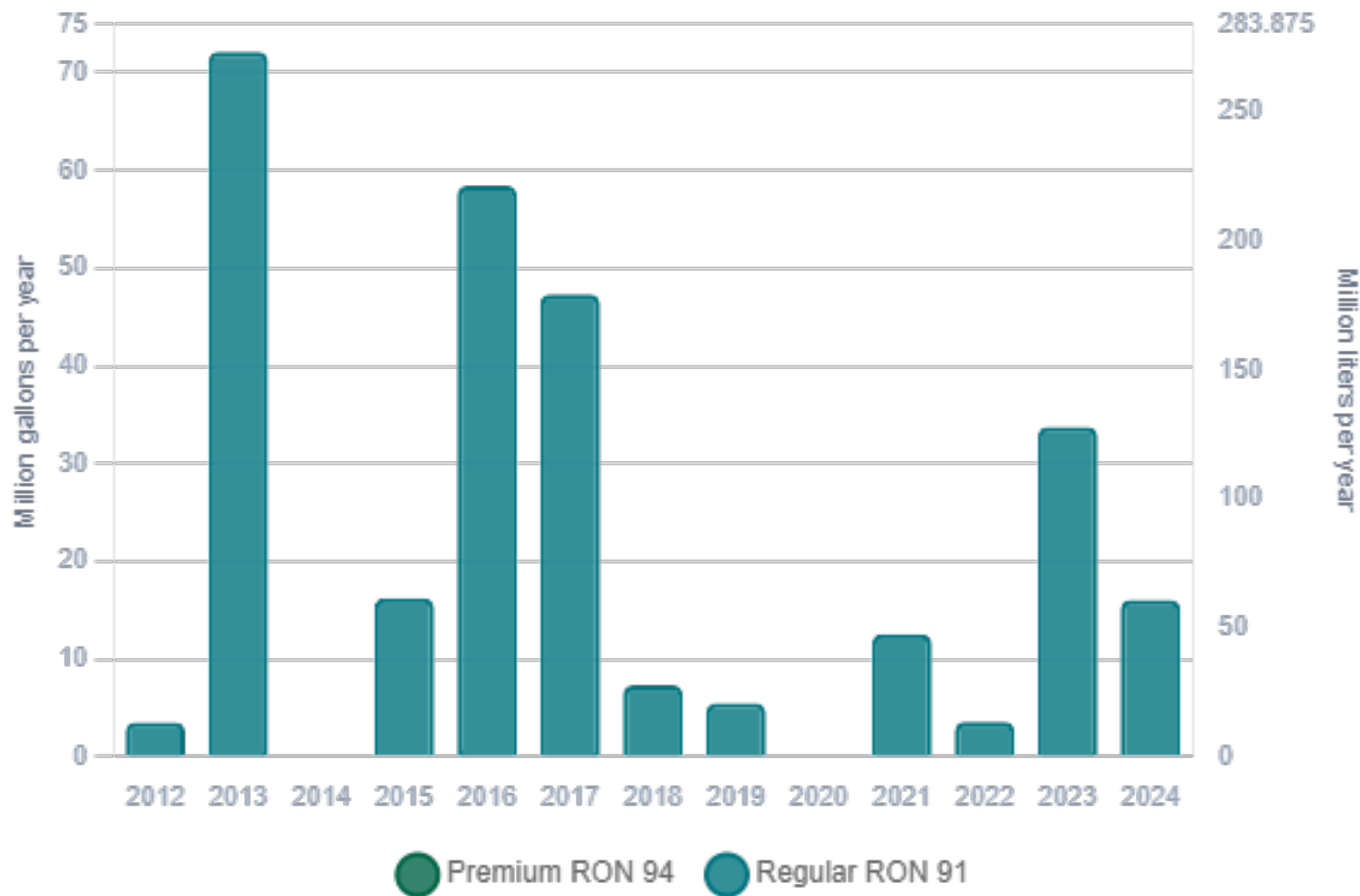
Source: Korean Energy Economics Institute

Gasoline Production in South Korea



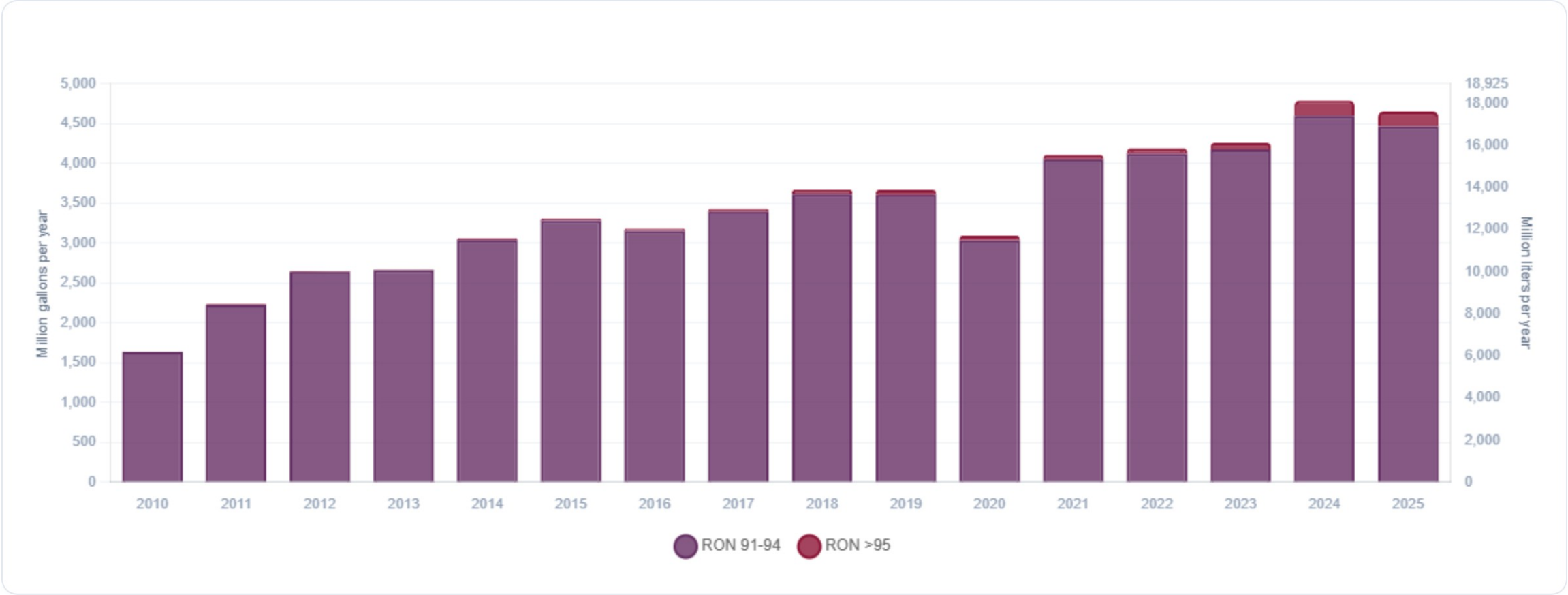
Source: Korean Energy Economics Institute

Gasoline Imports in South Korea



Source: Korean Energy Economics Institute

Gasoline Exports in South Korea



Source: Korean Energy Economics Institute

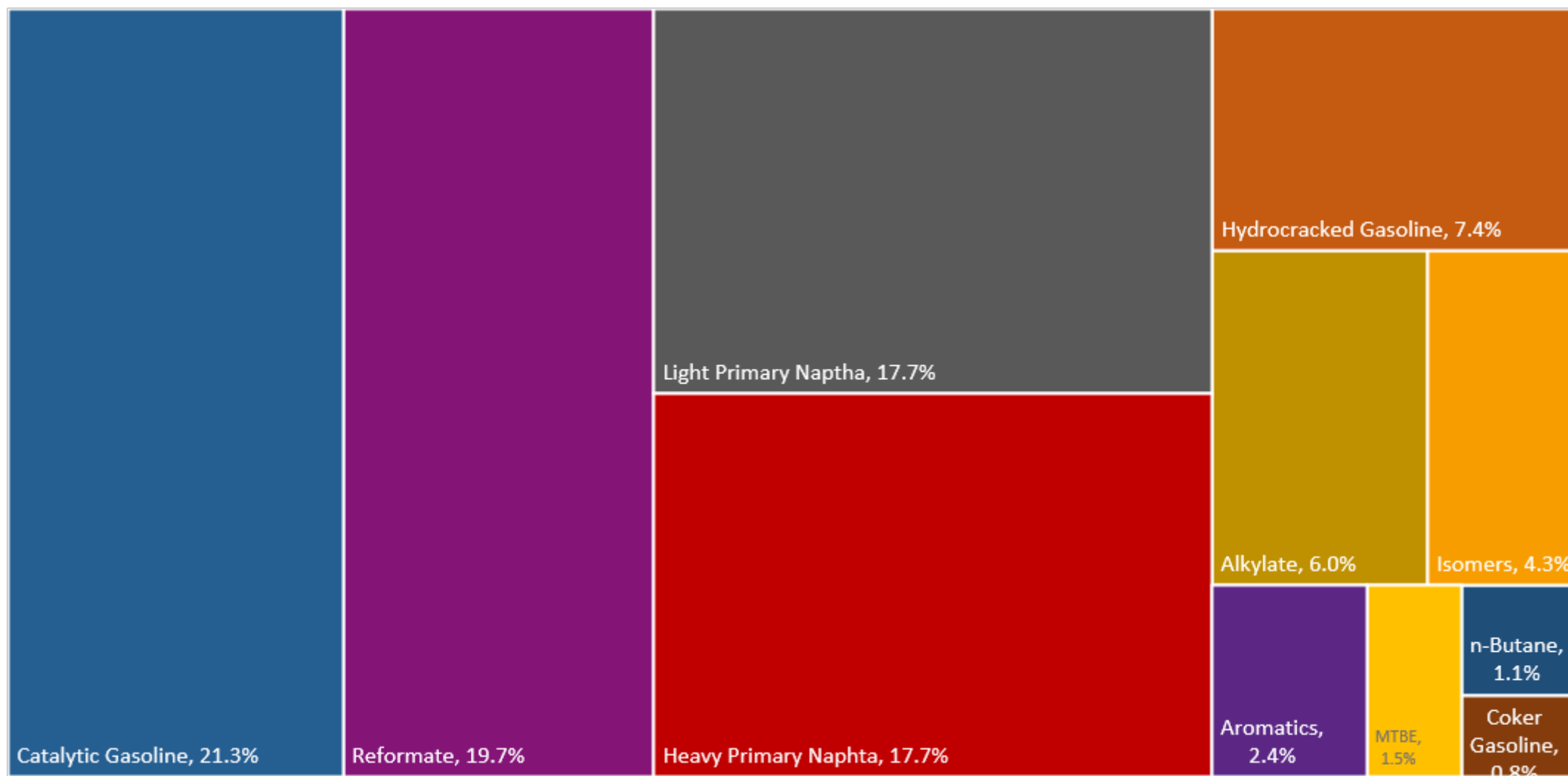
The FARO90 logo is shown in a blue box, followed by a chemical structure diagram of a substituted cyclohexadiene derivative. The structure features a six-membered ring with two double bonds. Substituents include a methyl group, a vinyl group, and a group labeled 'X' in orange. The letters 'H', 'C', and 'X' are also highlighted in orange in the diagram.

43

Gasoline Quality in South Korea

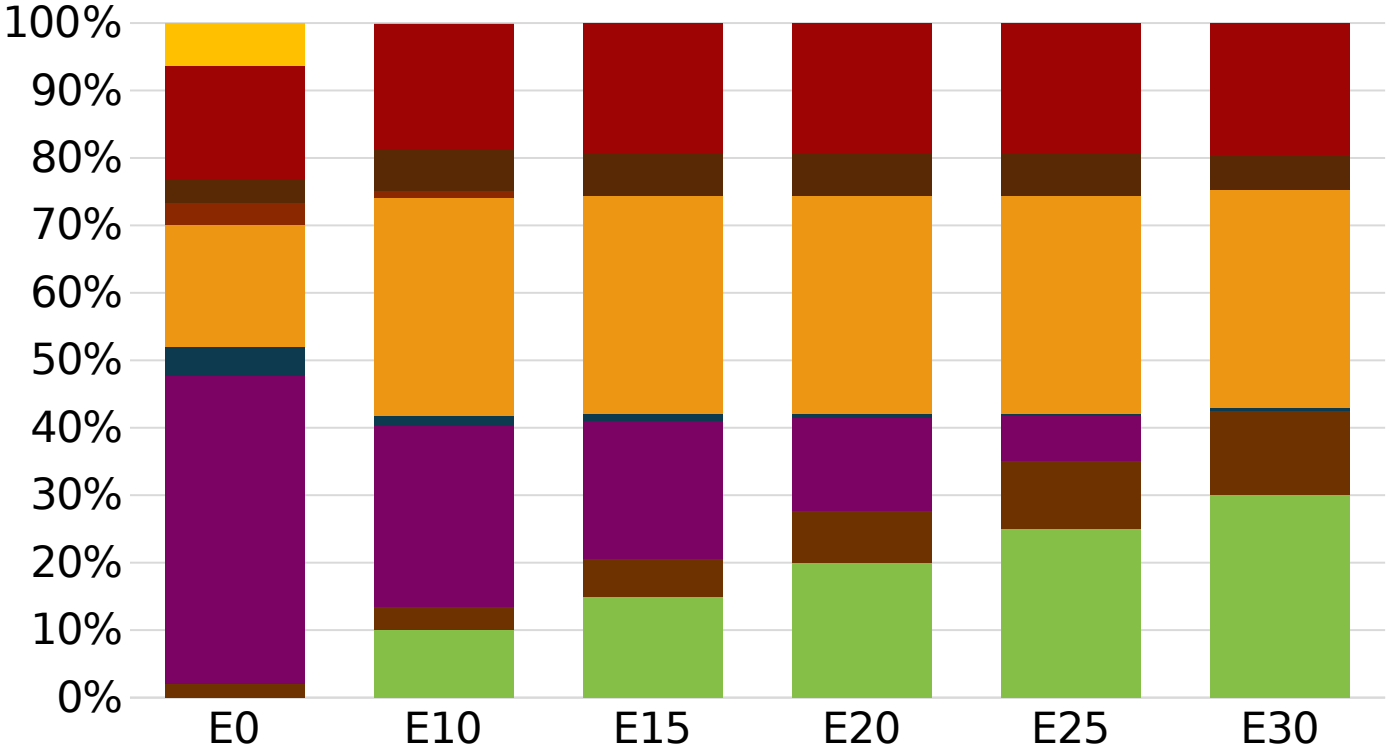
Name	Clean Air Conservation Act			
Implementation Year	2009			
Applicability	Nationwide			
Grade	RON 91		RON 94	
	Min	Max	Min	Max
RON	91.0		94.0	
MON				
AKI				
Benzene (%vol)		0.7		0.7
Aromatics (% vol)		24.0		24.0
Olefins (%vol)		18.0		18.0
Lead (g/L)		0.013		0.013
Manganese (mg/L)		-		-
Sulfur (ppm)		10		10
Oxygen (%v)	0.5	2.3	0.5	2.3
Ethanol (%vol)				
MTBE (%vol)				
RVP (psi)		8.7		8.7

South Korea Available Blending Components



Source:
Faro90

South Korea – Gasoline 91 – Constant Octane

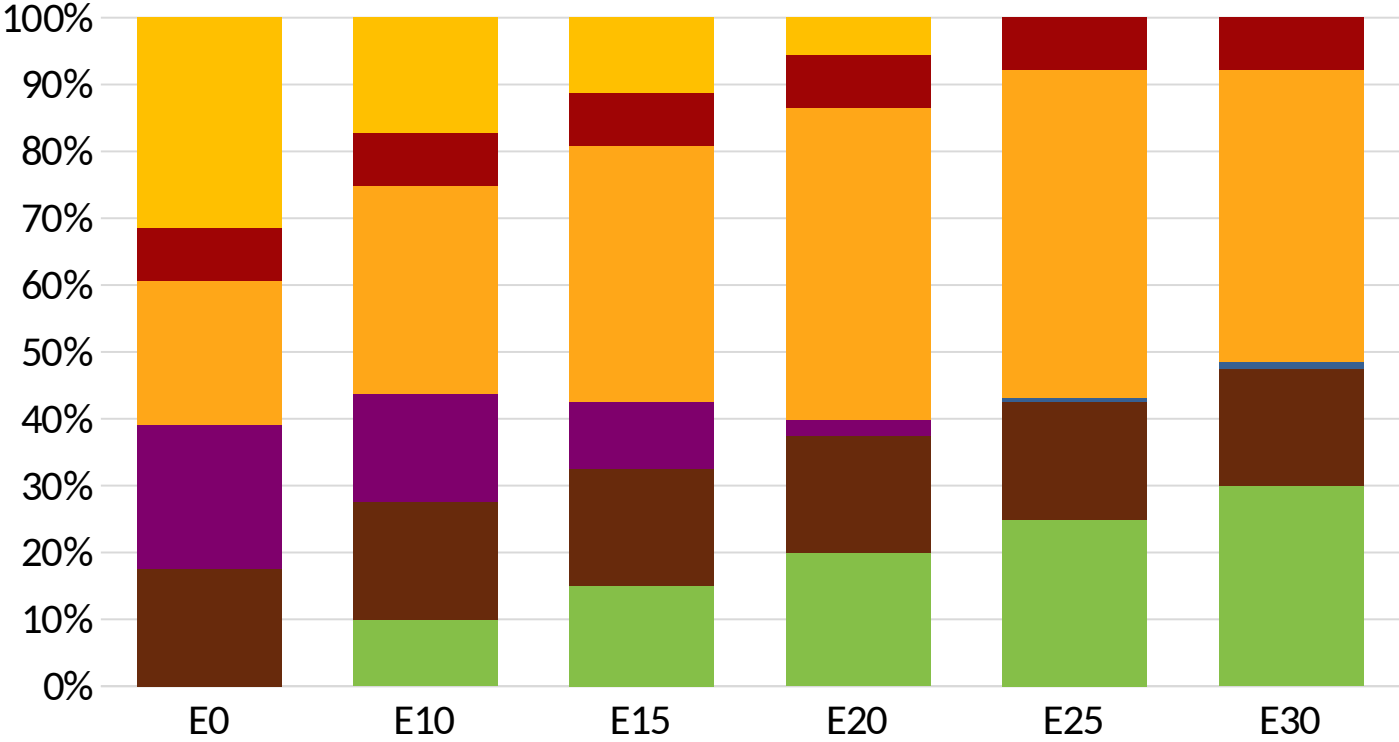


Average CIF 2025
Prices

Octane (RON)	91.0	91.0	92.1	93.2	94.3	95.3
Price (US\$/gal)	\$2.03	\$1.97	\$1.93	\$1.87	\$1.85	\$1.87

Source:
Faro90

South Korea - Gasoline 94 - Constant Octane

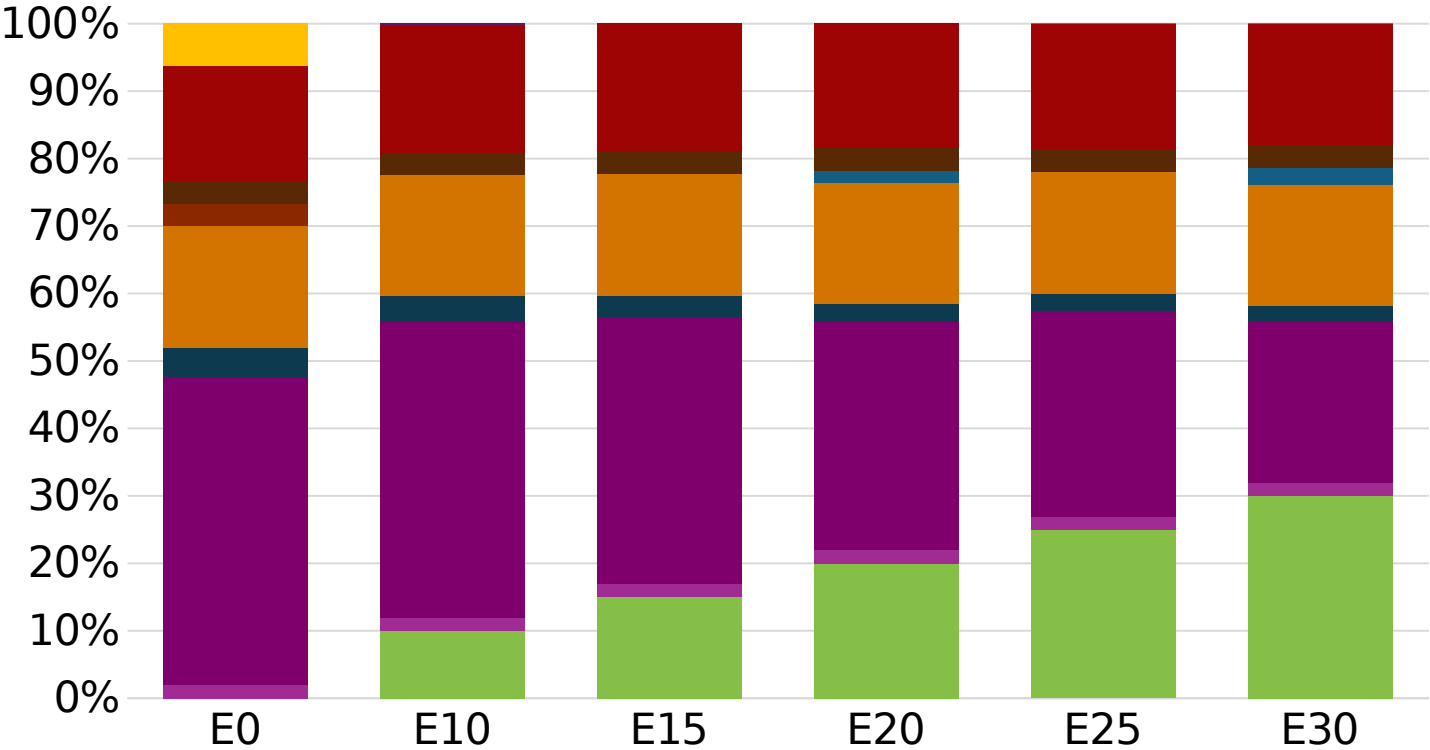


Octane (RON)	94.0	94.0	94.0	94.0	94.7	96.9
Price (US\$/gal)	\$2.04	\$1.92	\$1.88	\$1.84	\$1.79	\$1.74

Average CIF 2025 Prices

Source:
Faro90

South Korea – Gasoline 91 – Increased Octane

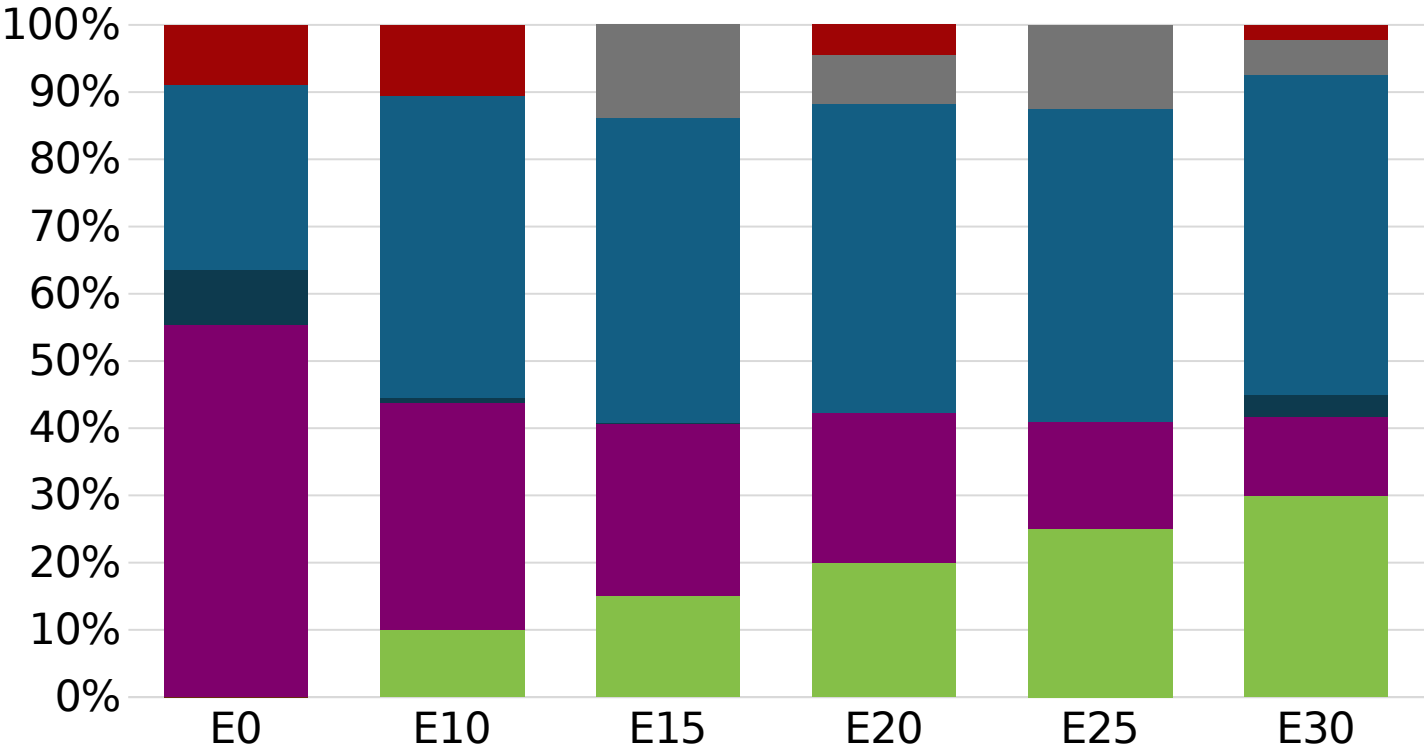


Average CIF 2025 Prices

Octane (RON)	91.0	93.7	95.4	97.1	98.8	100.3
Price (US\$/gal)	\$2.03	\$2.03	\$2.03	\$2.03	\$2.03	\$2.03

Source:
Faro90

South Korea – Gasoline 94 – Increased Octane

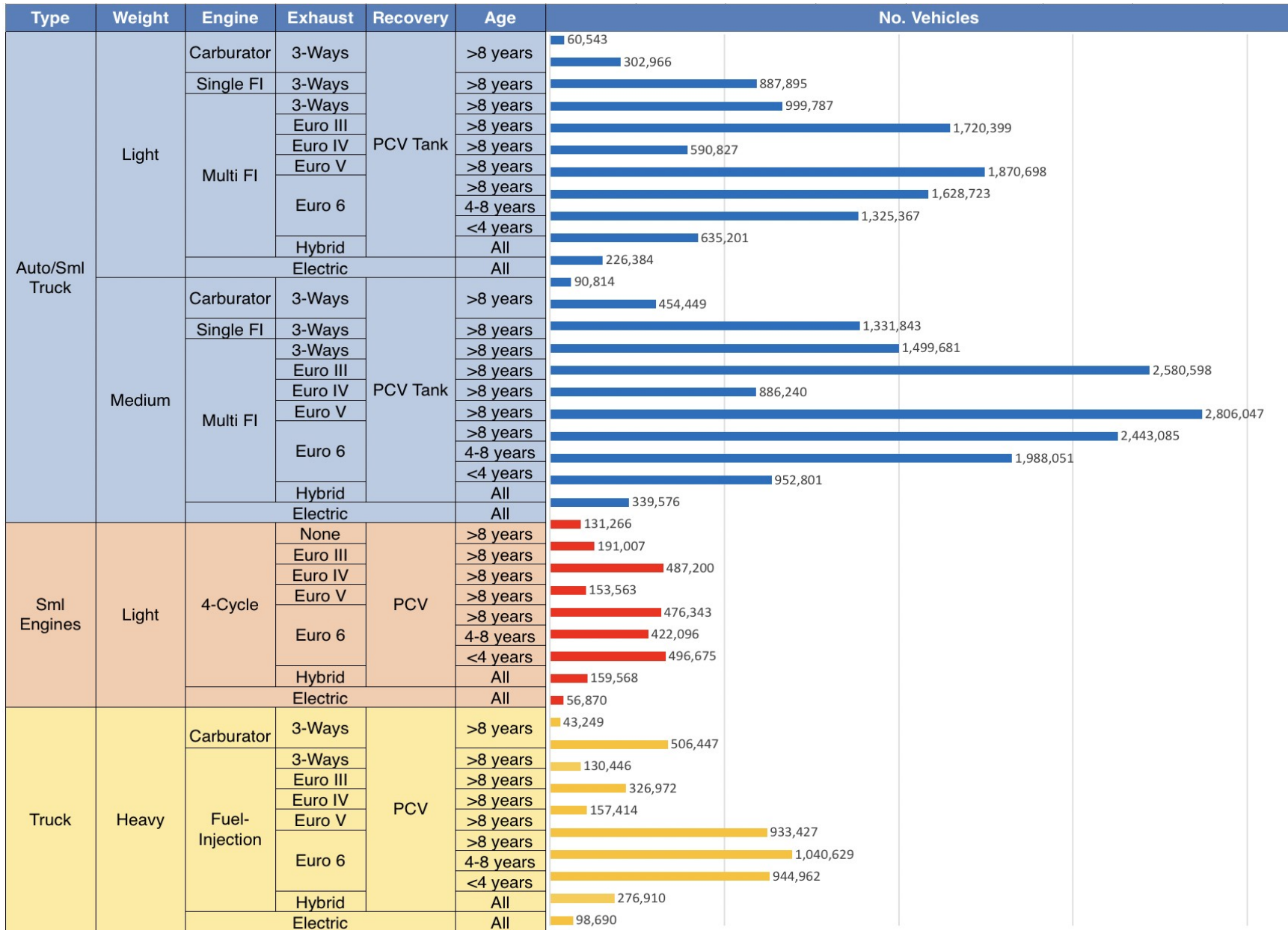


Average CIF 2025
Prices

Octane (RON)	94.0	94.0	94.2	95.6	97.3	99.5
Price (US\$/gal)	\$2.04	\$2.04	\$2.04	\$2.04	\$2.04	\$2.04

Source:
Faro90

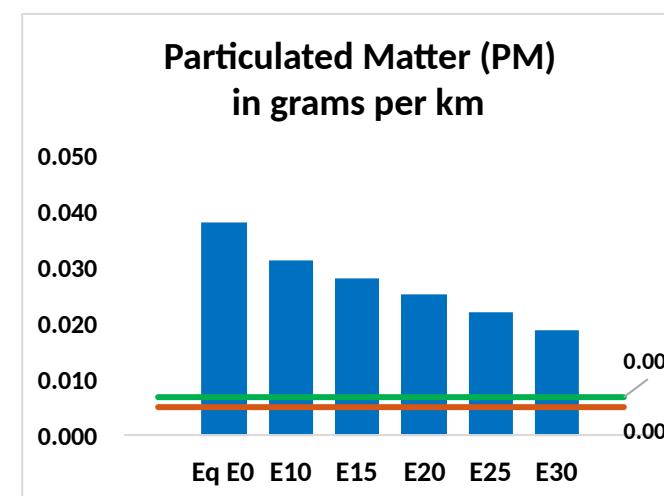
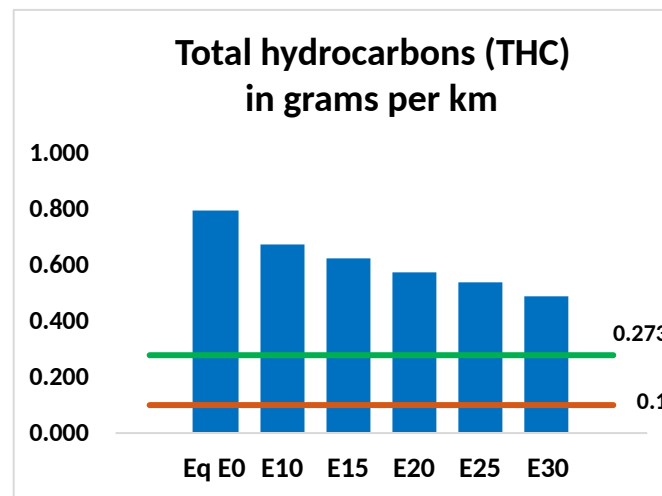
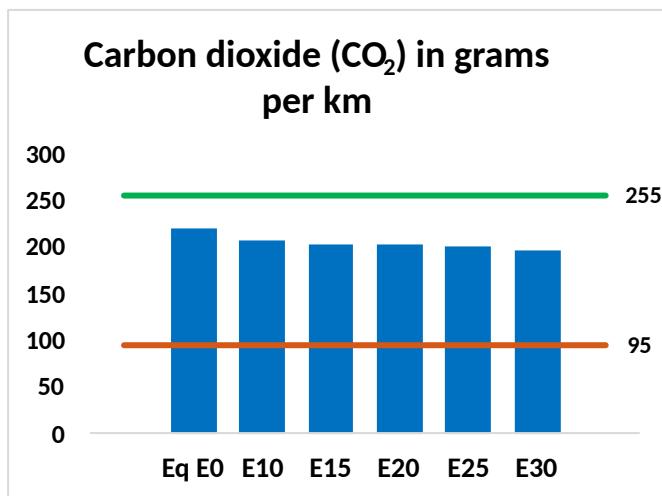
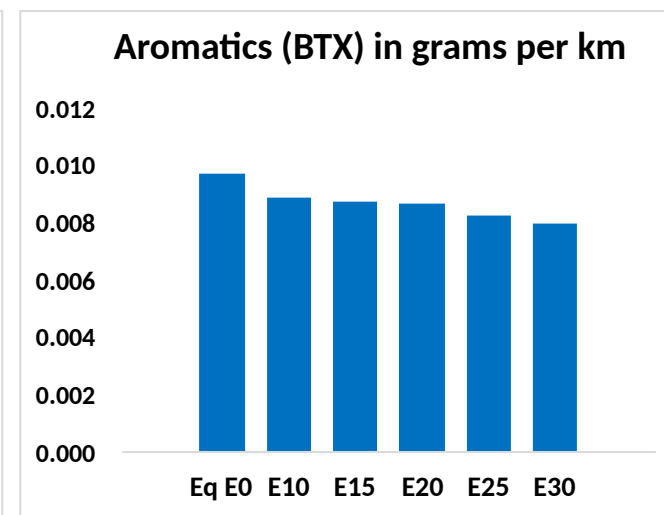
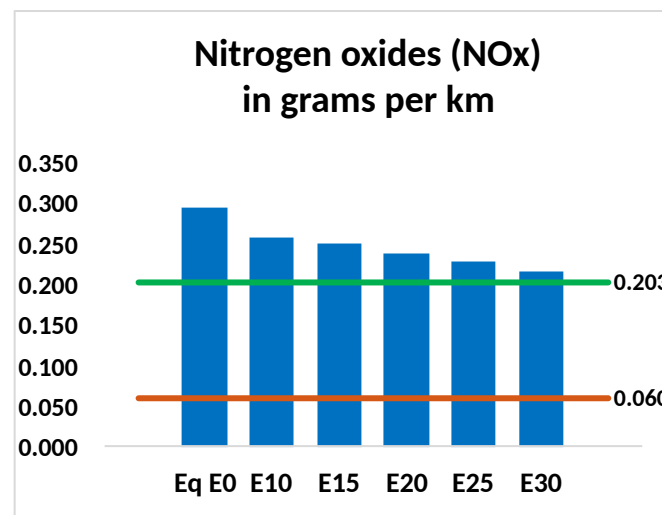
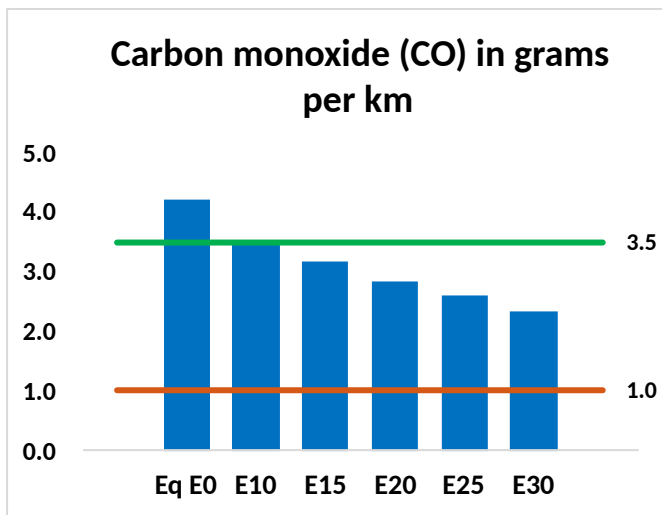
Gasoline Vehicle Fleet - South Korea



Vehicle Fleet: 32,655,710
Gasoline Vehicle Fleet: 27,483,451

Source: MOLIT

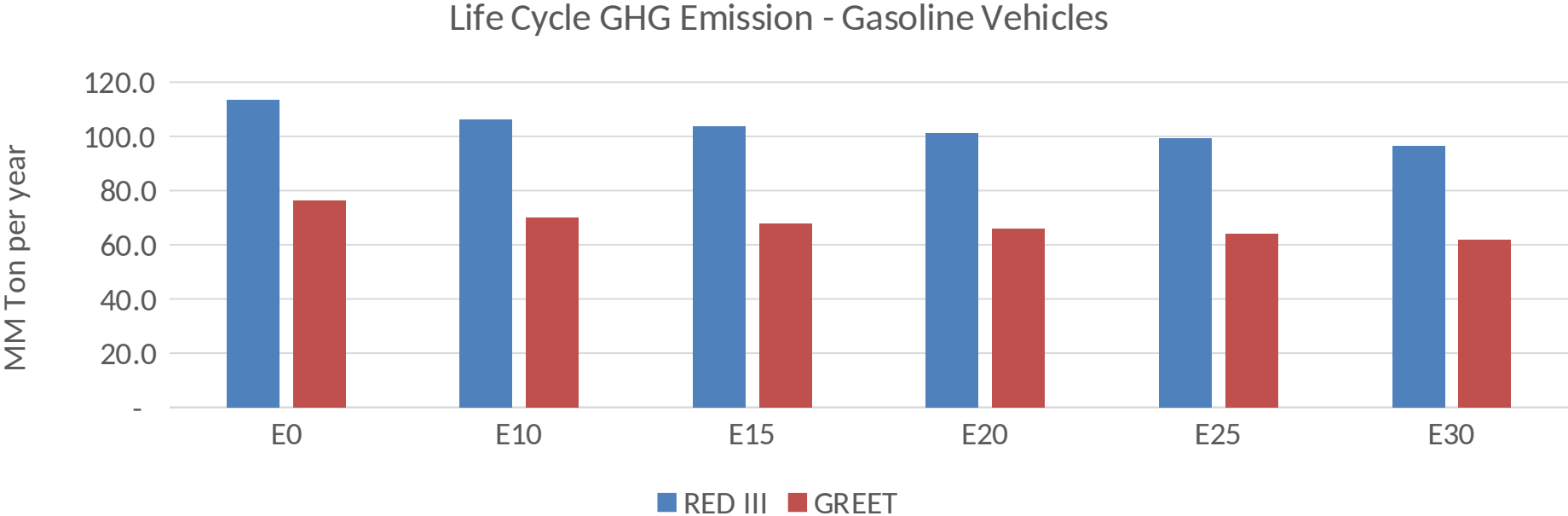
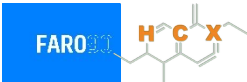
South Korea – Vehicular Emissions



South Korea – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	2,698,816	2,459,685	2,220,554	2,018,419	1,812,161	1,652,704	1,484,377	-18%	-33%
CO2	139,947,032	136,348,394	132,949,680	130,276,692	128,953,993	128,324,905	125,959,508	-5%	-8%
VOC	507,796	469,392	430,988	399,641	368,152	343,957	311,608	-15%	-27%
NOx	189,582	175,047	165,224	160,298	152,275	146,026	138,644	-13%	-20%
SOx	1,639	1,515	1,391	1,286	1,186	1,077	962	-15%	-28%
NH3	50,231	49,734	49,236	49,470	50,027	50,418	50,743	-2%	0%
N2O	2,899	2,883	2,872	2,885	2,945	2,978	3,012	-1%	2%
CH4	105,041	105,041	105,041	106,617	108,928	110,924	112,814	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	2,100	1,986	1,873	1,789	1,710	1,649	1,565	-11%	-19%
Acetaldehyde	6,839	7,849	11,517	17,539	23,895	27,304	32,263	68%	249%
Formaldehyde	26,080	25,355	29,557	34,706	36,360	40,224	43,788	13%	39%
Benzene	6,223	5,980	5,667	5,577	5,531	5,289	5,119	-9%	-11%
PM2.5	7,880	7,164	6,447	5,817	5,200	4,539	3,847	-18%	-34%
PM10	16,575	15,069	13,562	12,236	10,938	9,547	8,092	-18%	-34%

South Korea – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	682.3
Target @ 2035 (MMT/year)	332.3
Delta	349.9

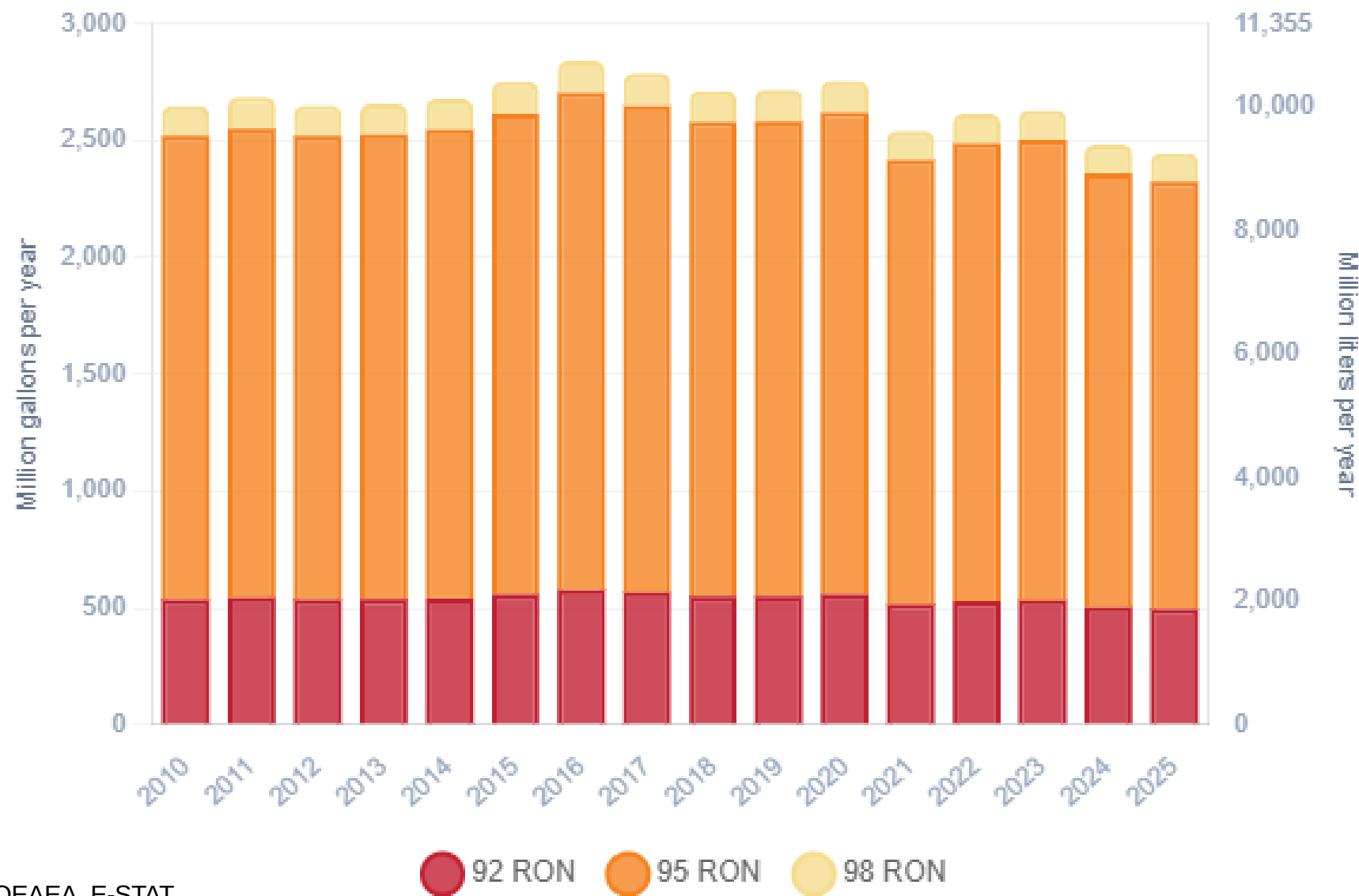
Ethanol Gasoline Blending in Taiwan



In 2025, gasoline consumption in Taiwan was 9,262 million liters (2,447 million gallons) and growth rate was -0.5%. Gasoline production and net imports were 13,152 and -4,064 million liters (3,475 and -1074 million gallons) correspondingly. Taiwan offers three main grades: RON 92, RON 95 and RON 98) with an average octane of 94.5 RON.

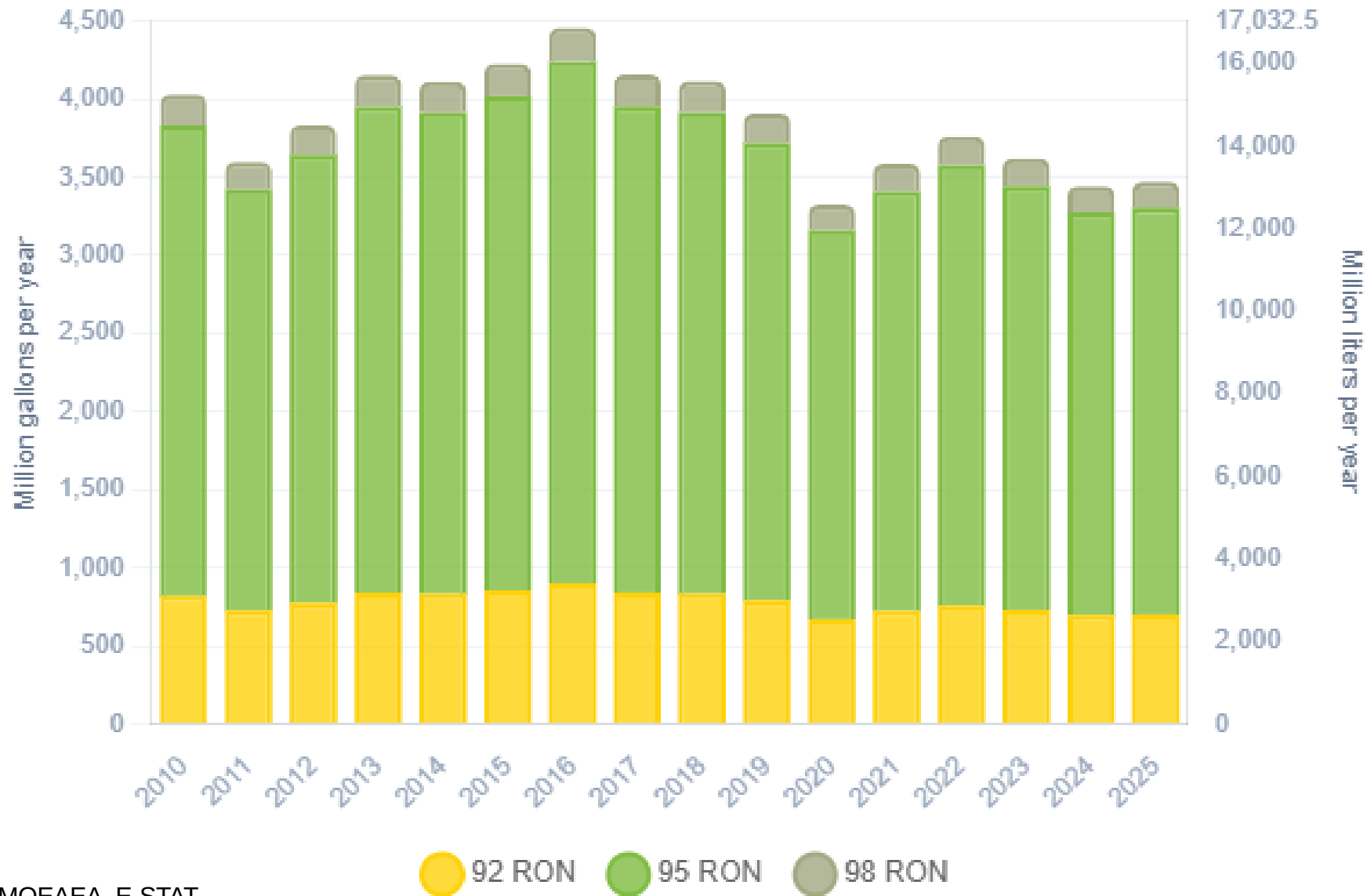
Taiwan has policies to promote biofuels in transportation to reduce imported fuel dependency and lower carbon emissions. While there was a limited trial of E3 gasohol in the mid-2000s, the lack of domestic production and insufficient incentives led to its discontinuation. Only a few gas stations offer it. Ethanol is supplied by imports and corresponds to an average content of less than 0.1%. CPC's 2026 budget indicates that the company plans to procure 96,000 liters of ethanol for E3 blends. Future bioethanol use depends on developing profitable cellulosic ethanol production from non-food waste like rice straw and food waste, rather than imported feedstocks.

Gasoline Demand in Taiwan



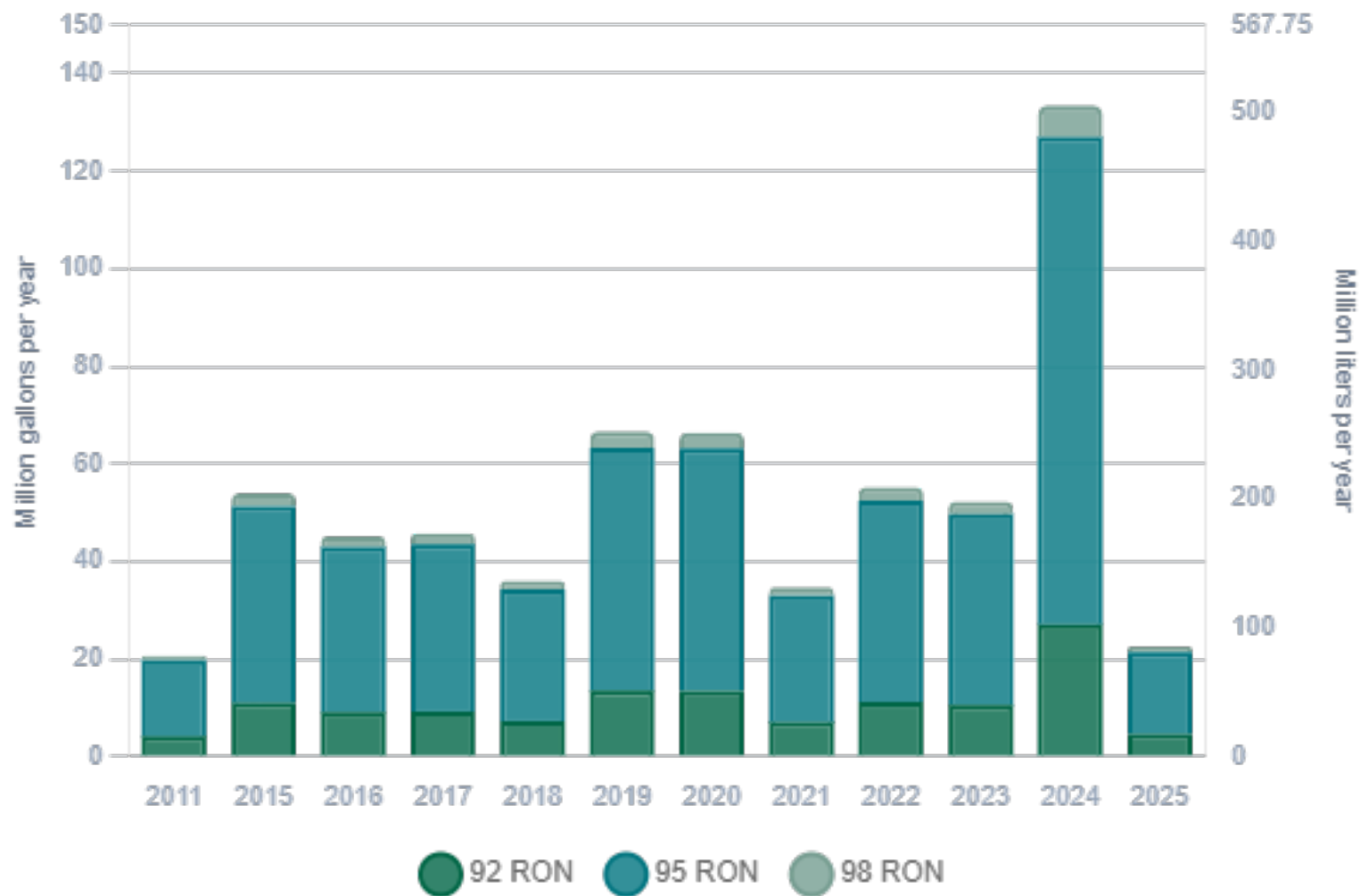
Source: MOEAEA, E-STAT

Gasoline Production in Taiwan



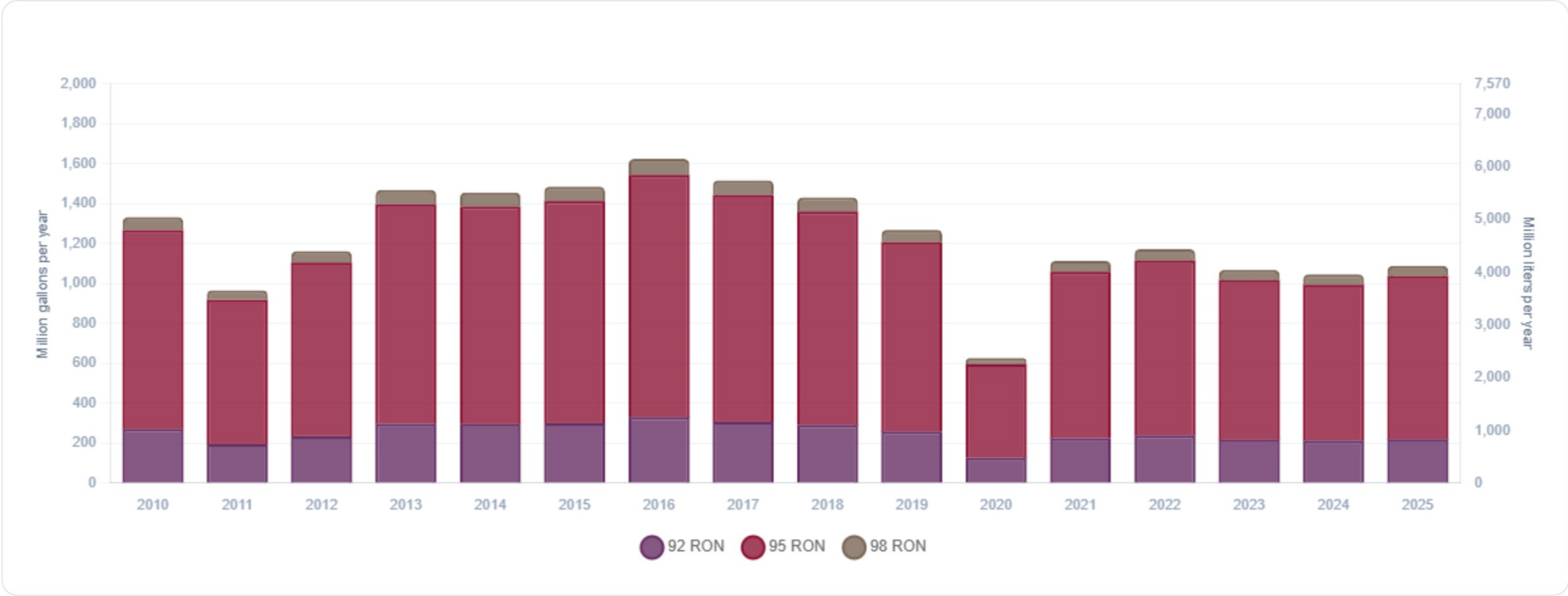
Source: MOEAEA, E-STAT

Gasoline Imports in Taiwan



Source: MOEAEA, E-STAT

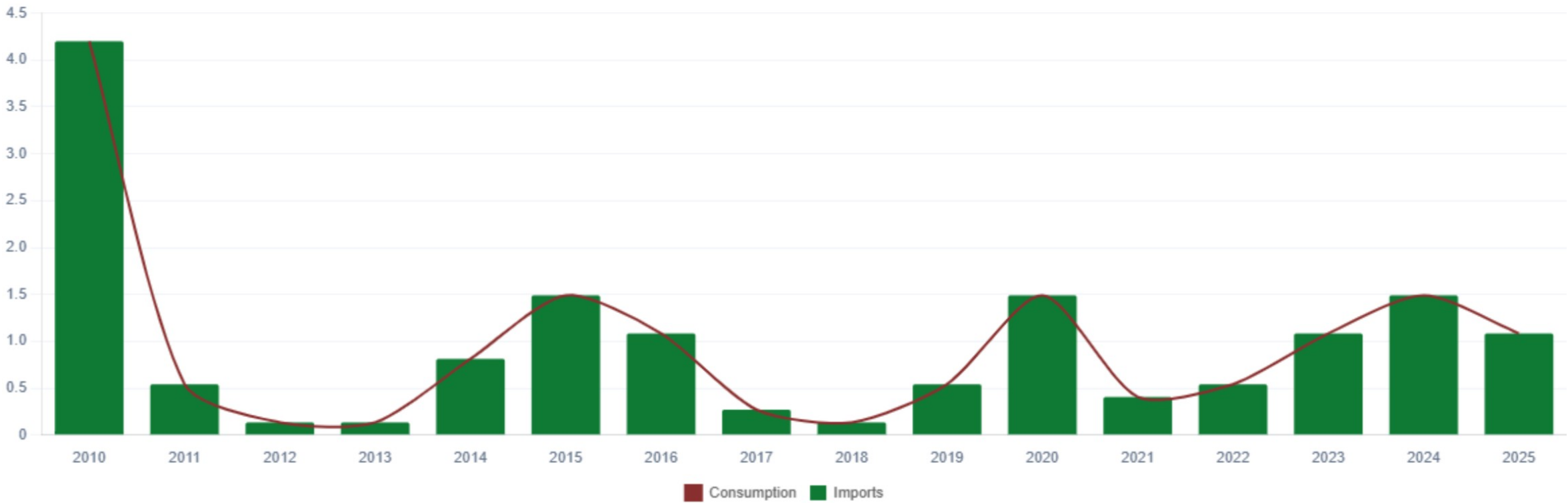
Gasoline Exports in Taiwan



Source: MOEAEA, E-STAT

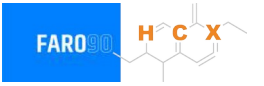
Ethanol Balance in Taiwan

Currently E3 blends are used in a small proportion of gas stations with a subsidy. There is no local production. Ethanol is supplied by imports. Country level ethanol content is 0.1%.



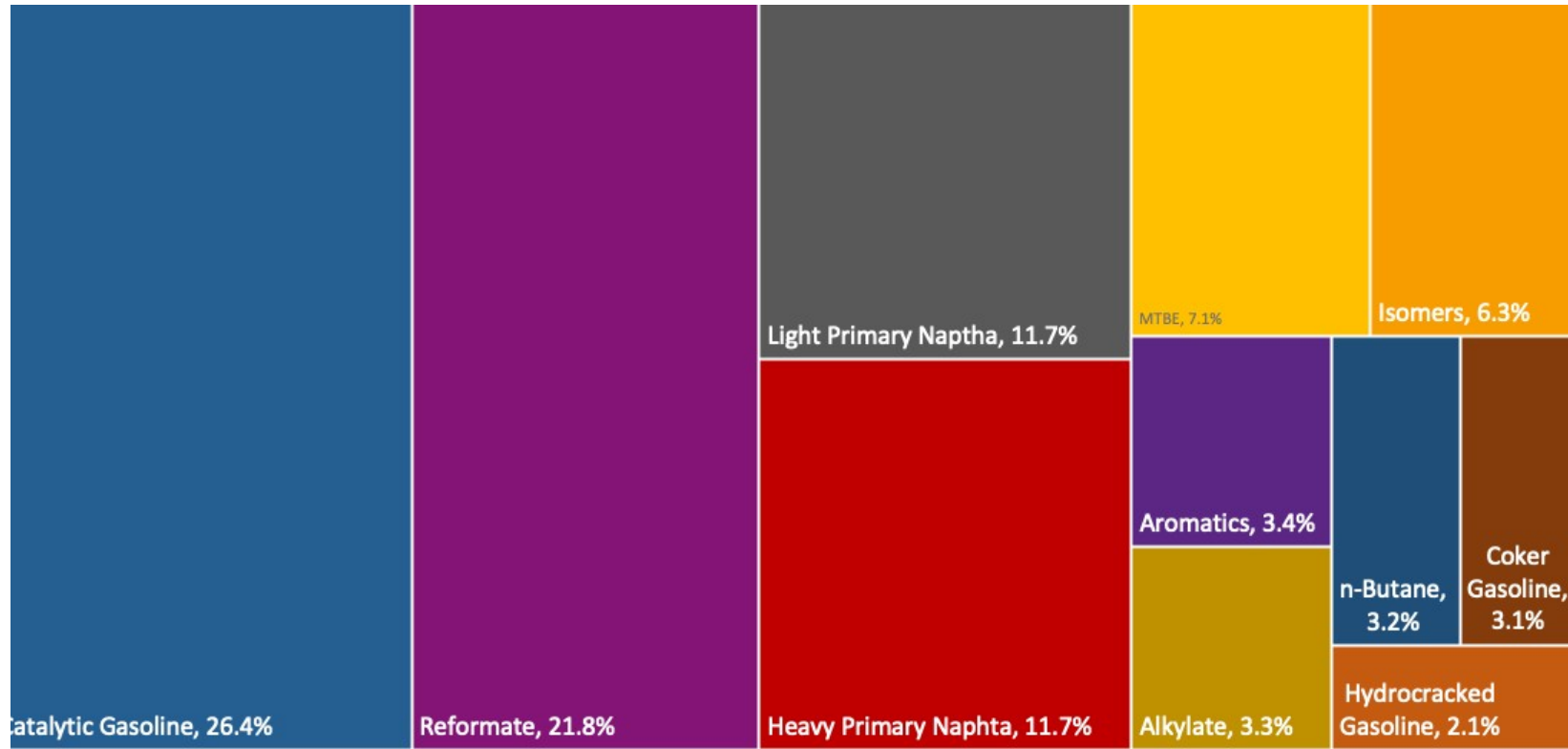
Source: MOEAEA, E-STAT

Gasoline Quality in Taiwan

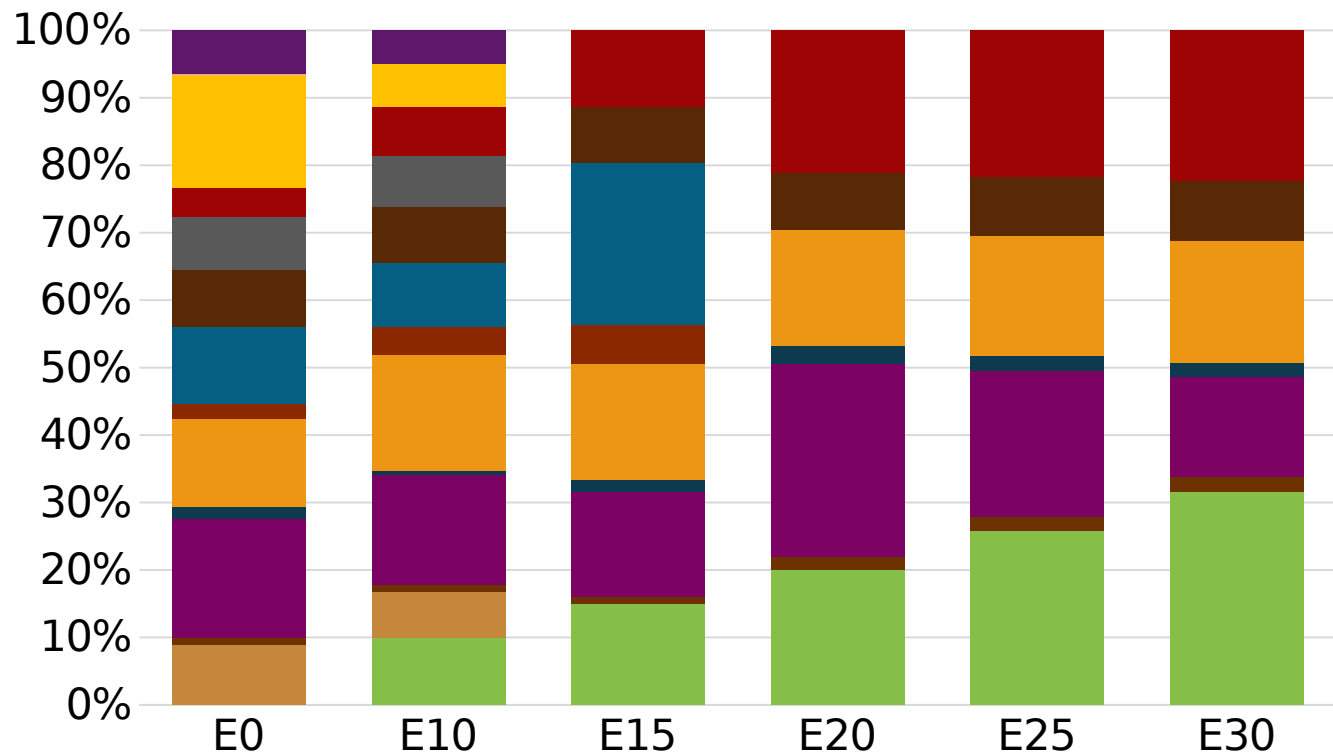


Source: MOEAEA, MOENV, Formosa Petrochemical

Taiwan Available Blending Components



Taiwan – Gasoline 95 – Constant Octane

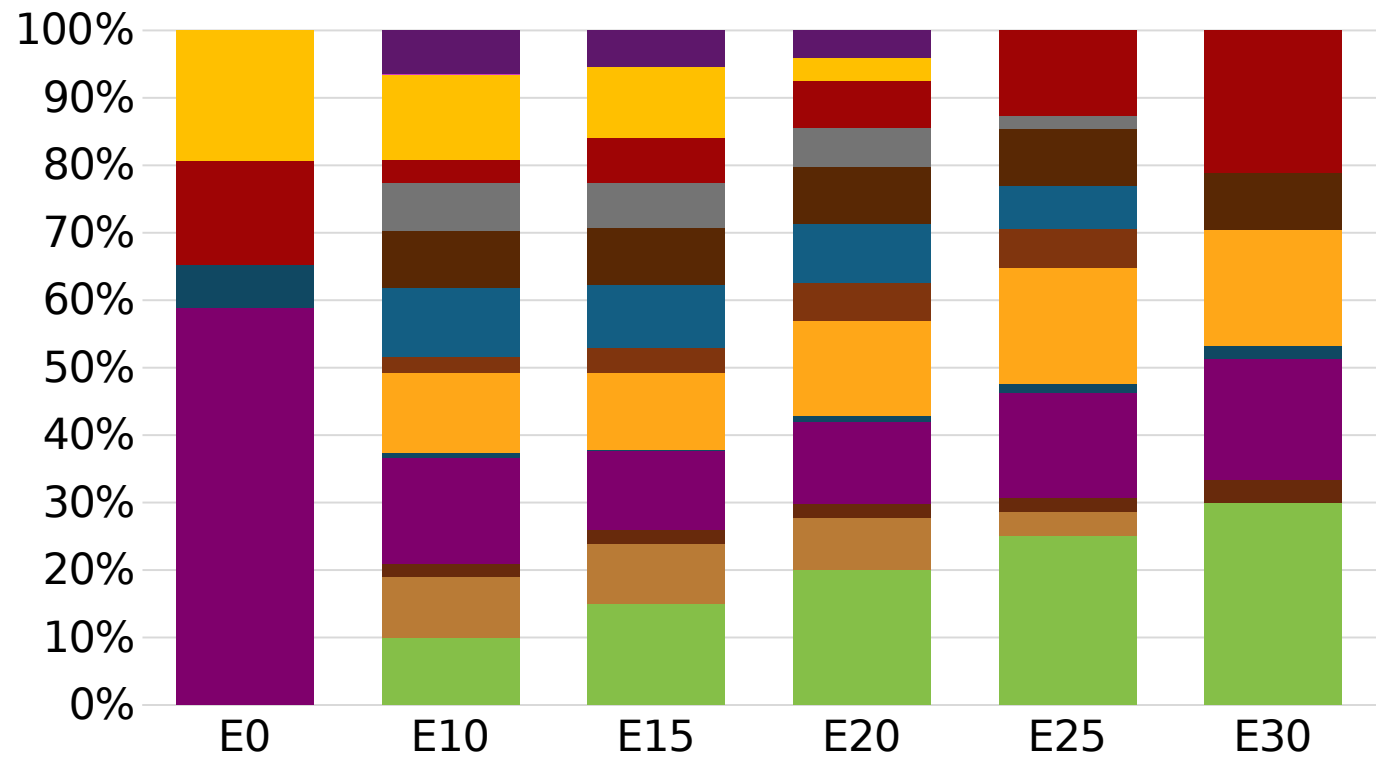


Octane (RON)	95.0	95.0	95.0	95.8	95.0	95.0
Price (US\$/gal)	\$2.17	\$2.09	\$2.01	\$1.94	\$1.85	\$1.78



Average CIF 2025
Prices

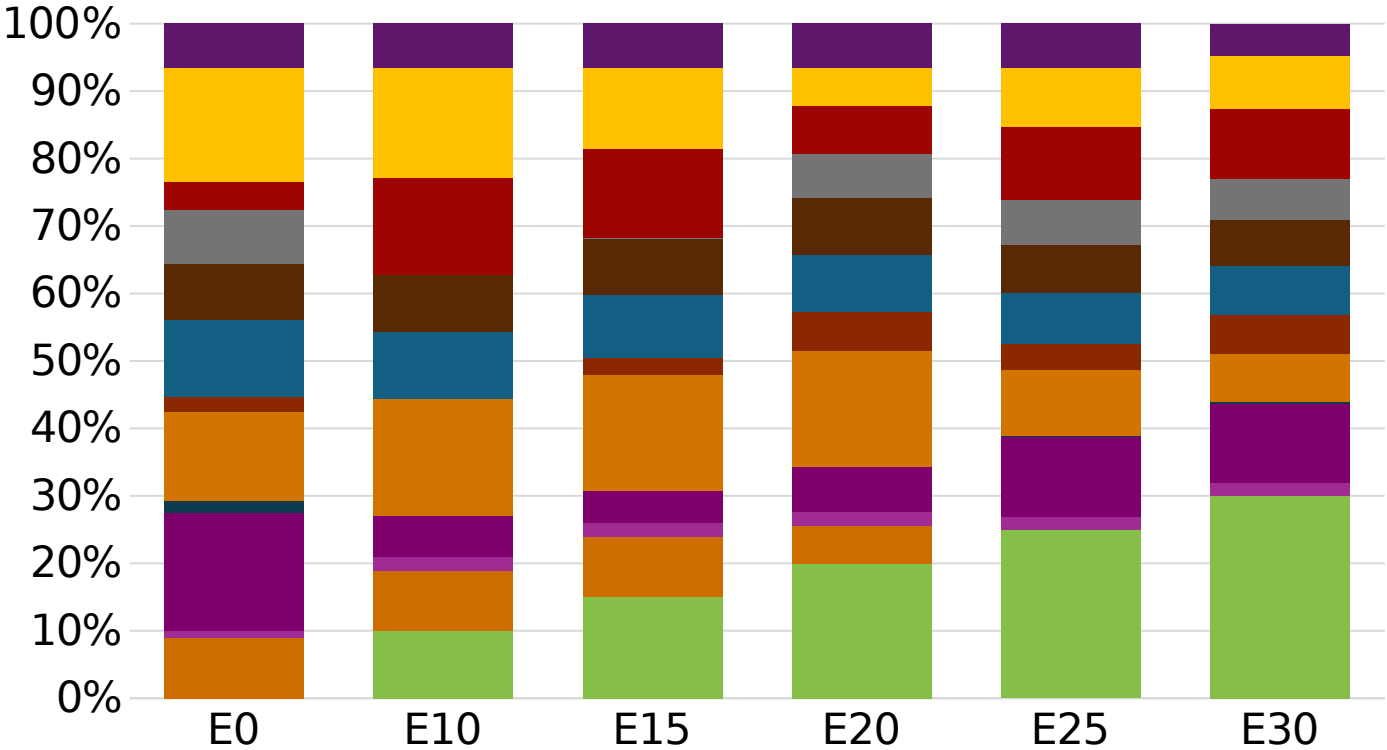
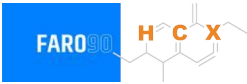
Taiwan - Gasoline 98 - Constant Octane



Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.6
Price (US\$/gal)	\$2.23	\$2.21	\$2.13	\$2.09	\$2.00	\$1.90

Average CIF 2025
Prices

Taiwan – Gasoline 95 – Increased Octane

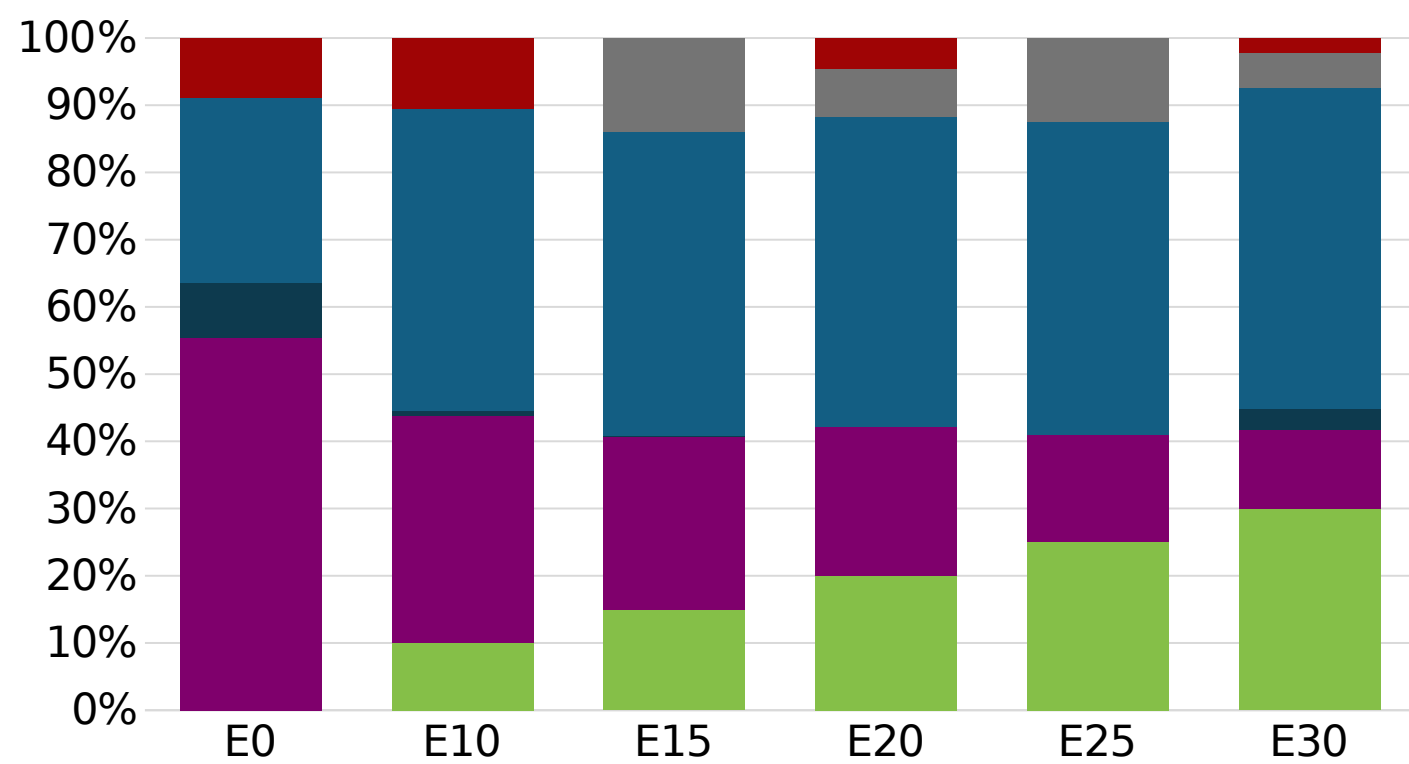


Average CIF 2025 Prices

Octane (RON)	95.0	96.4	97.4	98.1	100.6	102.0
Price (US\$/gal)	\$2.17	\$2.17	\$2.17	\$2.17	\$2.17	\$2.17

Source:
Faro90

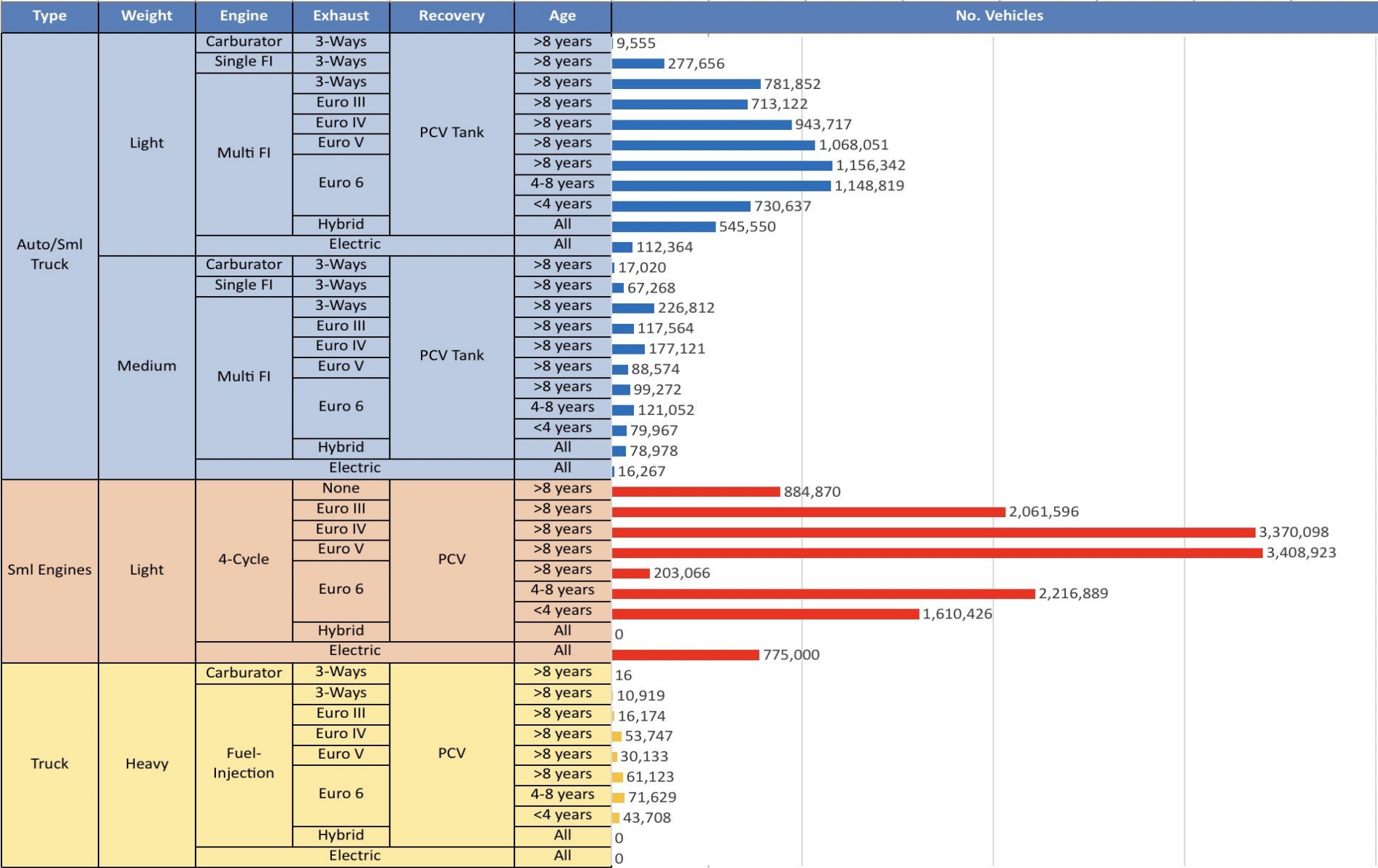
Taiwan – Gasoline 98 – Increased Octane



Average CIF 2025
Prices

Octane (RON)	98.0	99.8	99.9	101.3	102.6	103.9
Price (US\$/gal)	\$2.23	\$2.23	\$2.23	\$2.23	\$2.23	\$2.23

Gasoline Vehicle Fleet - Taiwan

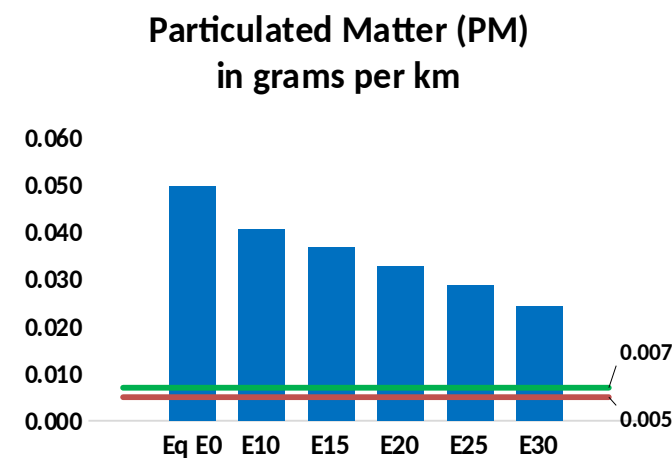
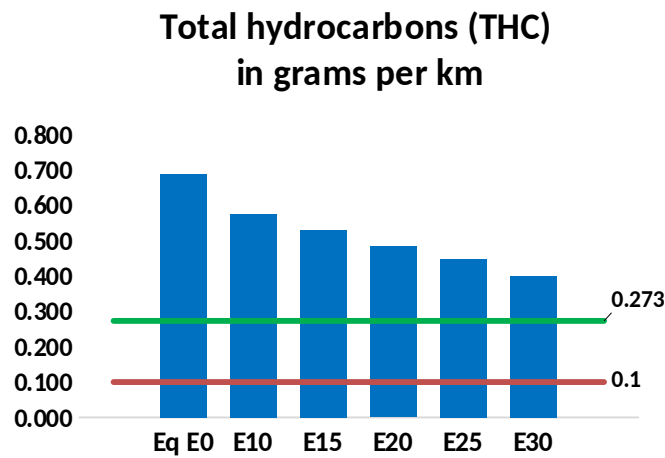
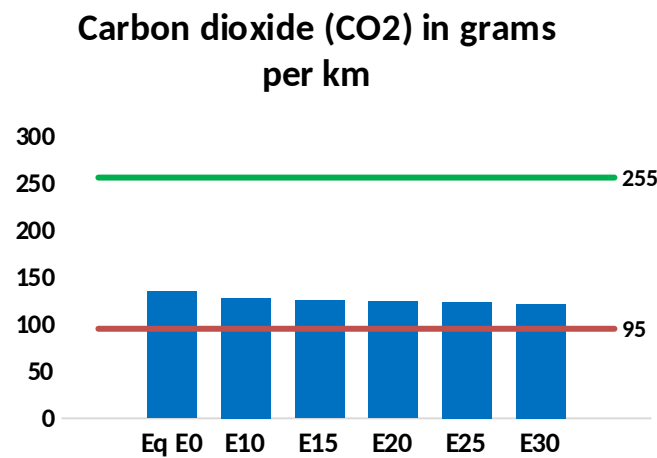
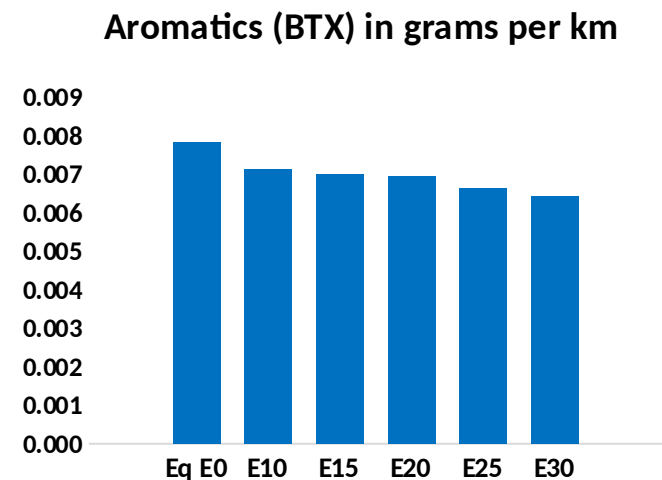
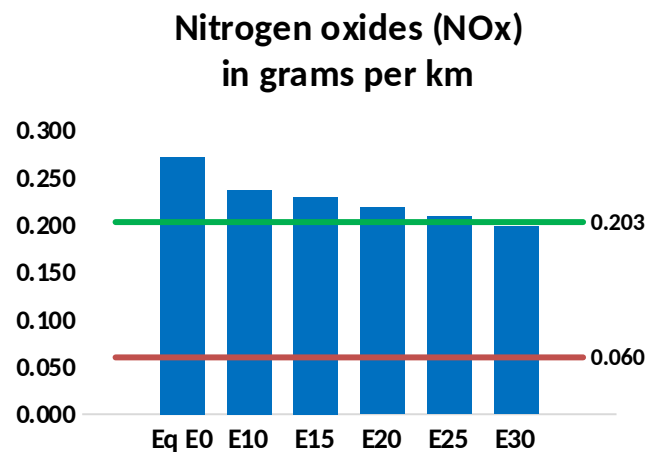
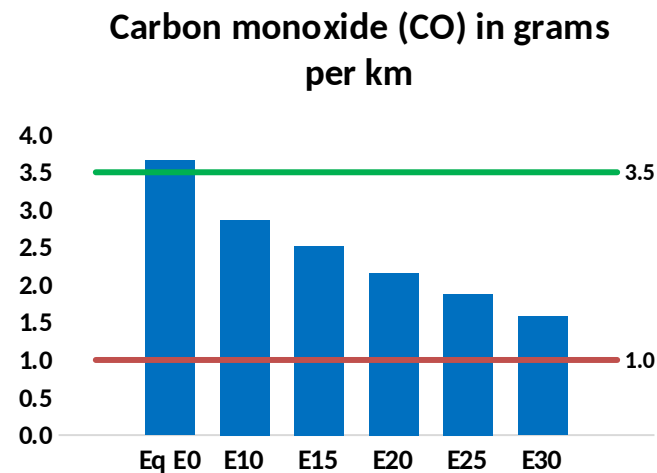
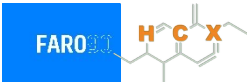


Vehicle Fleet: 23,395,876
Gasoline Vehicle Fleet: 20,722,999

Source: DATA-GOV.tw, CIECData

Taiwan – Vehicular Emissions

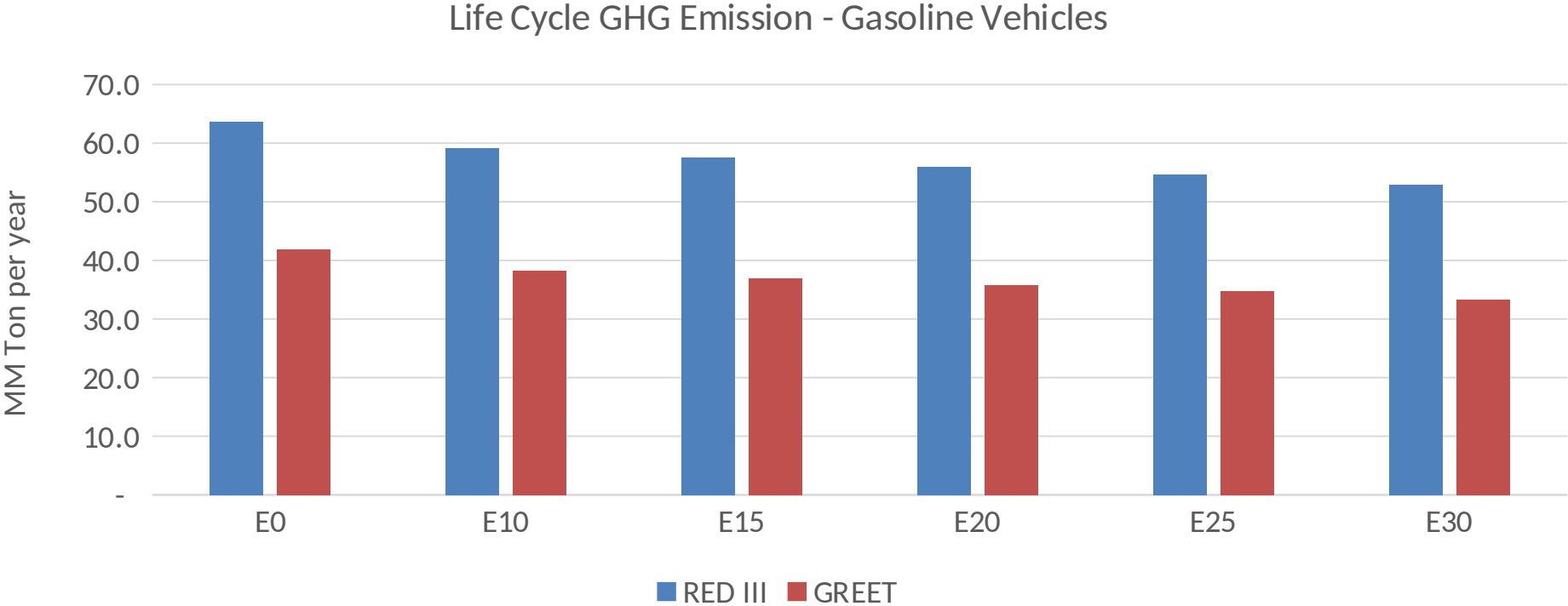
TIER III BIN 125 United States
Euro 6 Standard



Taiwan – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	1,784,176	1,590,911	1,397,645	1,228,325	1,052,895	916,478	772,360	-22%	-41%
CO2	65,534,716	63,849,538	62,257,981	61,006,268	60,386,871	60,108,716	59,000,740	-5%	-8%
VOC	335,531	308,344	281,156	258,605	235,786	218,202	194,870	-16%	-30%
NOx	132,659	122,489	115,583	112,103	106,455	102,047	96,844	-13%	-20%
SOx	716	662	608	562	518	471	421	-15%	-28%
NH3	27,896	27,620	27,344	27,474	27,783	28,000	28,181	-2%	0%
N2O	888	883	879	884	902	912	922	-1%	2%
CH4	88,450	88,450	88,450	89,777	91,722	93,403	94,995	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	2,196	2,040	1,885	1,760	1,635	1,539	1,410	-14%	-26%
Acetaldehyde	7,016	8,052	11,815	17,992	24,512	28,010	33,097	68%	249%
Formaldehyde	28,328	27,541	32,105	37,698	39,494	43,691	47,563	13%	39%
Benzene	3,818	3,669	3,476	3,421	3,393	3,245	3,140	-9%	-11%
PM2.5	4,571	4,156	3,740	3,374	3,017	2,633	2,232	-18%	-34%
PM10	19,701	17,910	16,119	14,543	13,001	11,348	9,618	-18%	-34%

Taiwan – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	291.7
Target @ 2035 (MMT/year)	204.2
Delta	87.5

Source: Greet, RED II/III, climateactiontracker.org, F90

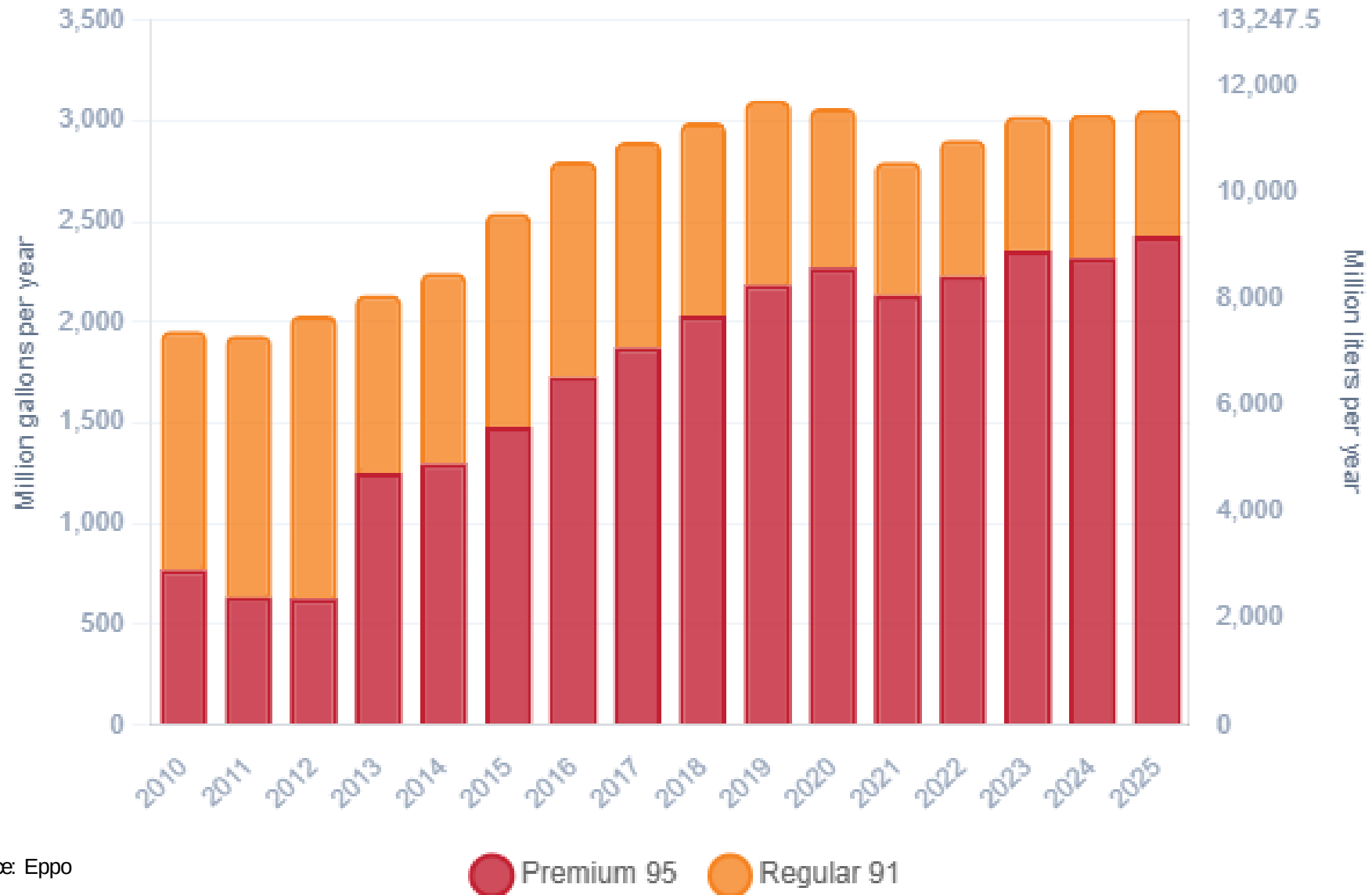
Ethanol Gasoline Blending in Thailand



In 2025, gasoline consumption in Thailand was 11,576 million liters (3,448 million gallons) and growth rate was 3.0%. Gasoline production and net imports were 13,051 and -1,312 million liters (3,448 and -347 million gallons) correspondingly. Thailand offers two main grades: RON 91 (E10) and RON 95 (E10 and E20) with an average octane of 94.2 RON.

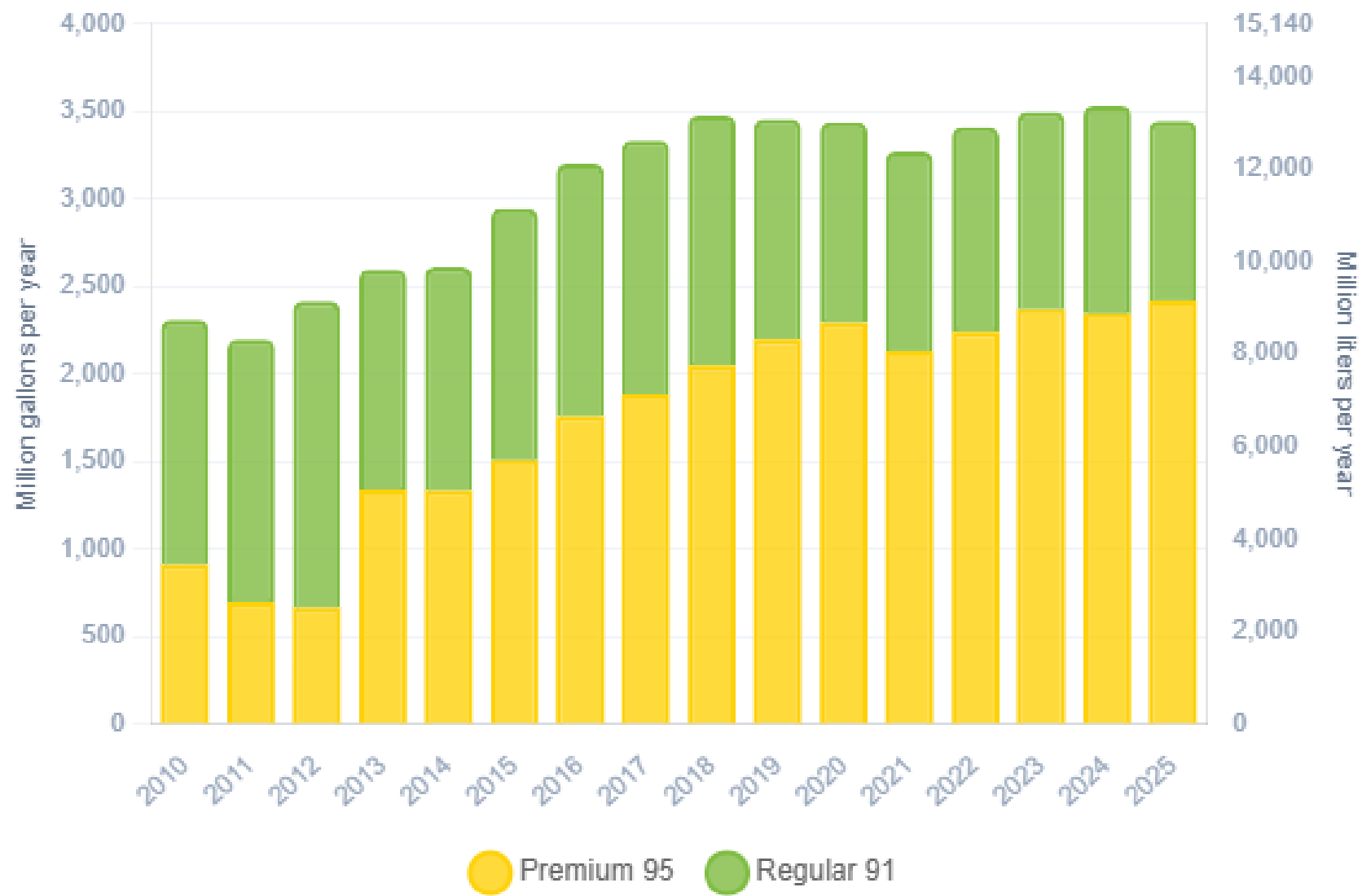
Thailand's bioethanol production primarily uses sugarcane and cassava as feedstocks, with approximately 30 facilities producing over 1 billion liters annually. The promotion of bio-based products like ethanol is part of Thailand's bioeconomy policy, supporting both the energy sector and domestic economic growth. In 2025, ethanol consumption supplied by local production in Thailand was 1,290 million liters (341 million gallons) representing 11.0% of gasoline demand.

Gasoline Demand in Thailand



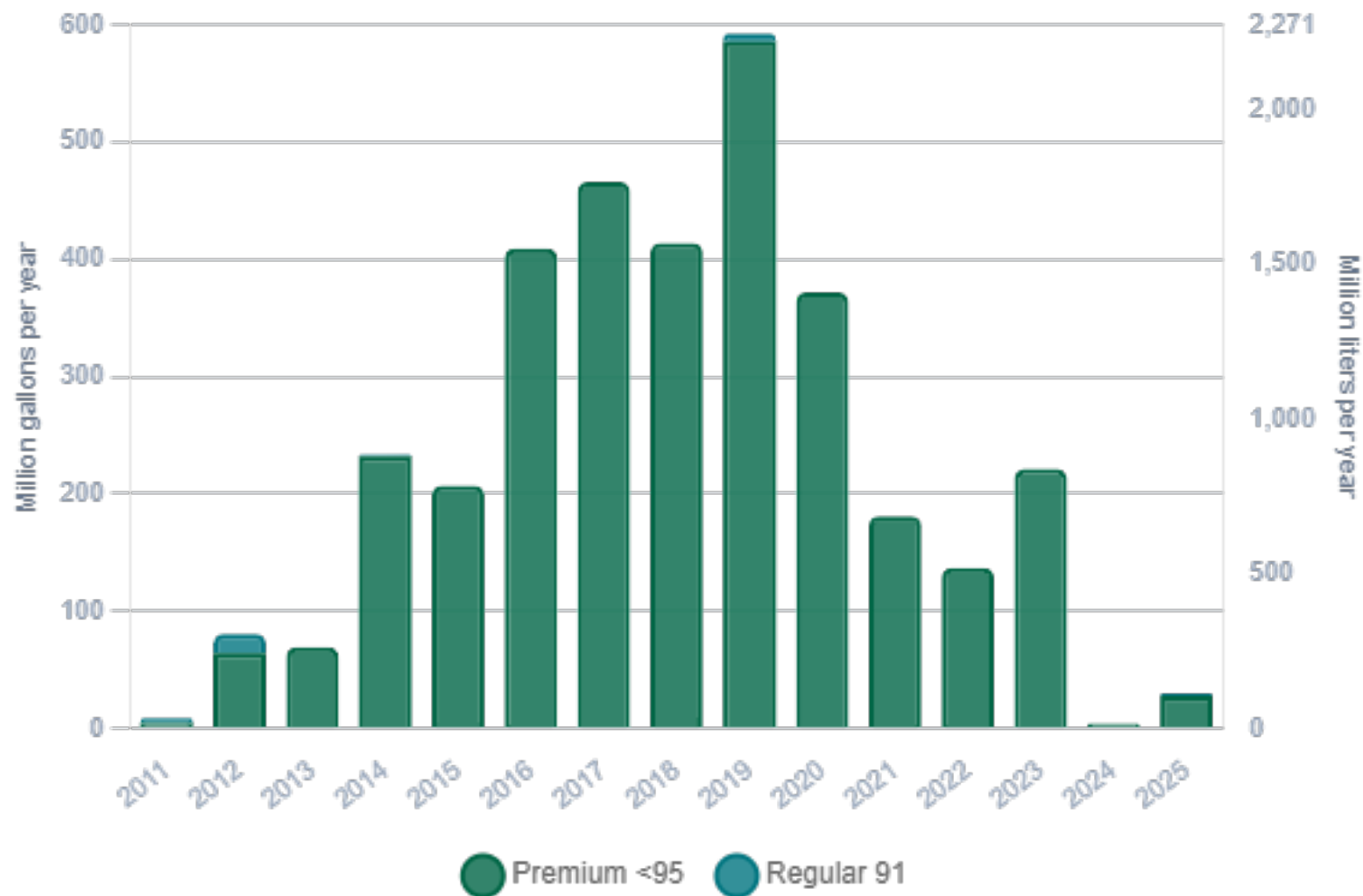
Source: Eppo

Gasoline Production in Thailand



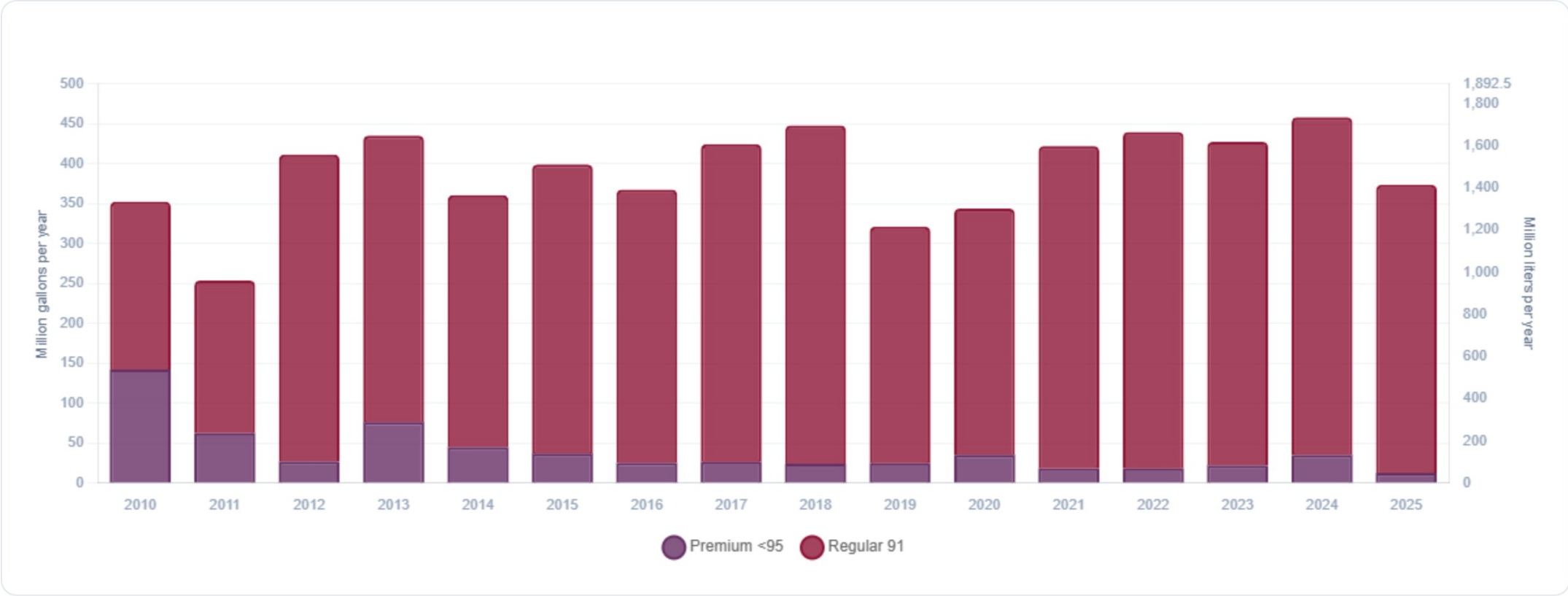
Source: Eppo

Gasoline Imports in Thailand



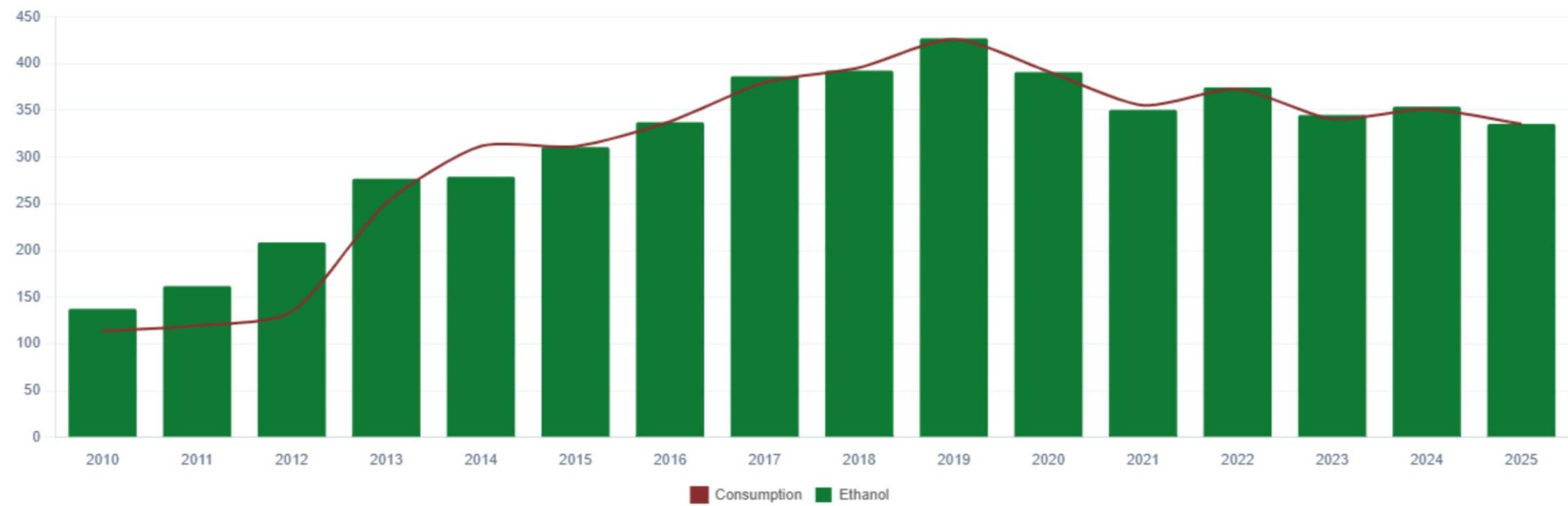
Source: Eppo

Gasoline Exports in Thailand



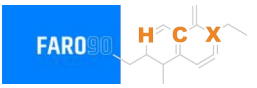
Source: Eppo

Ethanol Balance in Thailand



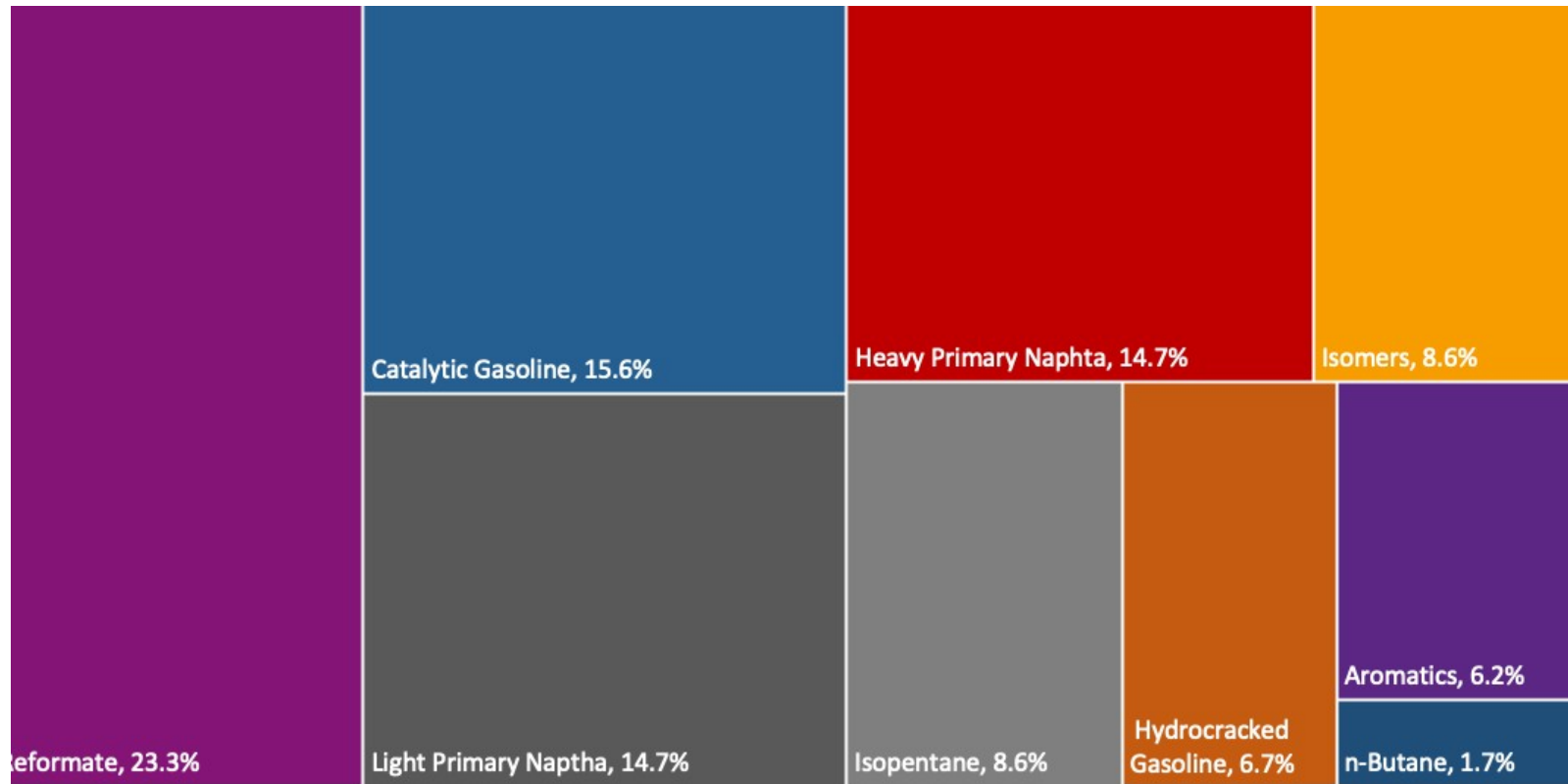
Source: Eppo, USDA

Gasoline Quality in Thailand

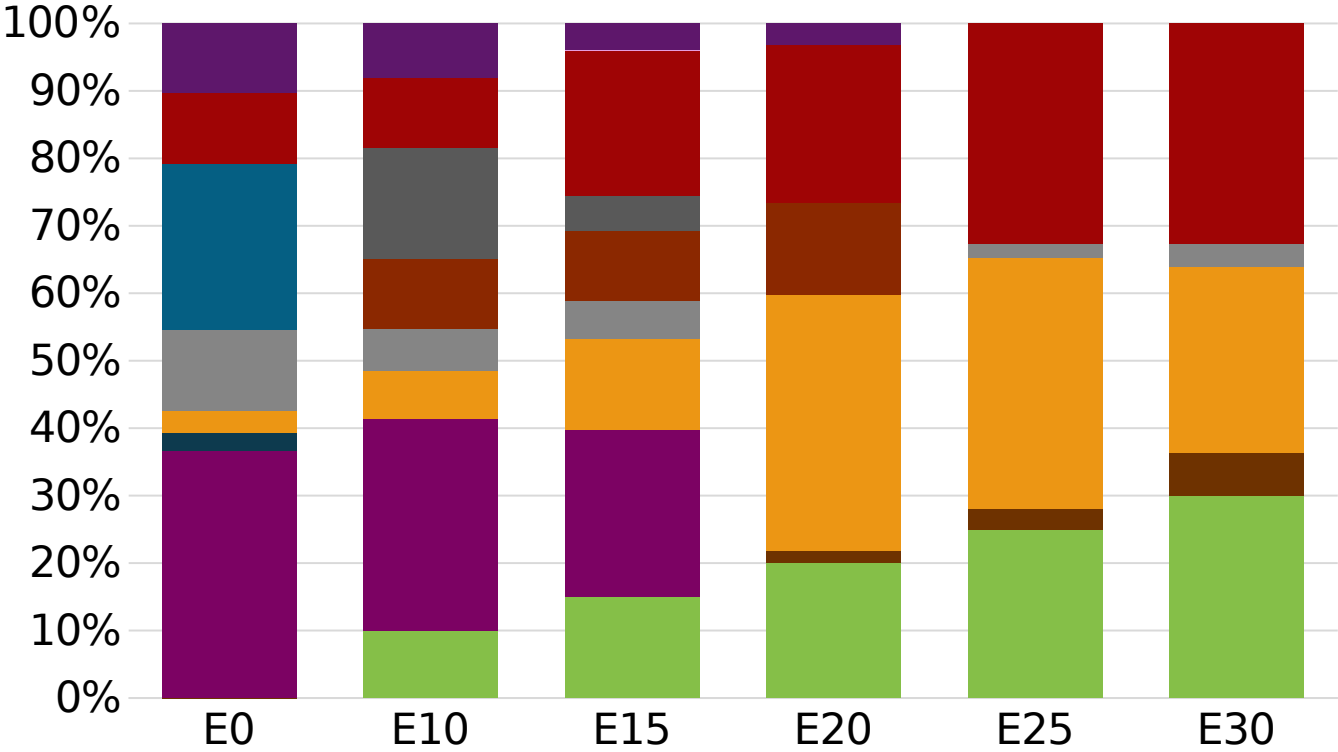


Source: Department of Energy Buisness

Thailand Available Blending Components



Thailand – Gasoline 91 – Constant Octane

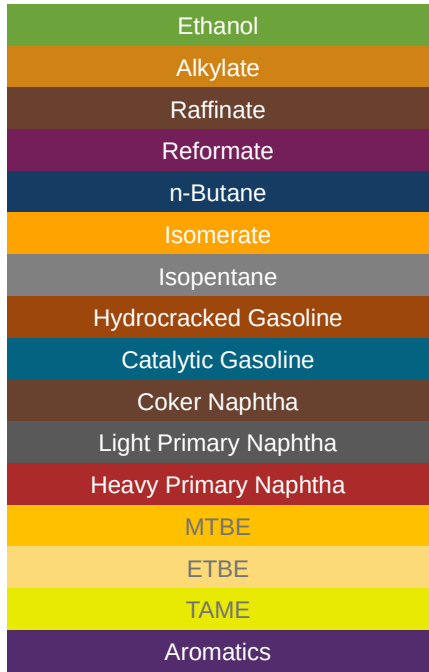
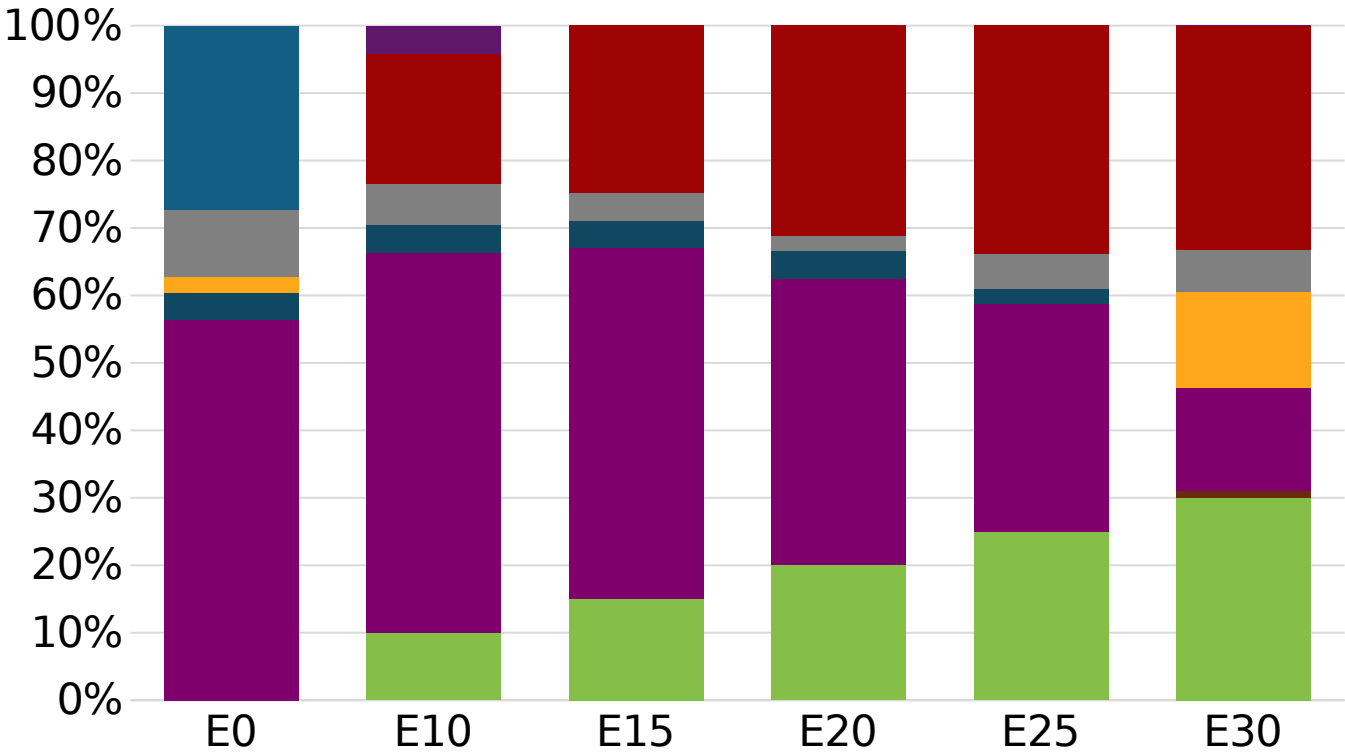


Average CIF 2025
Prices

Octane (RON)	91.0	91.0	91.0	91.4	91.6	93.1
Price (US\$/gal)	\$ 2.043	\$ 1.910	\$ 1.794	\$ 1.734	\$ 1.595	\$ 1.553

Source:
Faro90

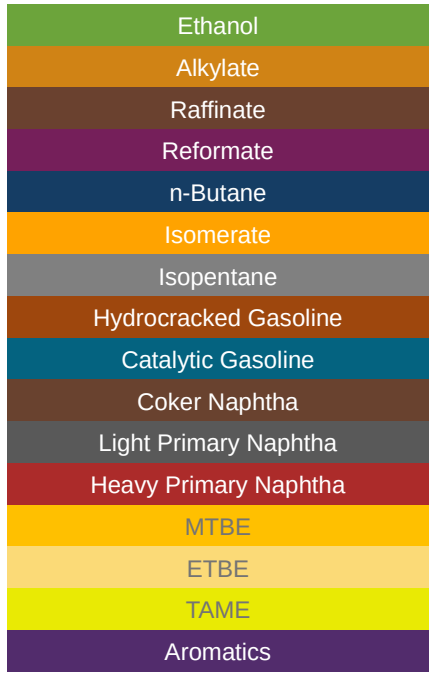
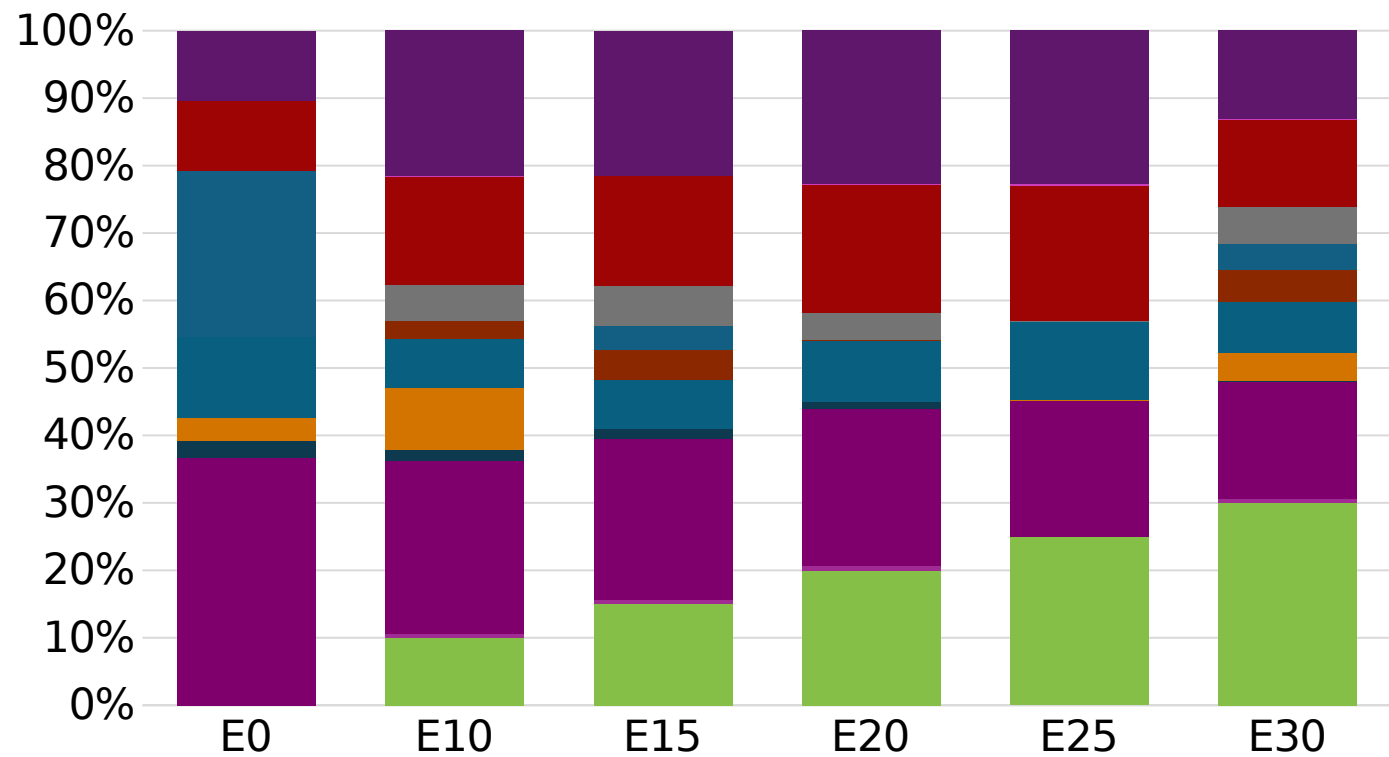
Thailand - Gasoline 95 - Constant Octane



Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.0
Price (US\$/gal)	\$ 2.243	\$ 2.063	\$ 1.966	\$ 1.862	\$ 1.755	\$ 1.660

Average CIF 2025
Prices

Thailand – Gasoline 91 – Increased Octane

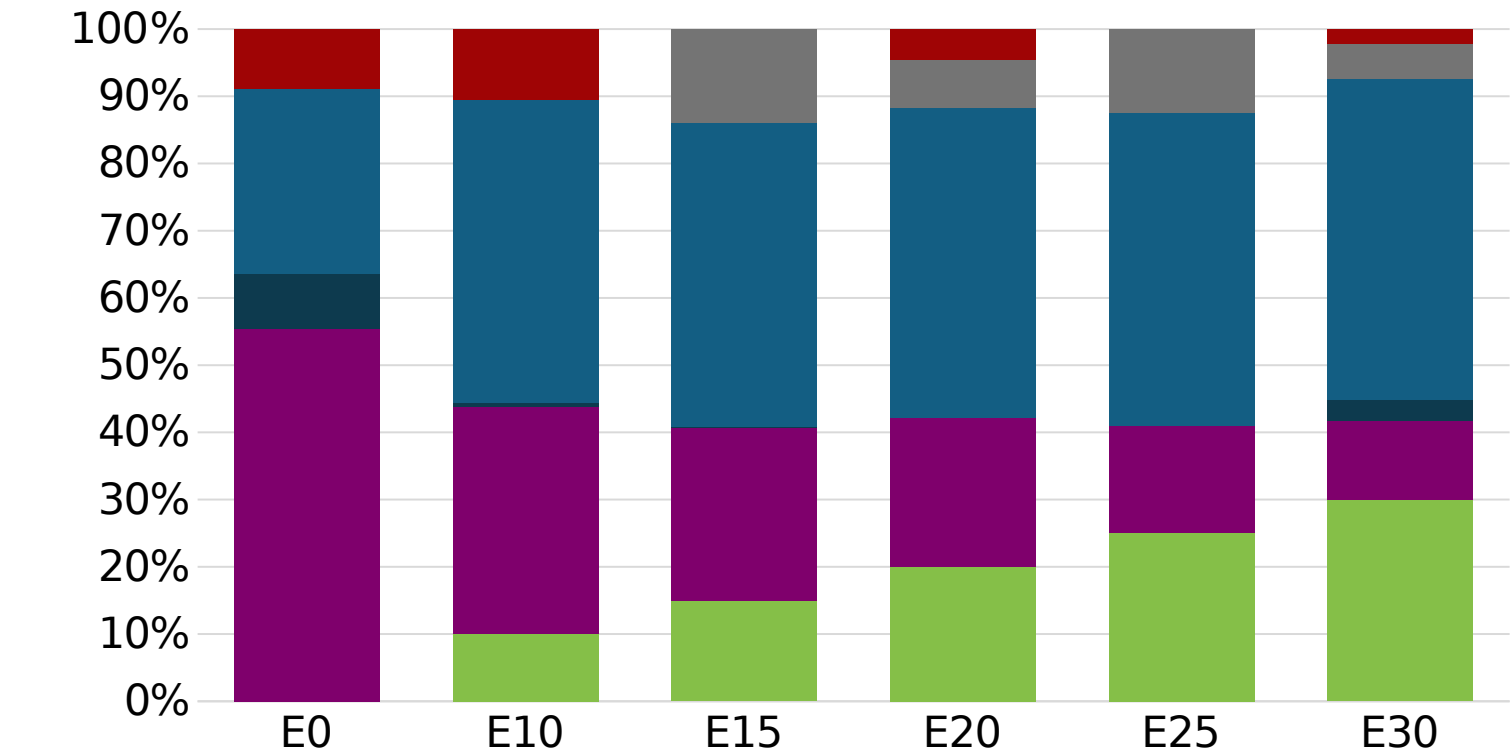


Average CIF 2025
Prices

Octane (RON)	91.0	93.8	95.7	97.5	99.2	100.7
Price (US\$/gal)	\$ 2.043	\$ 2.043	\$ 2.043	\$ 2.043	\$ 2.043	\$ 2.043

Source:
Faro90

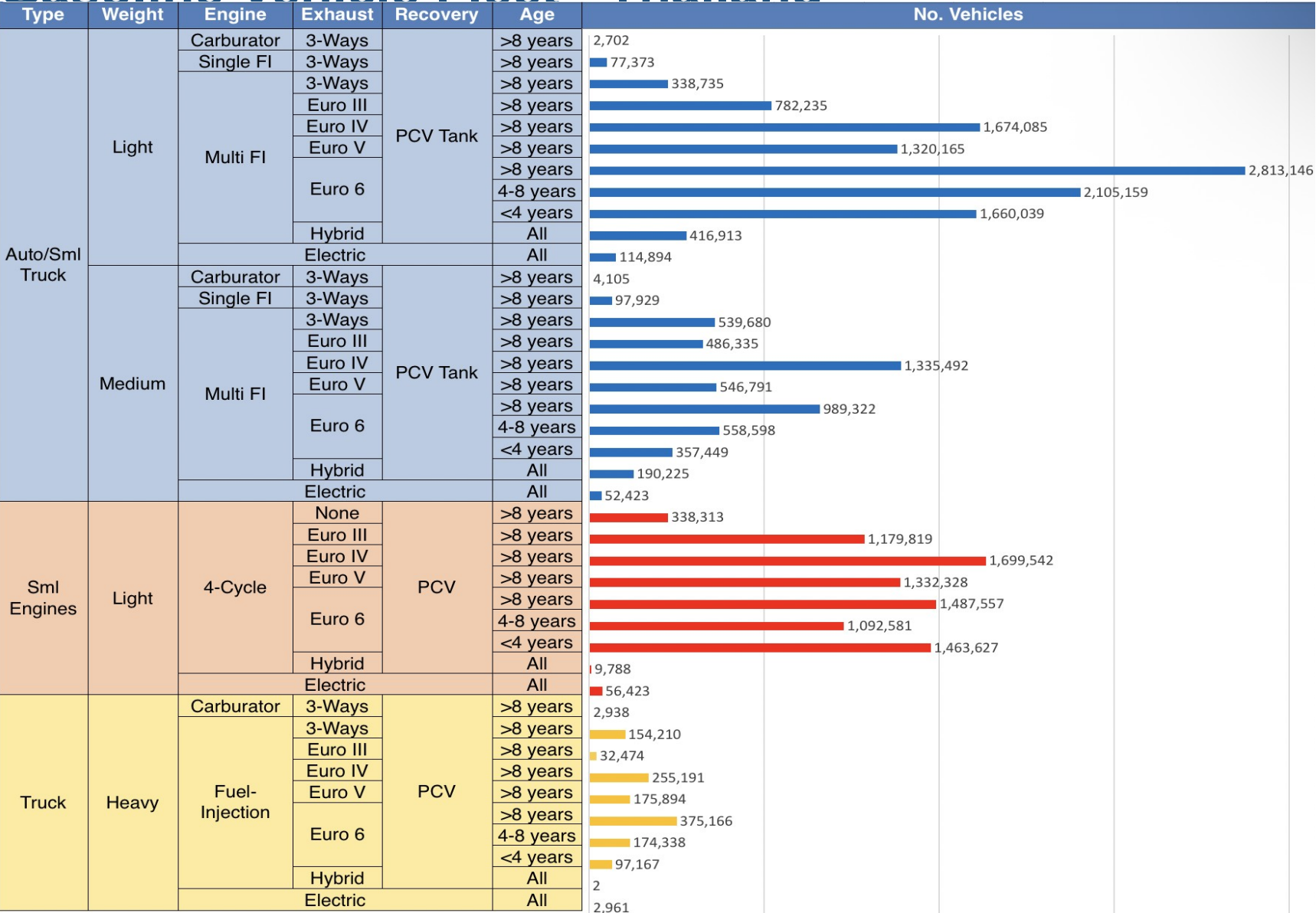
Thailand – Gasoline 95 – Increased Octane



Average CIF 2025
Prices

Octane (RON)	95.0	97.9	99.7	101.2	102.8	104.2
Price (US\$/gal)	\$ 2.292	\$ 2.292	\$ 2.292	\$ 2.292	\$ 2.292	\$ 2.292

Gasoline Vehicle Fleet - Thailand

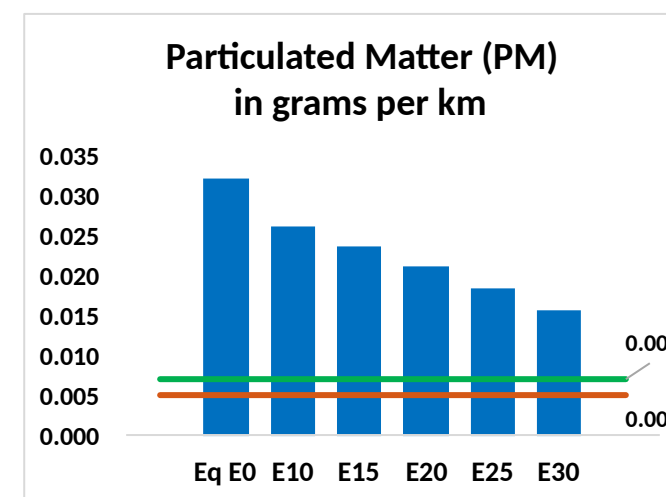
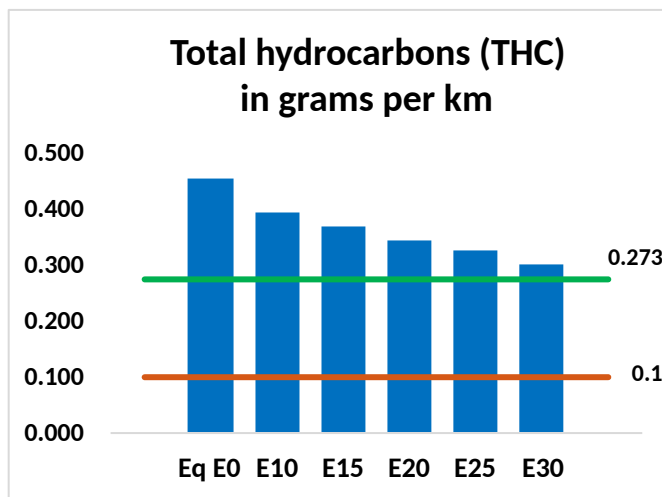
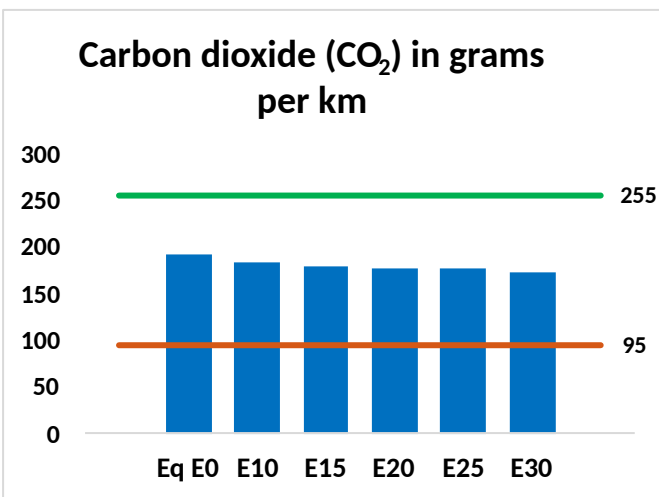
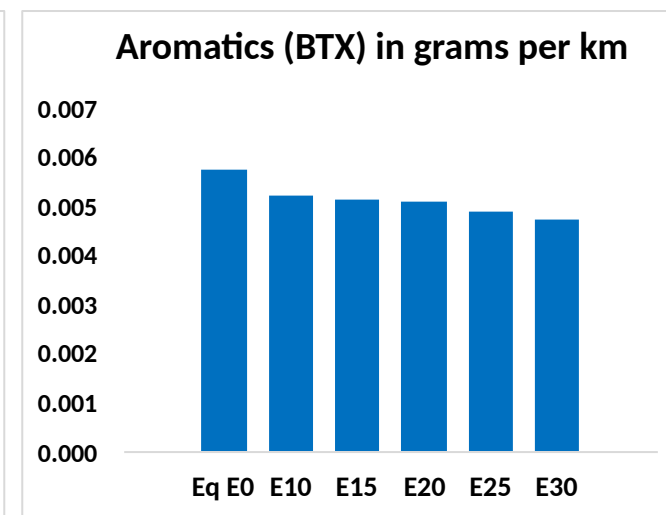
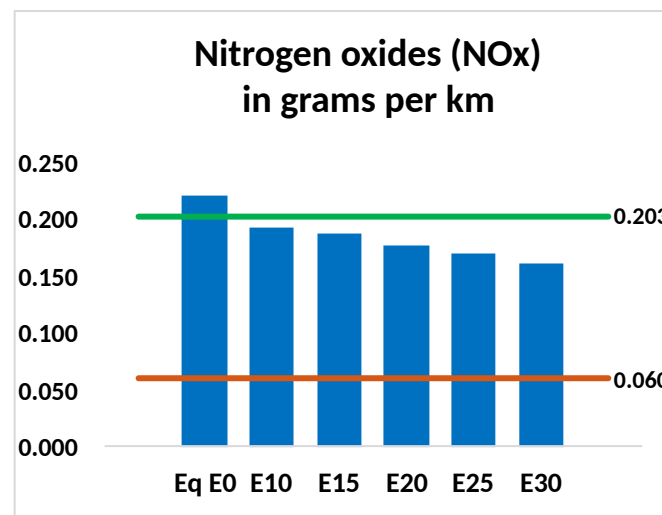
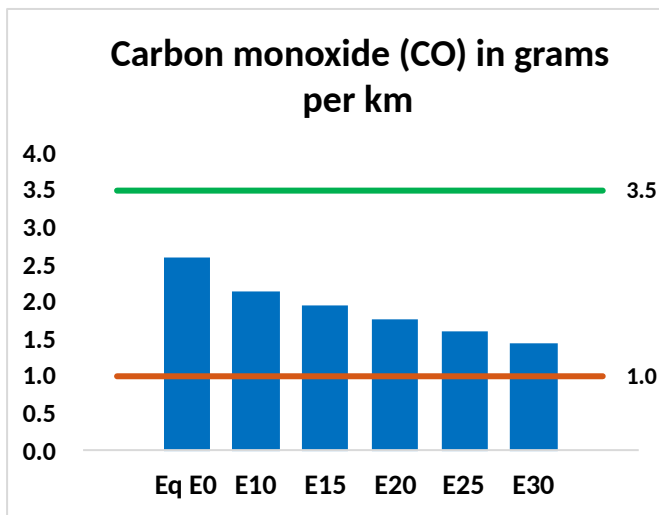


Vehicle Fleet: 26,394,115
Gasoline Vehicle Fleet: 23,593,936

Source: datagov.mot.go.th

Thailand – Vehicular Emissions

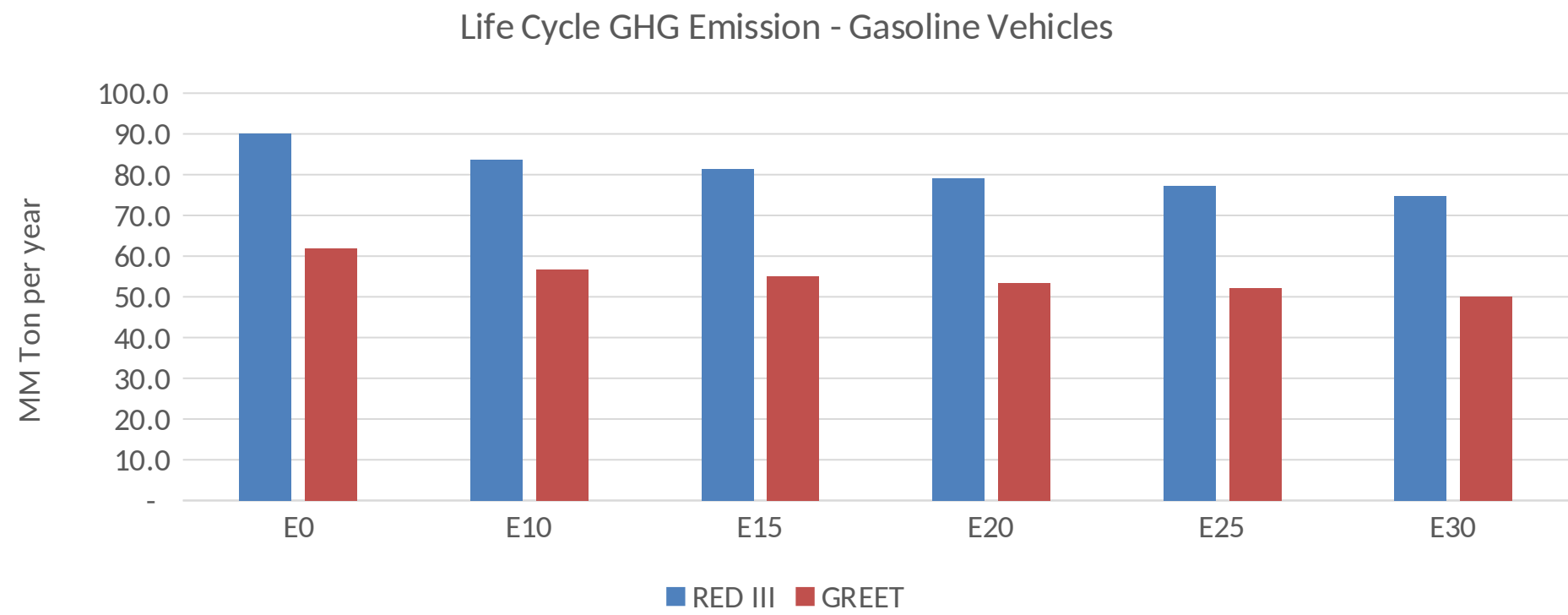
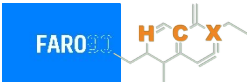
— TIER III BIN 125 United States
— Euro 6 Standard



Thailand – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	1,272,173	1,161,670	1,051,167	958,136	863,375	790,165	712,435	-17%	-32%
CO2	94,740,194	92,304,017	90,003,184	88,193,646	87,298,216	86,891,621	85,289,959	-5%	-8%
VOC	223,716	208,572	193,427	181,417	169,515	160,419	148,085	-14%	-24%
NOx	109,552	101,153	95,401	92,476	87,758	84,063	79,710	-13%	-20%
SOx	1,084	1,002	920	851	784	712	637	-15%	-28%
NH3	34,761	34,417	34,073	34,234	34,620	34,890	35,115	-2%	0%
N2O	1,377	1,370	1,364	1,371	1,399	1,415	1,431	-1%	2%
CH4	52,376	52,376	52,376	53,161	54,313	55,309	56,251	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	1,358	1,289	1,221	1,172	1,126	1,092	1,043	-10%	-17%
Acetaldehyde	3,940	4,522	6,636	10,105	13,768	15,732	18,589	68%	249%
Formaldehyde	14,999	14,582	16,998	19,960	20,911	23,133	25,183	13%	39%
Benzene	2,835	2,724	2,581	2,540	2,519	2,409	2,332	-9%	-11%
PM2.5	5,761	5,237	4,713	4,252	3,802	3,318	2,812	-18%	-34%
PM10	10,064	9,149	8,234	7,429	6,641	5,797	4,913	-18%	-34%

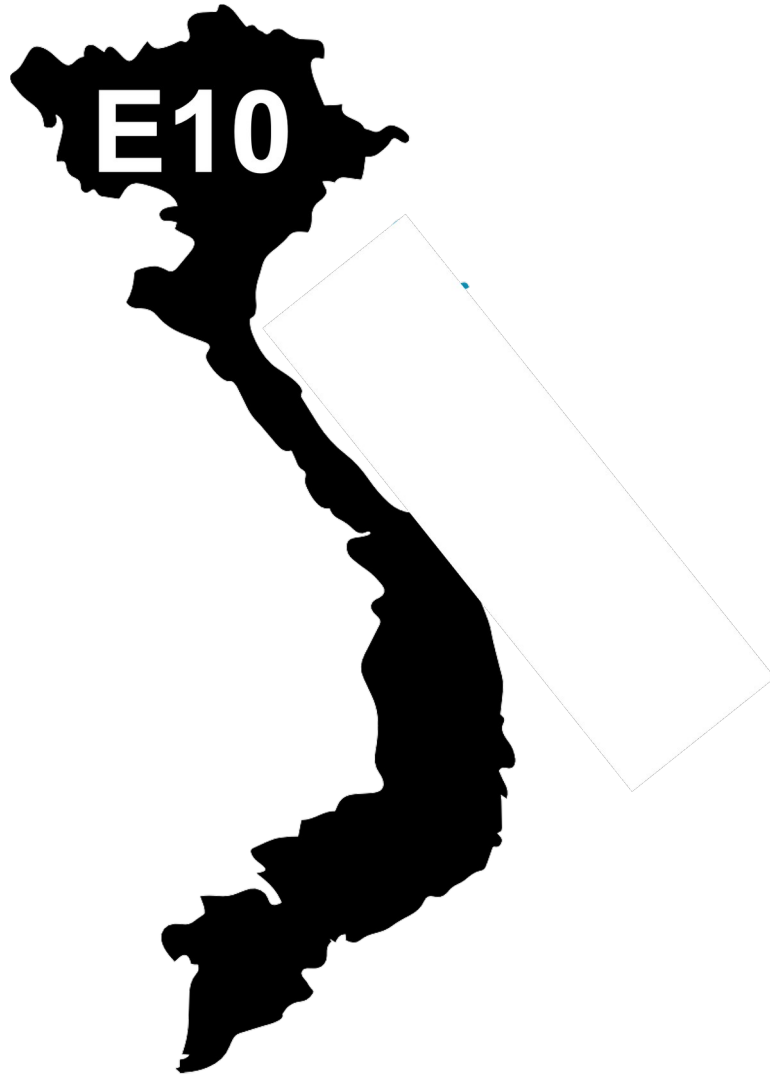
Thailand – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2025 (MMT/año)	356.2
Target @ 2035 (MMT/year)	211.6
Delta	144.7

Source: Greet, RED II/III, climateactiontracker.org, F90

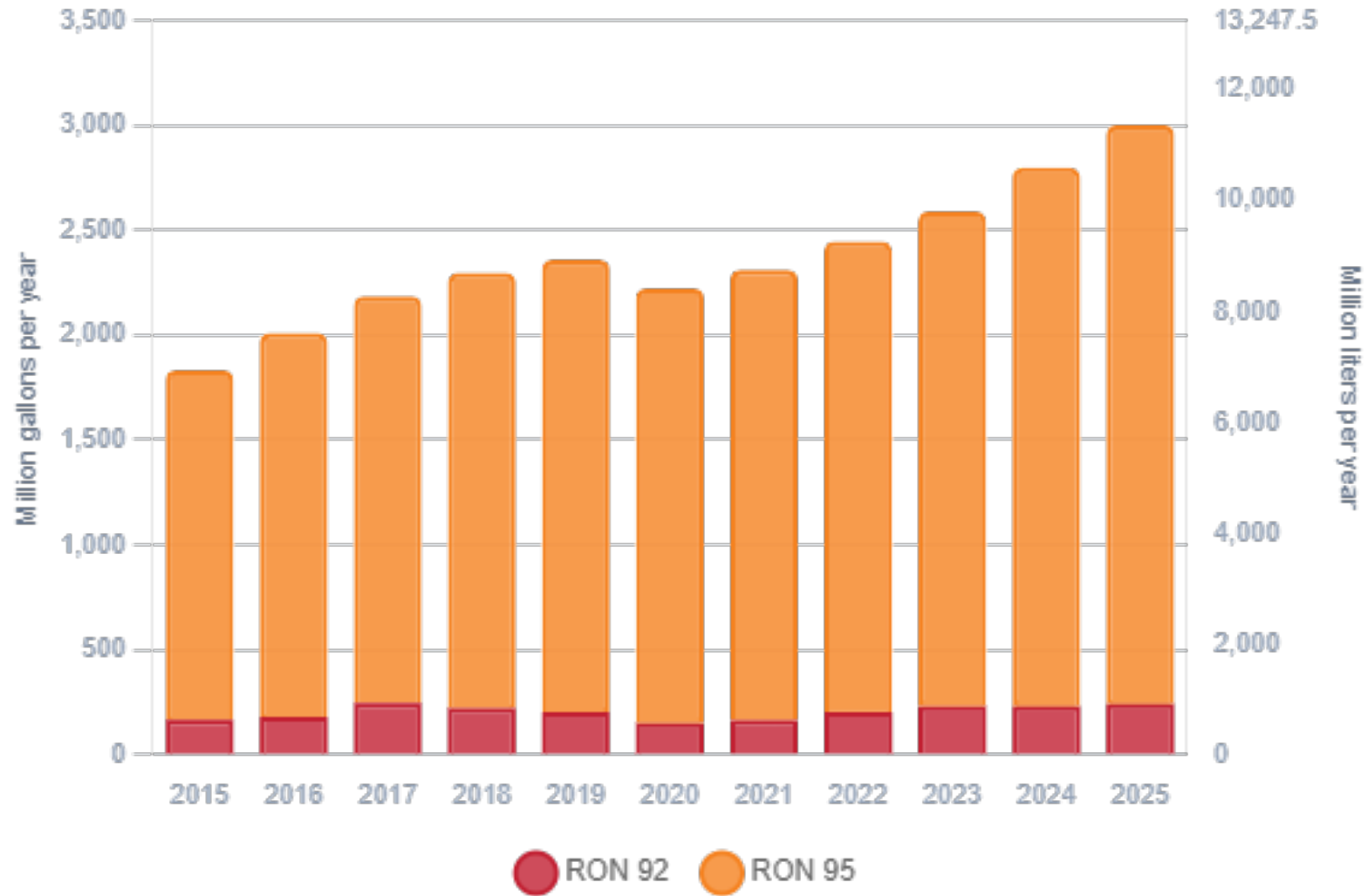
Ethanol Gasoline Blending in Vietnam



In 2025, gasoline consumption in Vietnam was 11,373 million liters (3,005 million gallons) and growth rate was 3.3%. Gasoline production and net imports were 7,949 and 3,423 million liters (2,100 and 904 million gallons) correspondingly. Vietnam complies offers three grades: RON 92 and RON 95 with an average octane of 94.7 RON.

Vietnam's bioethanol demand is growing rapidly, driven by government policies like a planned nationwide E10 gasoline mandate set to begin in June 2026. Ethanol consumption in Vietnam was 369 million liters (97 million gallons) representing 3.2% of gasoline demand. Ethanol production and Net Imports were 250 and 119 million liters (66 and 31 million gallons) correspondingly.

Gasoline Demand in Vietnam

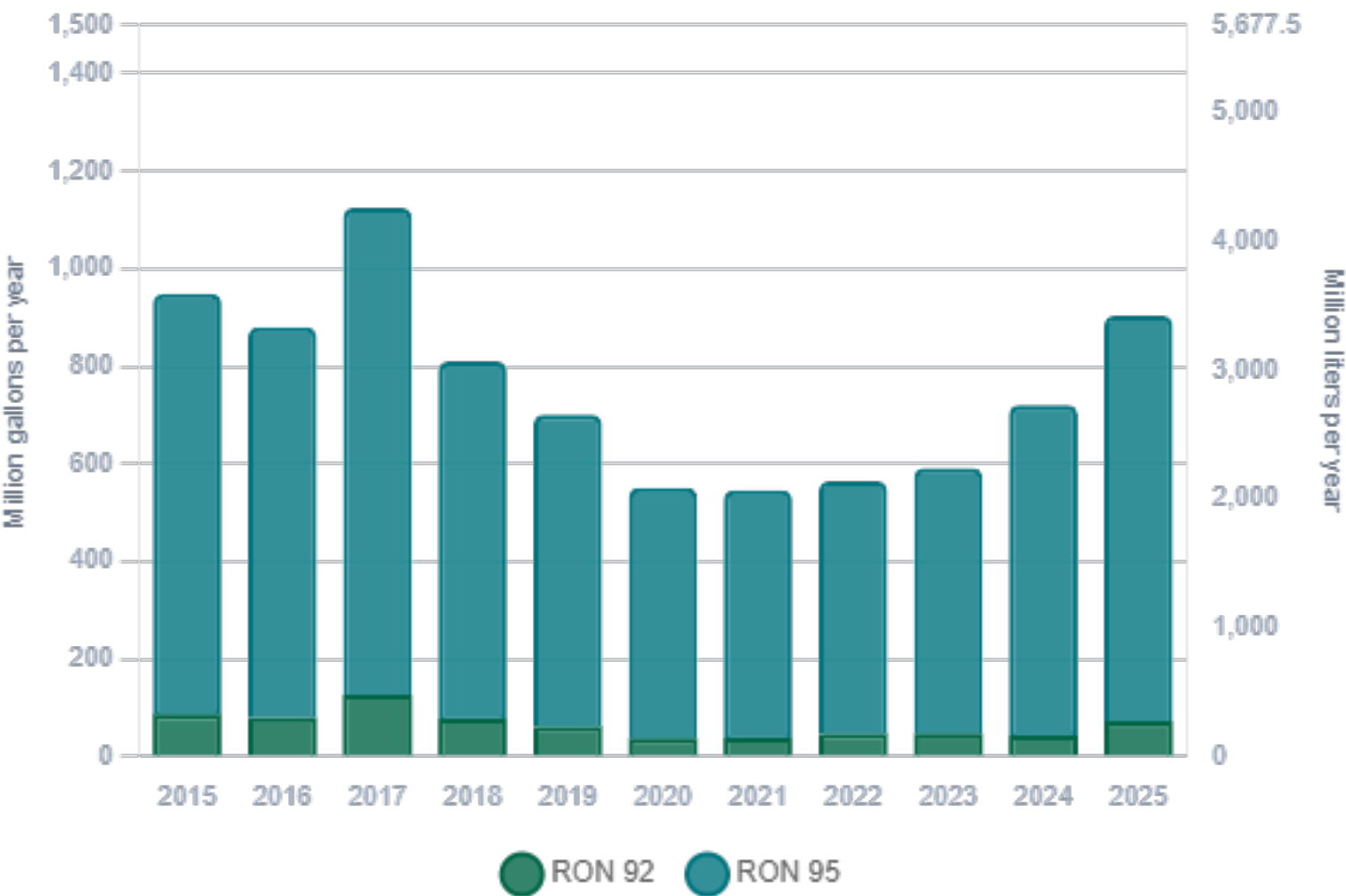


Source: Petro Vietnam, EIA, EIT data

The FARO90 logo is shown in a blue box, followed by a chemical structure of a substituted cyclohexadiene. The structure features a six-membered ring with two double bonds. Substituents include a hydrogen atom (H), a carbon atom (C), and a group labeled 'X' (which is a vinyl group, -CH=CH₂). The ring also has methyl groups at various positions.

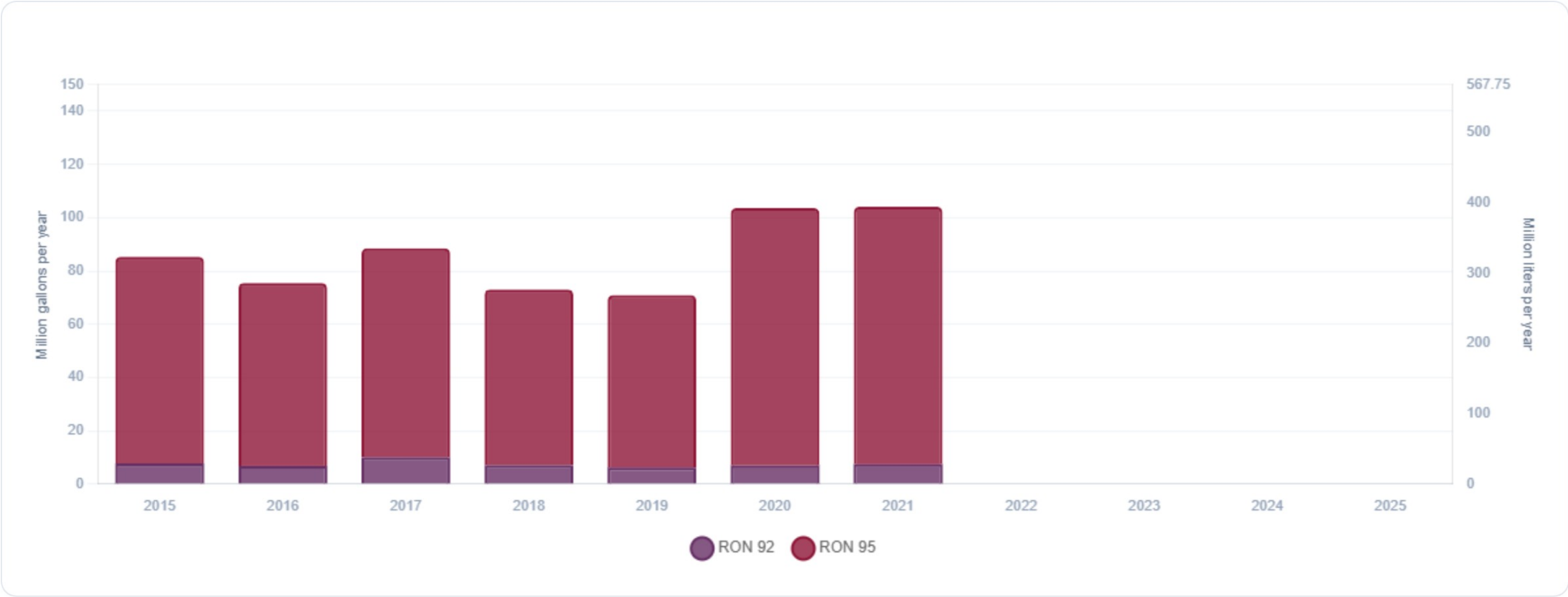


Gasoline Imports in Vietnam



Source: Petro Vietnam, EIA, EIT data

Gasoline Exports in Vietnam



Source: Petro Vietnam, EIA, EIT data

Ethanol Balance in Vietnam



Source: Petro Vietnam, USDA

Gasoline Quality in Vietnam

Name	QCVN 01:2022/BKHCN					
Implementation Year	2023					
Applicability	All territory					
Grade	RON 92		RON 95		RON 97	
	Min	Max	Min	Max	Min	Max
RON	92.0		95.0		97.0	
MON	-		-		-	
AKI	-		-		-	
Benzene (%vol)		1.0		1.0		1.0
Aromatics (% vol)		35.0		35.0		35.0
Olefins (%vol)		18.0		18.0		18.0
Lead (g/L)		0.005		0.005		0.005
Manganese (mg/L)		5.0		5.0		5.0
Sulfur (ppm)		10		10		10
		2.7		2.7		2.7
Oxygen (%w)						
Ethanol (%vol)		-		-		-
MTBE (%vol)		-		-		-
RVP (psi)		8.7		8.7		8.7

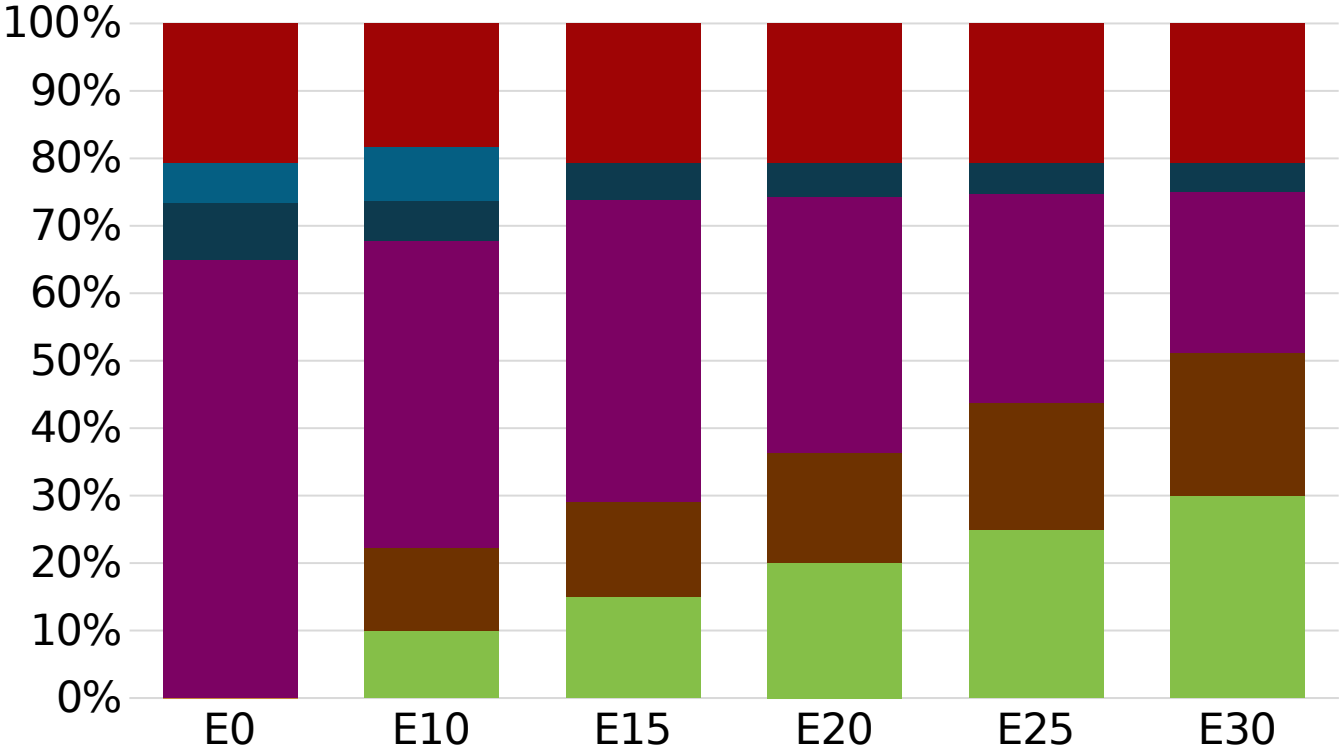
Source: Caselaw

Vietnam Available Blending Components



Source:
Faro90

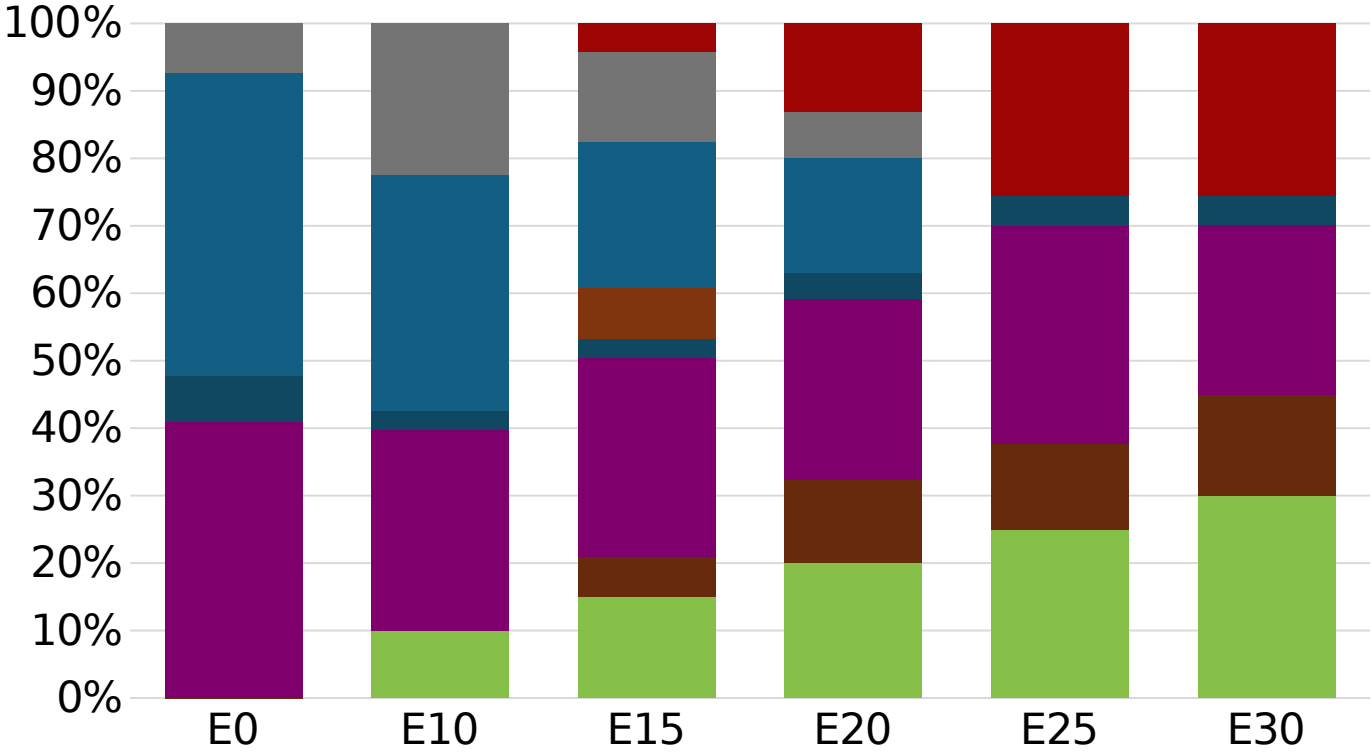
Vietnam – Gasoline 92 – Constant Octane



Average CIF 2025
Prices

Source:
Faro90

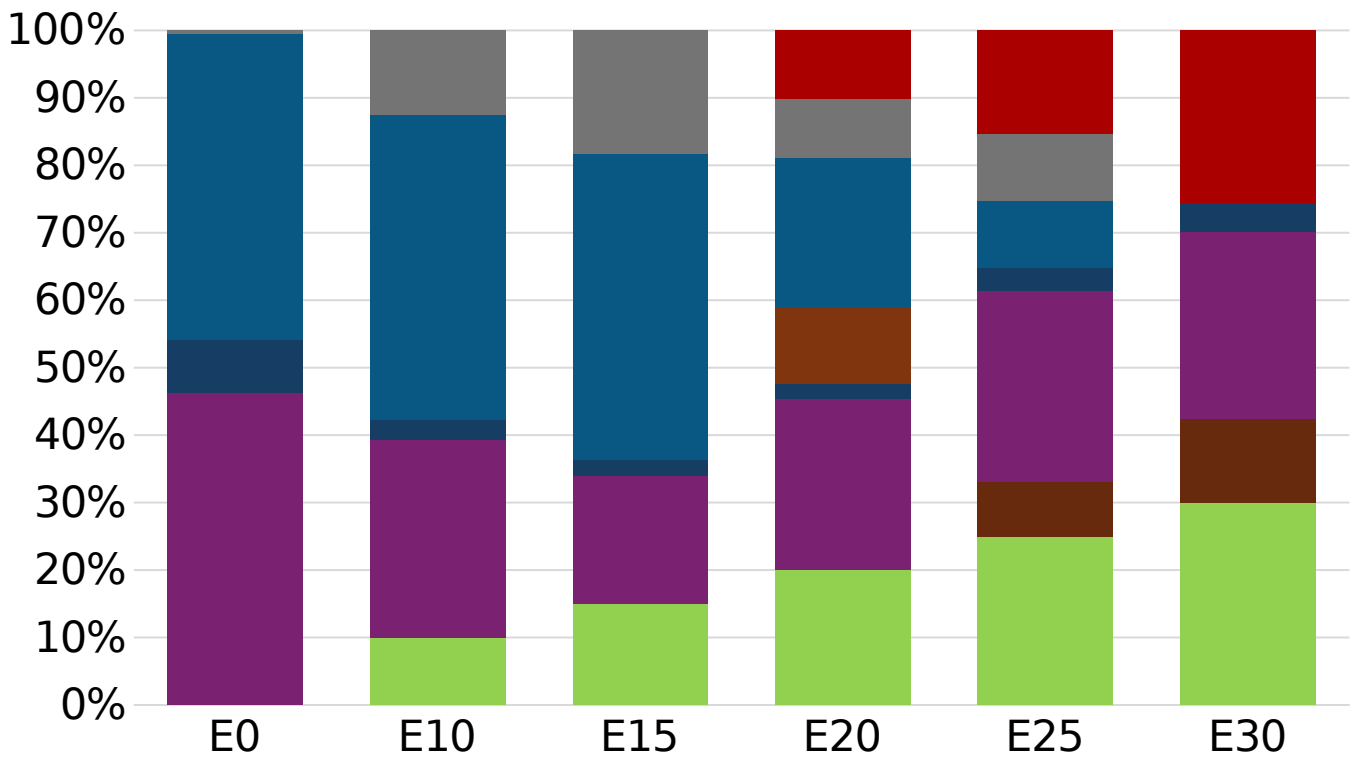
Vietnam – Gasoline 95 – Constant Octane



Average CIF 2025
Prices

Source:
Faro90

Vietnam - Gasoline 97 - Constant Octane

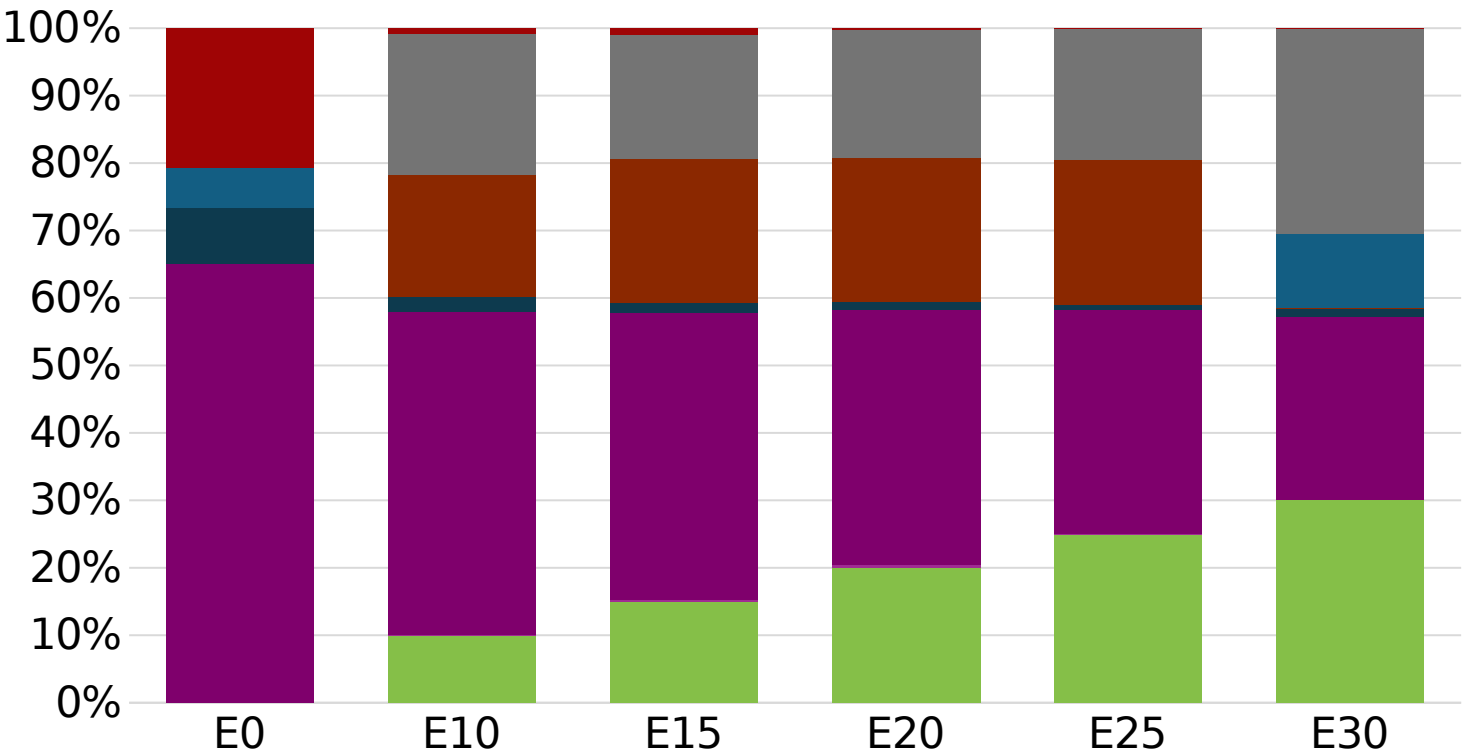


Octane (RON)	97.0	97.0	97.0	97.0	97.0	97.0
Price (US\$/gal)	\$ 2.329	\$ 2.203	\$ 2.108	\$ 2.015	\$ 1.892	\$ 1.770

Average CIF 2025 Prices

Source:
Faro90

Vietnam – Gasoline 92 – Increased Octane

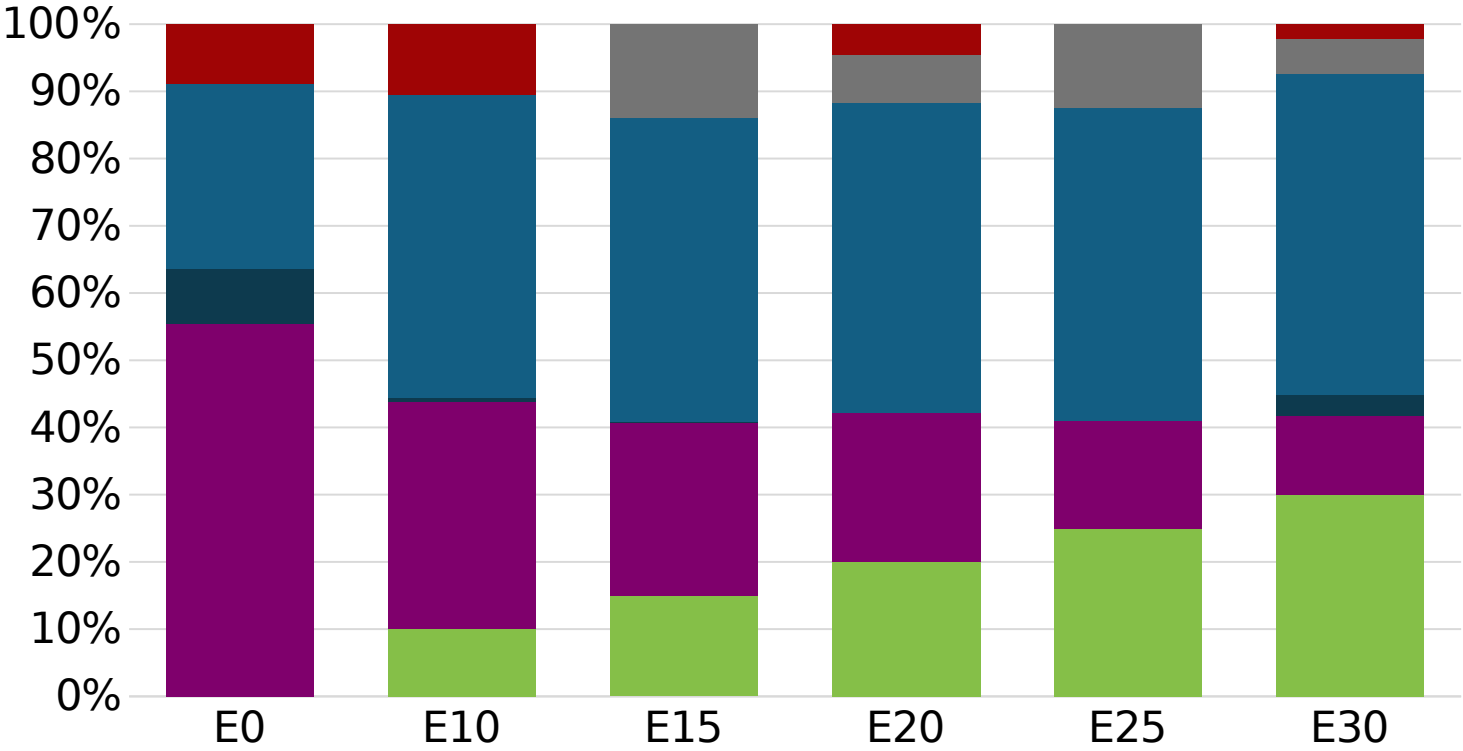


Average CIF 2025
Prices

Octane (RON)	92.0	94.1	95.7	97.4	99.0	101.0
Price (US\$/gal)	\$ 2.067	\$ 2.067	\$ 2.067	\$ 2.067	\$ 2.067	\$ 2.067

Source:
Faro90

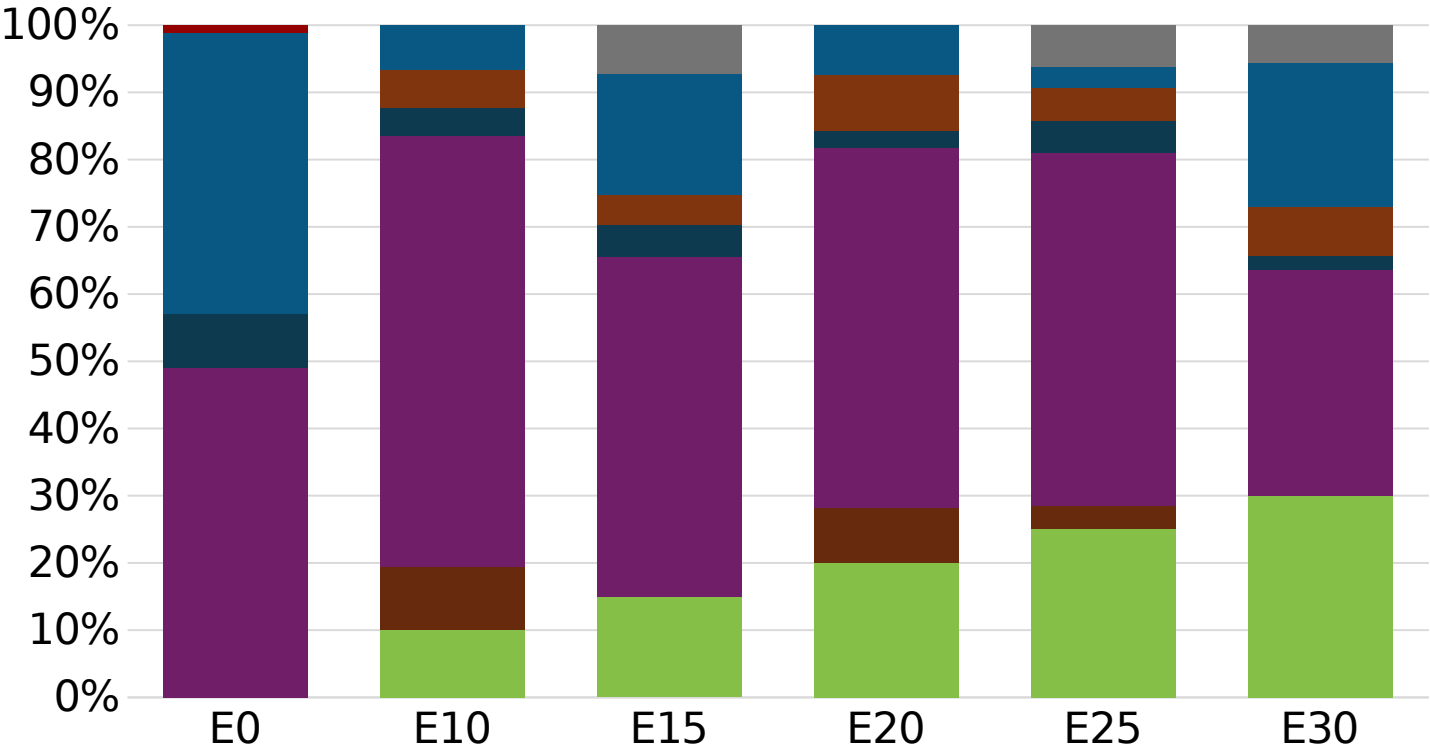
Vietnam – Gasoline 95 – Increased Octane



Average CIF 2025
Prices

Octane (RON)	95.0	96.8	98.4	100.0	101.6	103.7
Price (US\$/gal)	\$ 2.223	\$ 2.223	\$ 2.223	\$ 2.223	\$ 2.223	\$ 2.223

Vietnam – Gasoline 97 – Increased Octane



Average CIF 2025
Prices

Octane (RON)	97.0	99.1	101.1	101.9	103.9	104.9
Price (US\$/gal)	\$ 2.327	\$ 2.327	\$ 2.327	\$ 2.327	\$ 2.327	\$ 2.327

Source:
Faro90

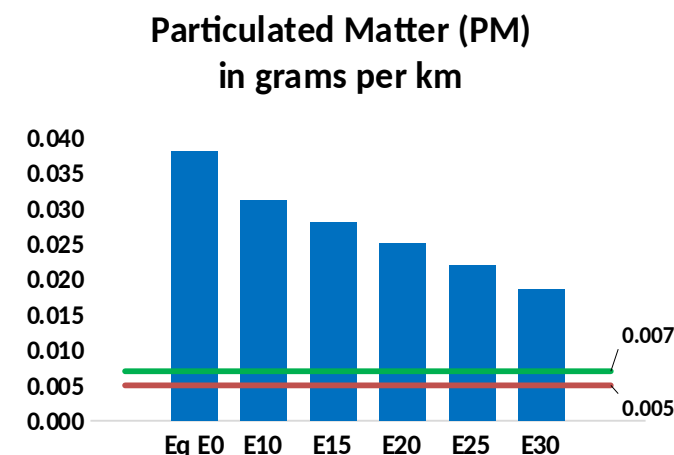
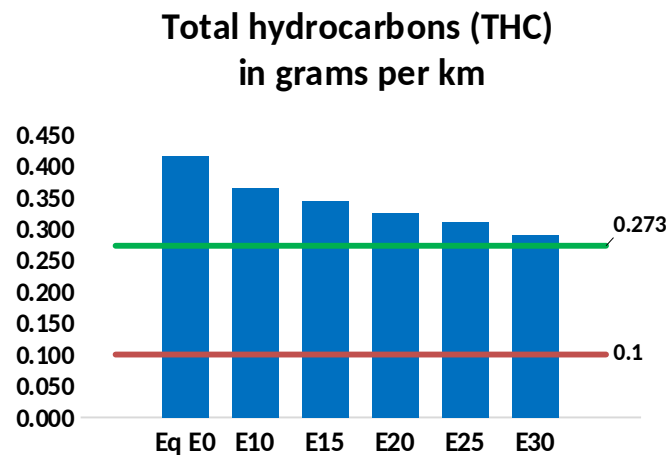
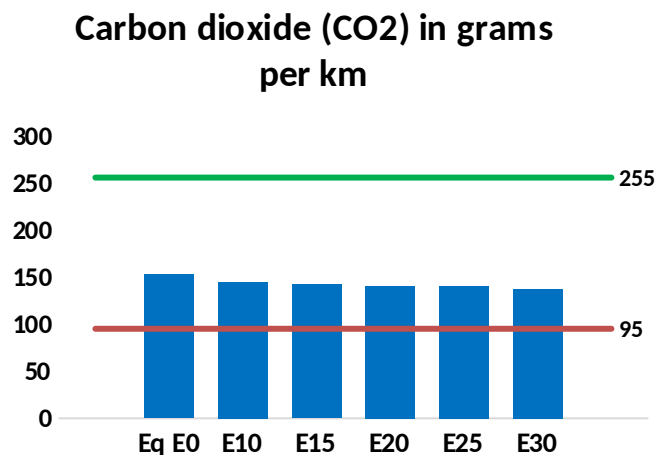
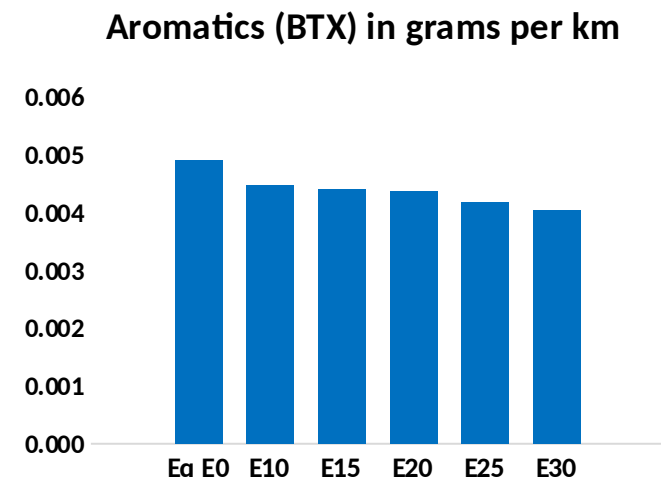
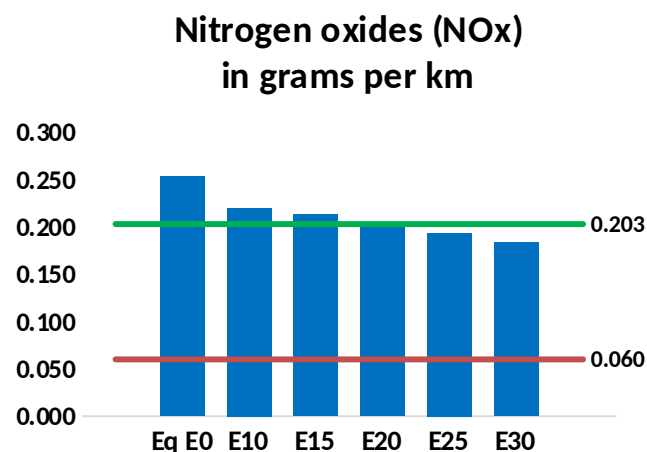
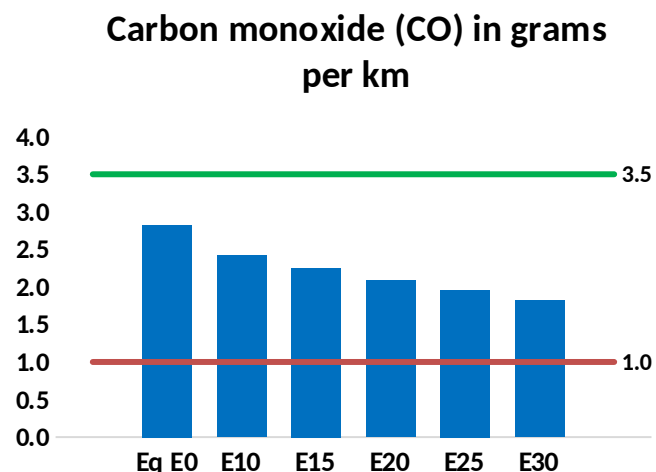
Gasoline Vehicle Fleet - Vietnam

Type	Weight	Engine	Exhaust	Recovery	Age	No. Vehicles					
Auto/Sml Truck	Light	Carburator	3-Ways	PCV Tank	>8 years	611					
		Single FI	3-Ways		>8 years	18,721					
		Multi FI	3-Ways		>8 years	75,441					
			Euro III		>8 years	96,476					
			Euro IV		>8 years	298,553					
			Euro V		>8 years	198,570					
			Euro 6		>8 years	469,620					
					4-8 years	590,965					
			<4 years		660,304						
		Hybrid	All		41,100						
	Electric			All	123,300						
	Medium	Carburator	3-Ways	PCV Tank	>8 years	281					
		Single FI	3-Ways		>8 years	8,602					
		Multi FI	3-Ways		>8 years	34,664					
			Euro III		>8 years	44,329					
			Euro IV		>8 years	137,179					
			Euro V		>8 years	91,239					
			Euro 6		>8 years	215,780					
					4-8 years	271,535					
			<4 years		303,396						
		Hybrid	All		18,900						
		Electric			All	56,700					
Sml Engines		Light	4-Cycle		None	PCV	>8 years	1,449,187			
	Euro III			>8 years	1,475,228						
	Euro IV			>8 years	4,377,632						
	Euro V			>8 years	2,727,054						
	Euro 6			>8 years	7,854,549						
				4-8 years	19,522,119						
	<4 years			7,715,447							
	Hybrid		All	0							
	Electric			All	2,300,000						
Truck	Heavy	Carburator	3-Ways	PCV	>8 years	530					
		Fuel- Injection	3-Ways		>8 years	119,365					
			Euro III		>8 years	131,538					
			Euro IV		>8 years	417,544					
			Euro V		>8 years	245,806					
			Euro 6		>8 years	665,533					
					4-8 years	837,500					
			<4 years		935,767						
		Hybrid	All		0						
		Electric			All	1,534					

Vehicle Fleet: 54,532,599
Gasoline Vehicle Fleet: 41,162,034

Source: MoIT

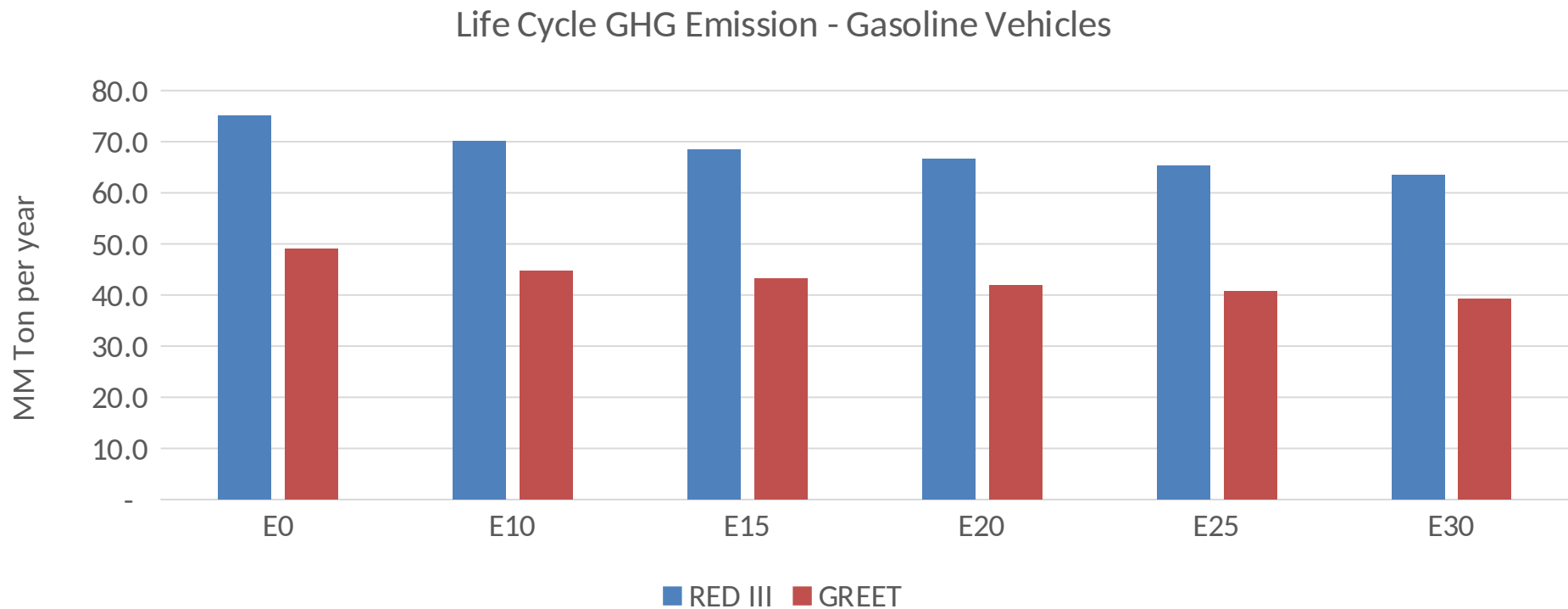
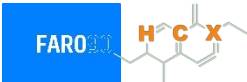
Vietnam – Vehicular Emissions



Vietnam – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	2,141,325	1,986,421	1,831,517	1,706,474	1,581,513	1,485,689	1,382,999	-14%	-26%
CO2	115,669,385	112,695,029	109,885,916	107,676,630	106,583,389	106,165,054	104,208,127	-5%	-8%
VOC	315,136	295,758	276,380	261,447	246,855	235,766	220,505	-12%	-22%
NOx	191,893	177,181	167,018	161,805	153,447	146,877	139,153	-13%	-20%
SOx	1,241	1,147	1,053	974	898	815	729	-15%	-28%
NH3	51,080	50,575	50,069	50,306	50,872	51,270	51,601	-2%	0%
N2O	1,830	1,820	1,812	1,821	1,859	1,879	1,901	-1%	2%
CH4	81,580	81,580	81,580	82,803	84,598	86,148	87,617	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	2,443	2,324	2,204	2,119	2,040	1,981	1,895	-10%	-17%
Acetaldehyde	6,540	7,506	11,013	16,772	22,850	26,111	30,853	68%	249%
Formaldehyde	25,725	25,010	29,155	34,234	35,865	39,677	43,192	13%	39%
Benzene	3,729	3,584	3,396	3,342	3,314	3,170	3,067	-9%	-11%
PM2.5	8,263	7,512	6,761	6,100	5,453	4,760	4,034	-18%	-34%
PM10	20,587	18,715	16,844	15,197	13,585	11,858	10,050	-18%	-34%

Vietnam – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2025 (MMT/año)	489.9
Target @ 2035 (MMT/year)	330.6
Delta	159.3

Source: Greet, RED II/III, climateactiontracker.org, F90